Biochemistry



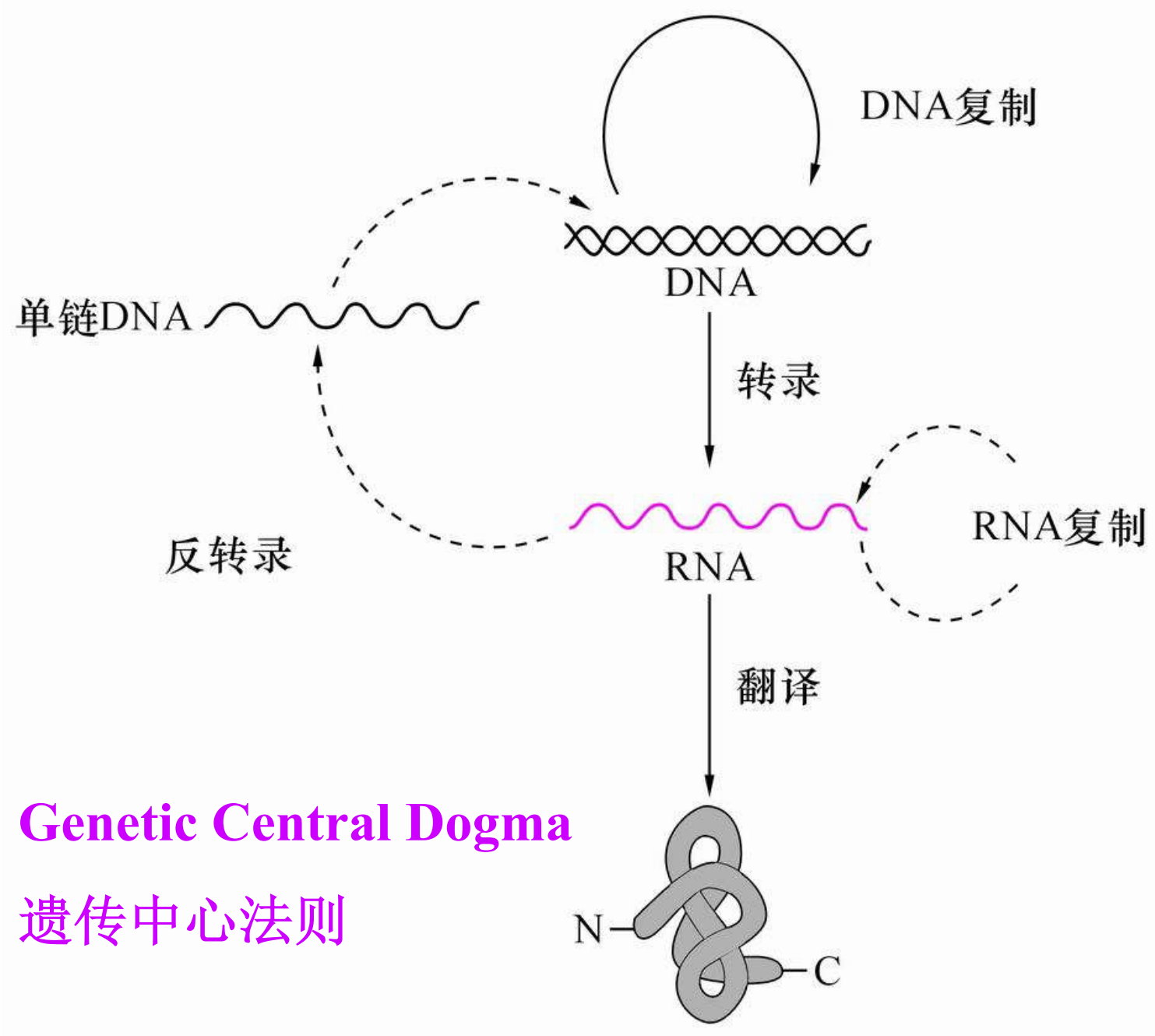
Section G
RNA synthesis and processing

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Biochemistry and Molecular Biology,
College of Life Sciences

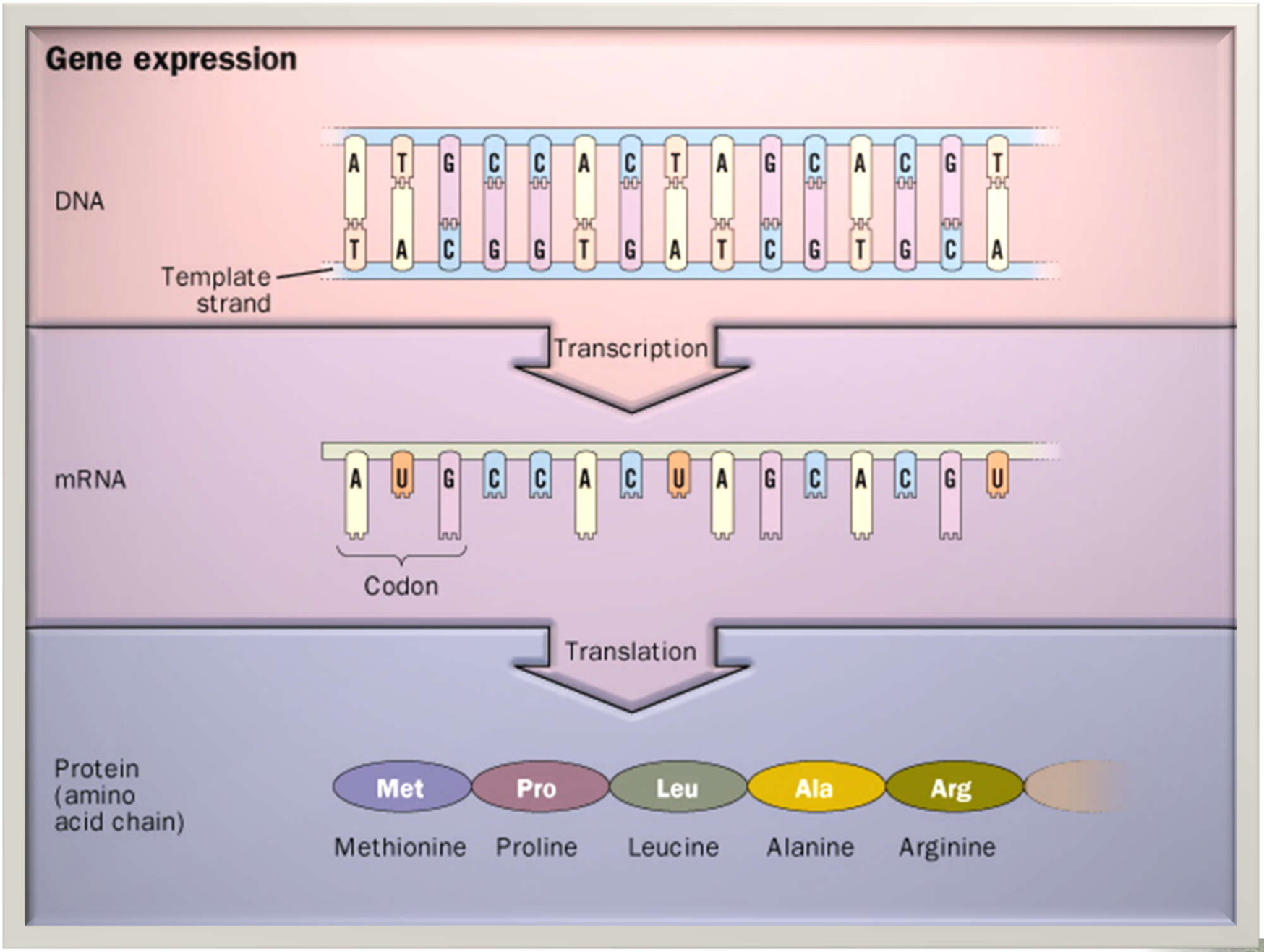
gdyang@sdau.edu.cn

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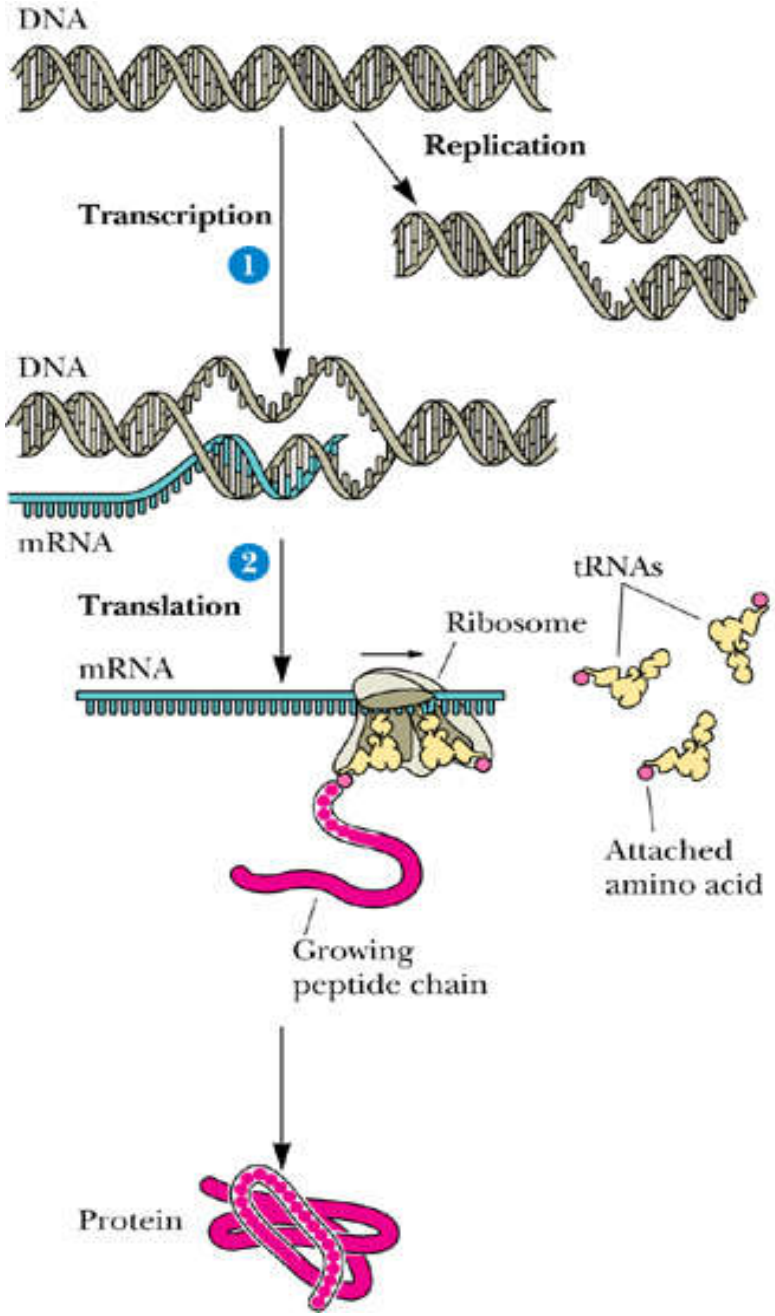


Genetic Central Dogma

遗传中心法则



Biochemistry



Replication
DNA replication yields two DNA molecules identical to the original one, ensuring transmission of genetic information to daughter cells with exceptional fidelity.

Transcription
The sequence of bases in DNA is recorded as a sequence of complementary bases in a single-stranded mRNA molecule.

Translation
Three-base codons on the mRNA corresponding to specific amino acids direct the sequence of building a protein. These codons are recognized by tRNAs (transfer RNAs) carrying the appropriate amino acids. Ribosomes are the "machinery" for protein synthesis.



G1 RNA structure: tRNA cloverleaf

G2 RNA Transcription in Prokaryotes

G3 The Lac operon

G5,G6 Transcription in Eukaryotes

G7 Processing of eukaryotic pre-mRNA

G8,G9 Processing of Ribosomal RNA, tRNA

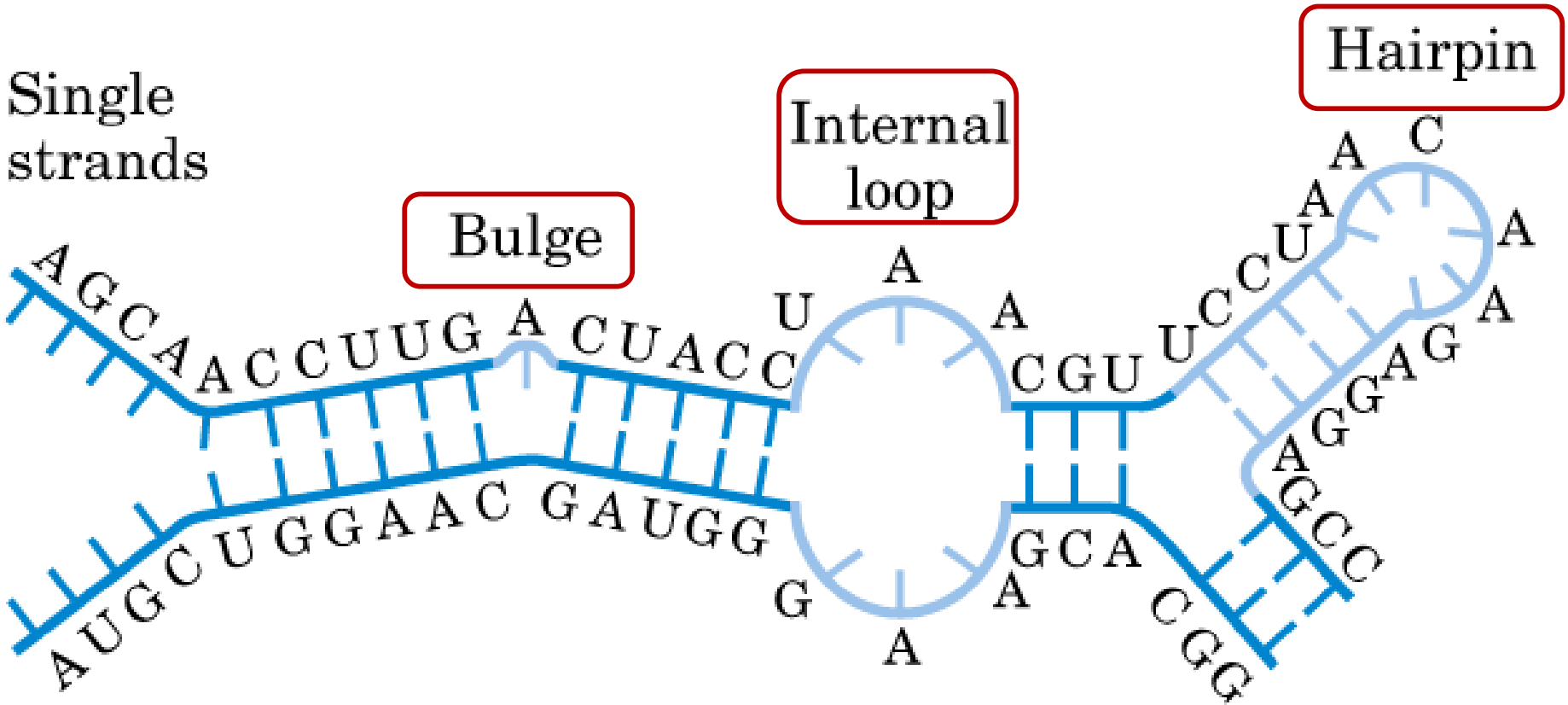


- **G1 RNA structure**

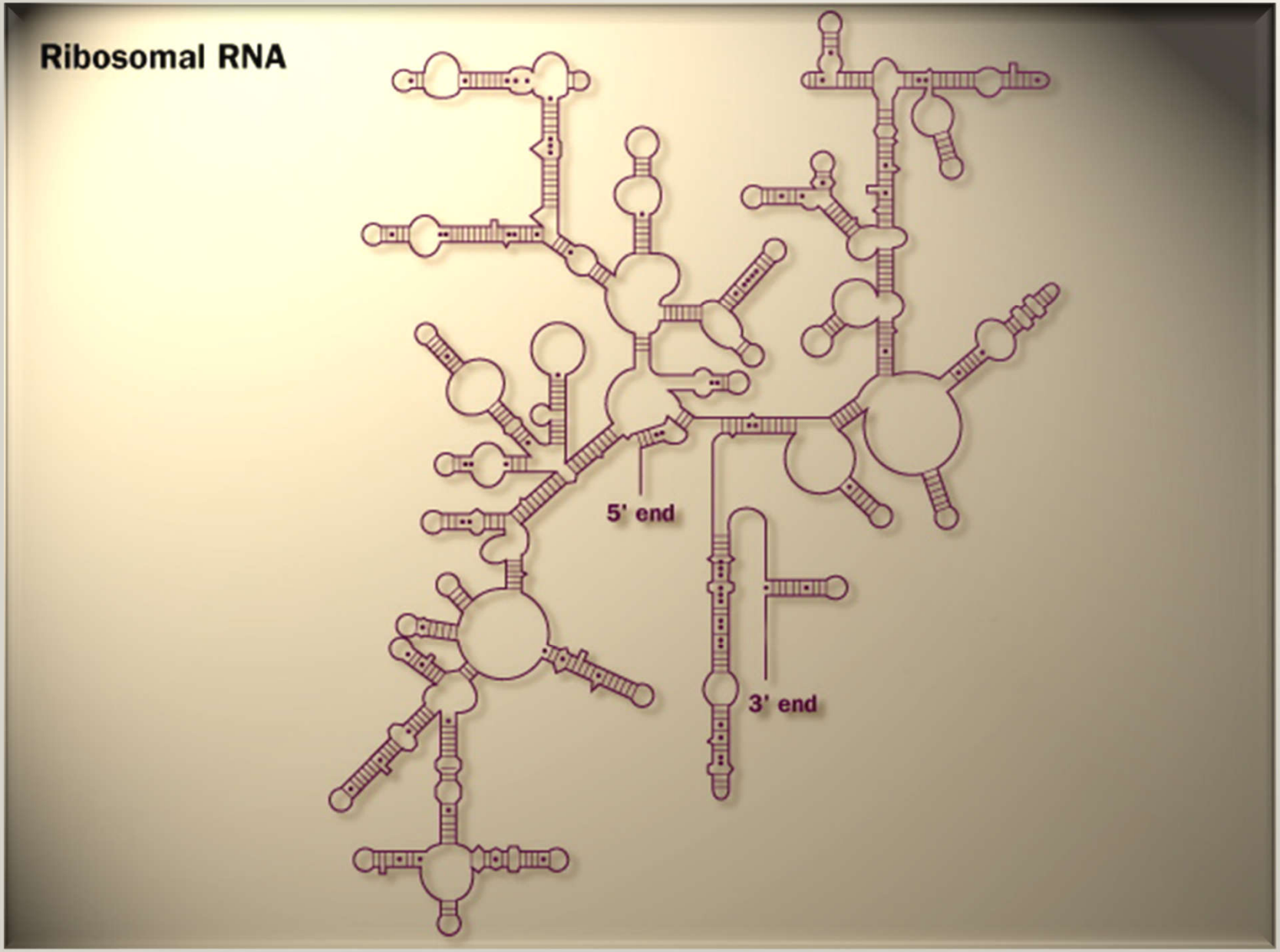
RNA molecules do not form simple, regular secondary structure but many form complex and unique three dimensional structures. (G1)

1. Single strand RNA tends to take right-handed helical conformation.
2. Self-complementary sequences on a RNA molecule lead to specific structures.
3. The standard base-pairing rules are followed (i.e., A with U, G with C).
4. Intrastrand base pairing makes structures including internal loops, and hairpins (helical).




Single strands
Bulge
Internal loop
Hairpin
AGCAACCUUG A CUACC
AUGCUGGAAC GAUGG
U A A CGU UCCU A C A A A G G A G G AGCC
G A GCA CGG

(a)

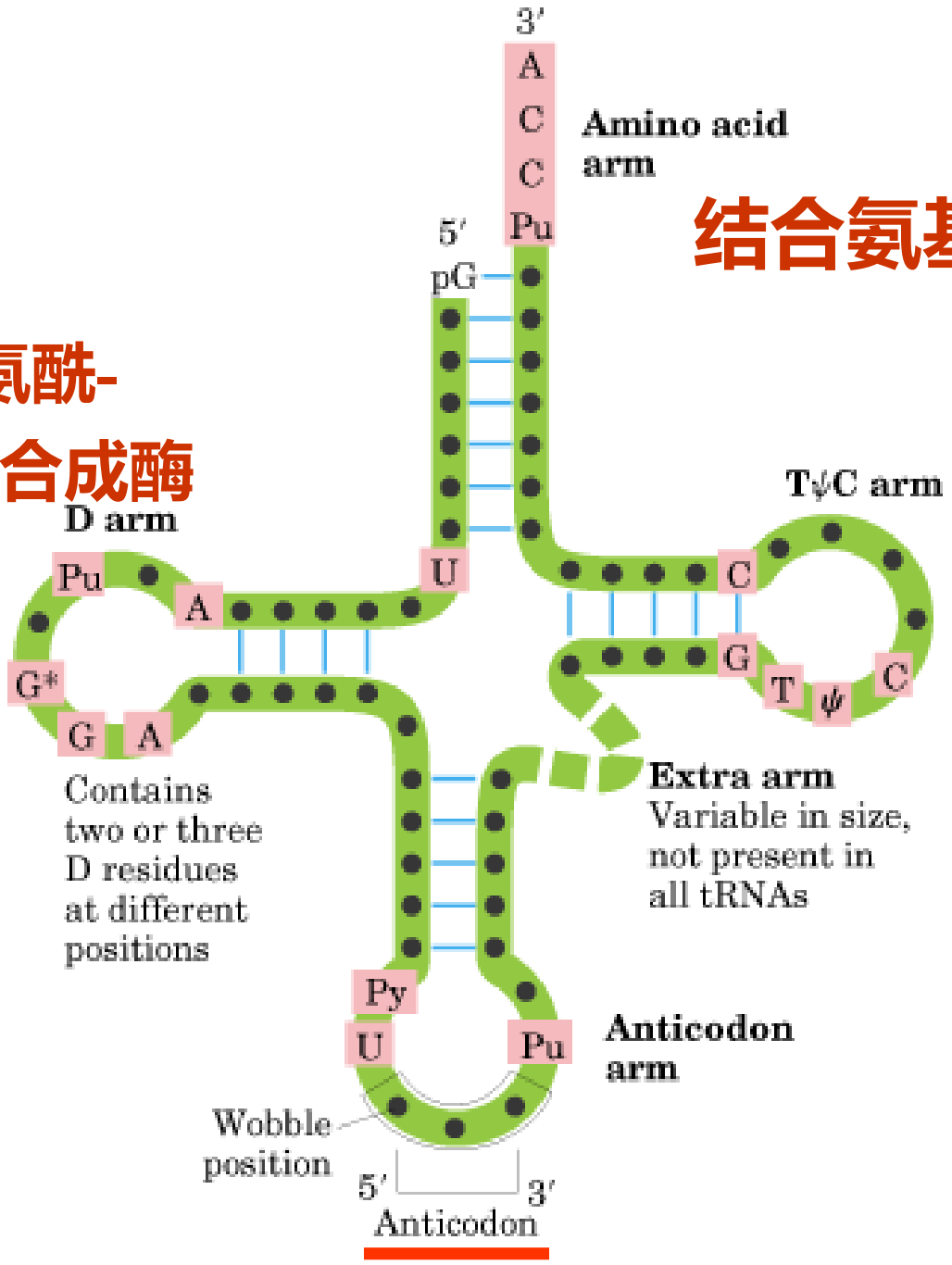

Ribosomal RNA
5' end
3' end

The structure and function of tRNA

结合氨基酸

识别氨酰-tRNA合成酶

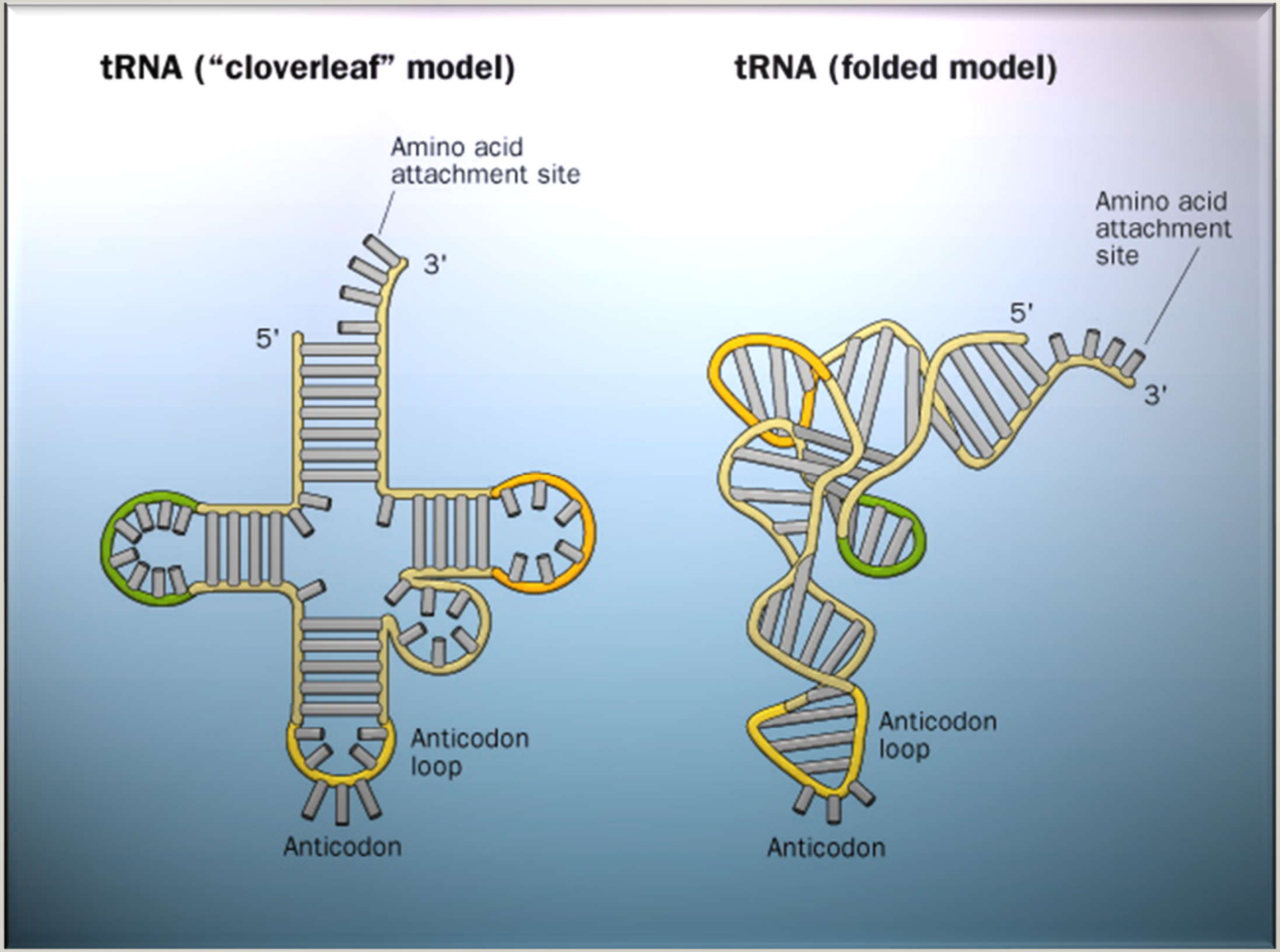
识别核糖体



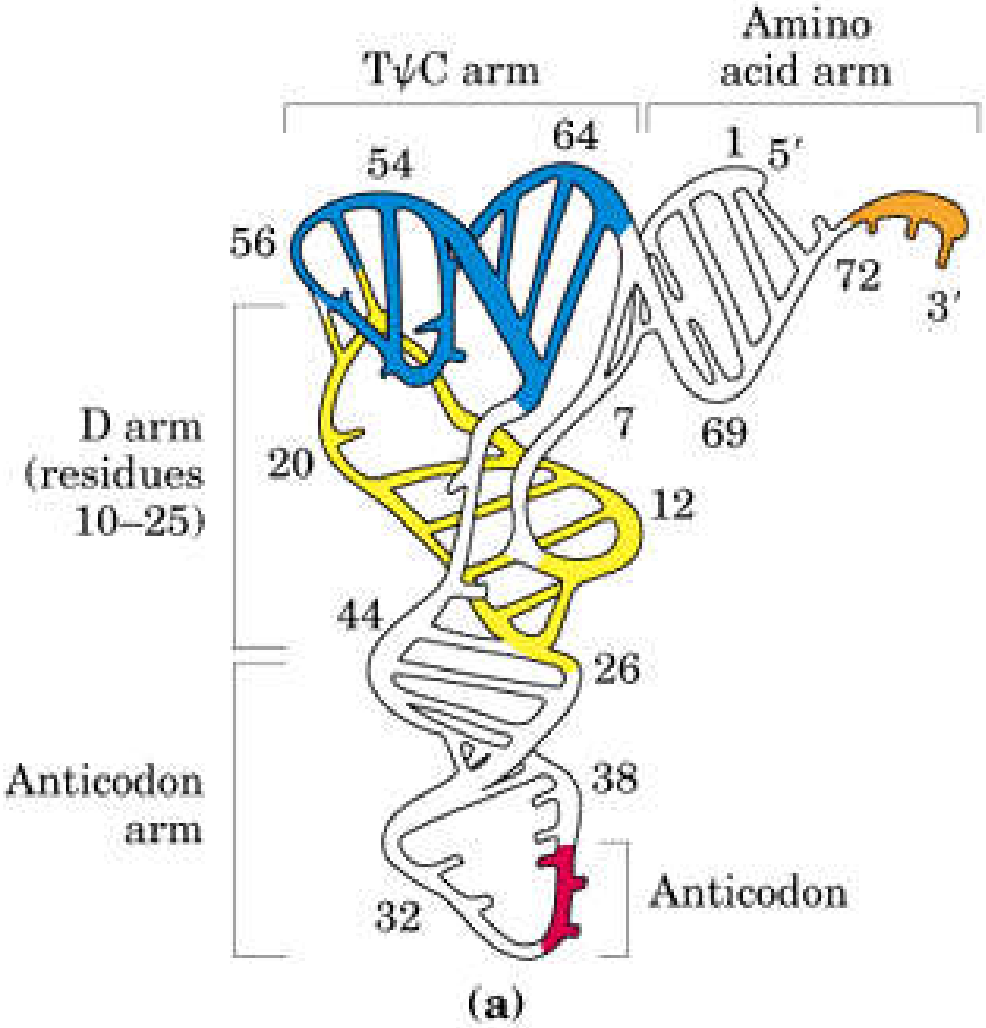


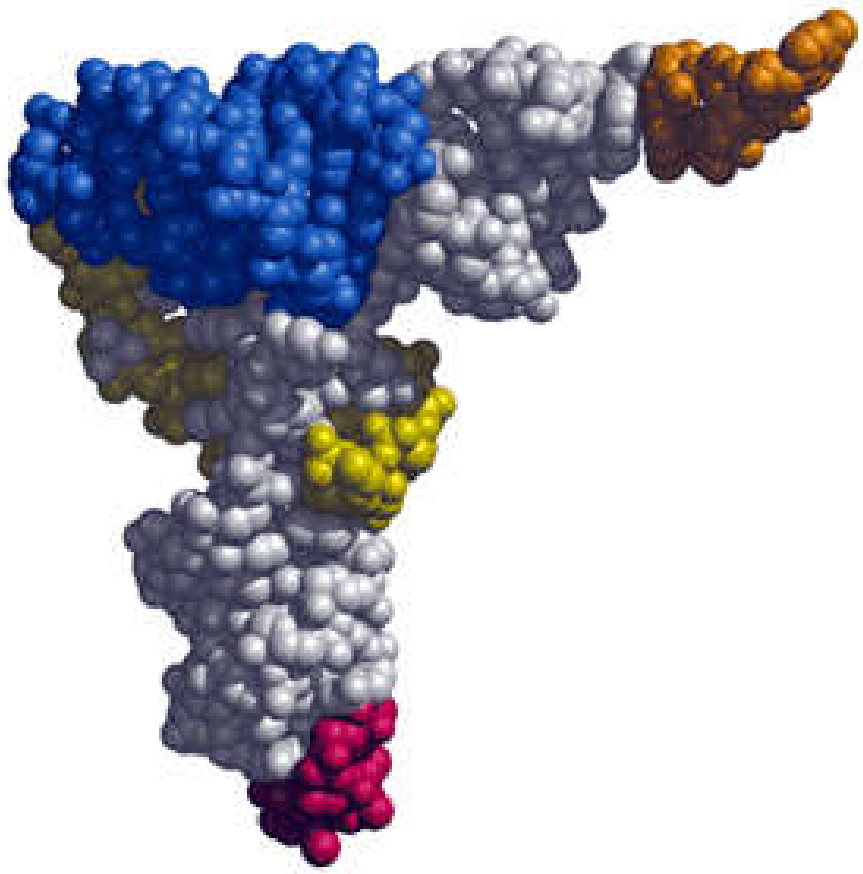
反密码区正中的3个核苷酸残基可以与mRNA的三联体密码子形成碱基互补配对，解读遗传密码，称为反密码子（anticodon）。

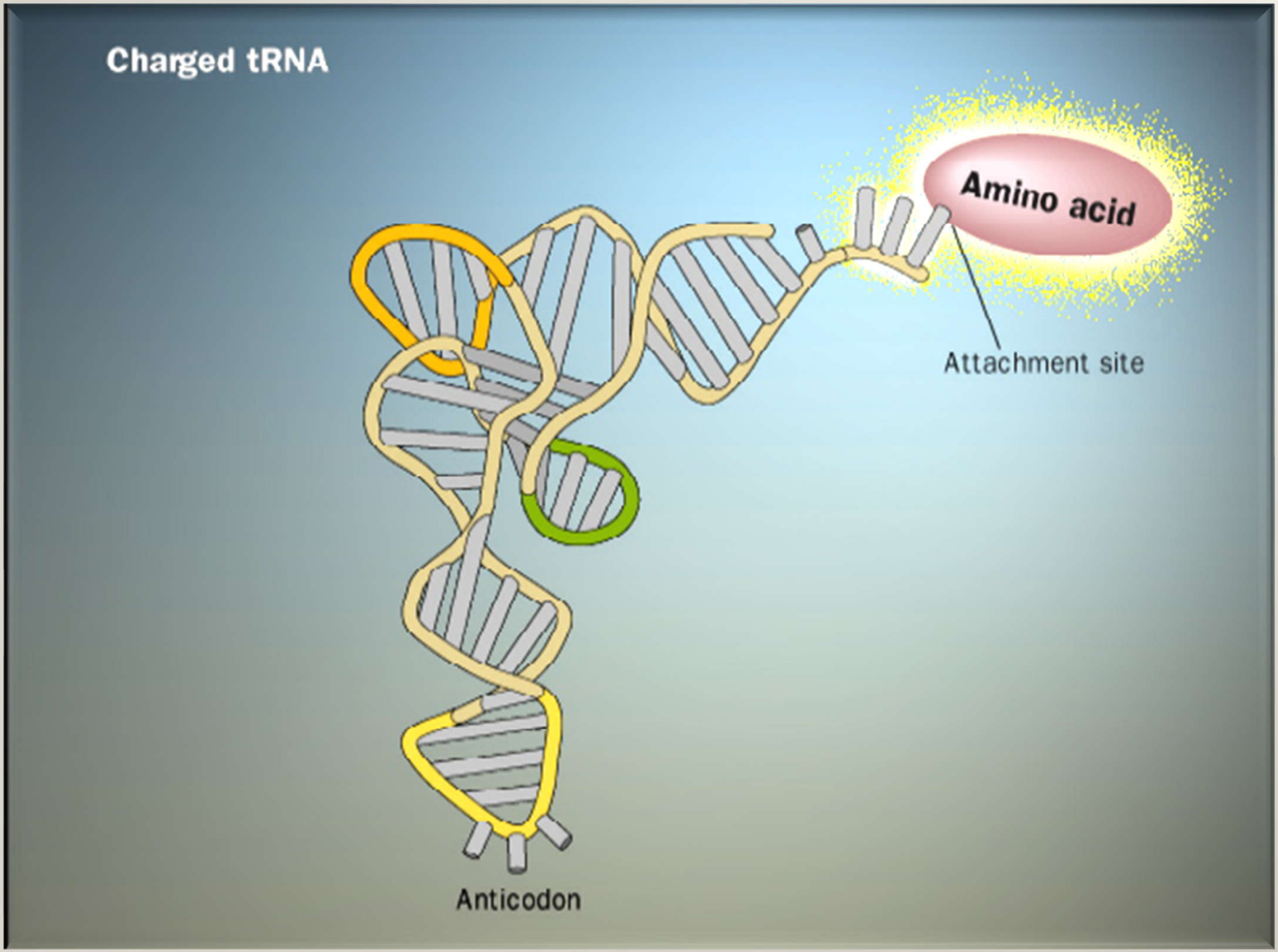
cloverleaf in secondary


tRNA ("cloverleaf" model)
Amino acid attachment site
3'
5'
Anticodon loop
Anticodon
tRNA (folded model)
Amino acid attachment site
5'
3'
Anticodon loop
Anticodon

Tertiary structure of tRNA






Charged tRNA
Amino acid
Attachment site
Anticodon

RNA Transcription in Prokaryotes (G2)

Biochemistry



1. DNA and RNA syntheses are similar in some aspects but different in others

Similar in fundamental chemical mechanism:

- both are guided by a **template**;
- both have the same polarity in strand extension (**5` to 3`**);
- both use triphosphate nucleotides (**dNTP or NTP**).



Different aspects:

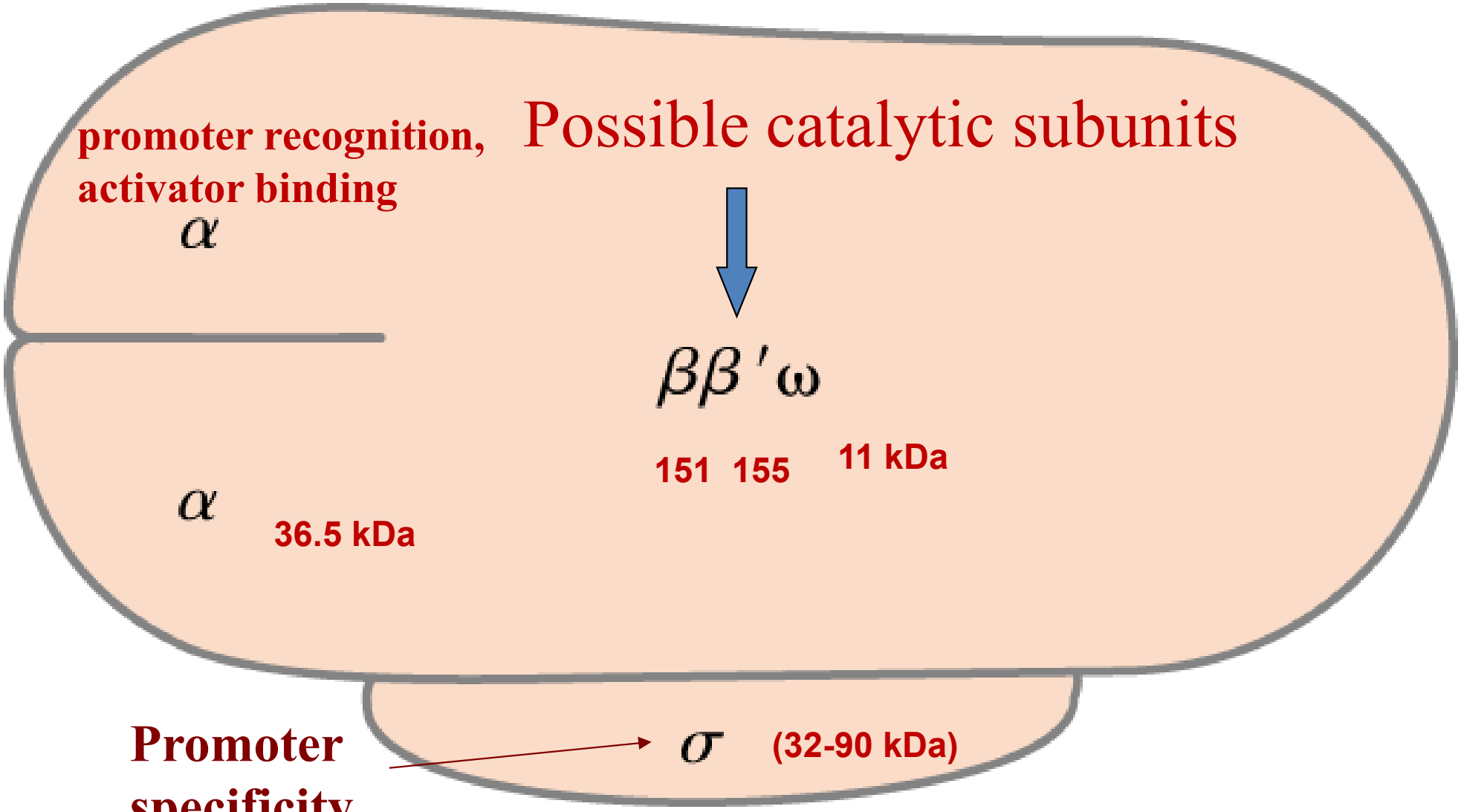
- Needs **no primers**;
- Only involves a **short segment** of a large DNA molecule;
- Uses only **one** of the two complementary DNA strands as the template strand;
- Has **no proofreading** activities;
- Be subject to **great variation**.



2. The multimeric RNA polymerase in *E.coli* has multiple functions

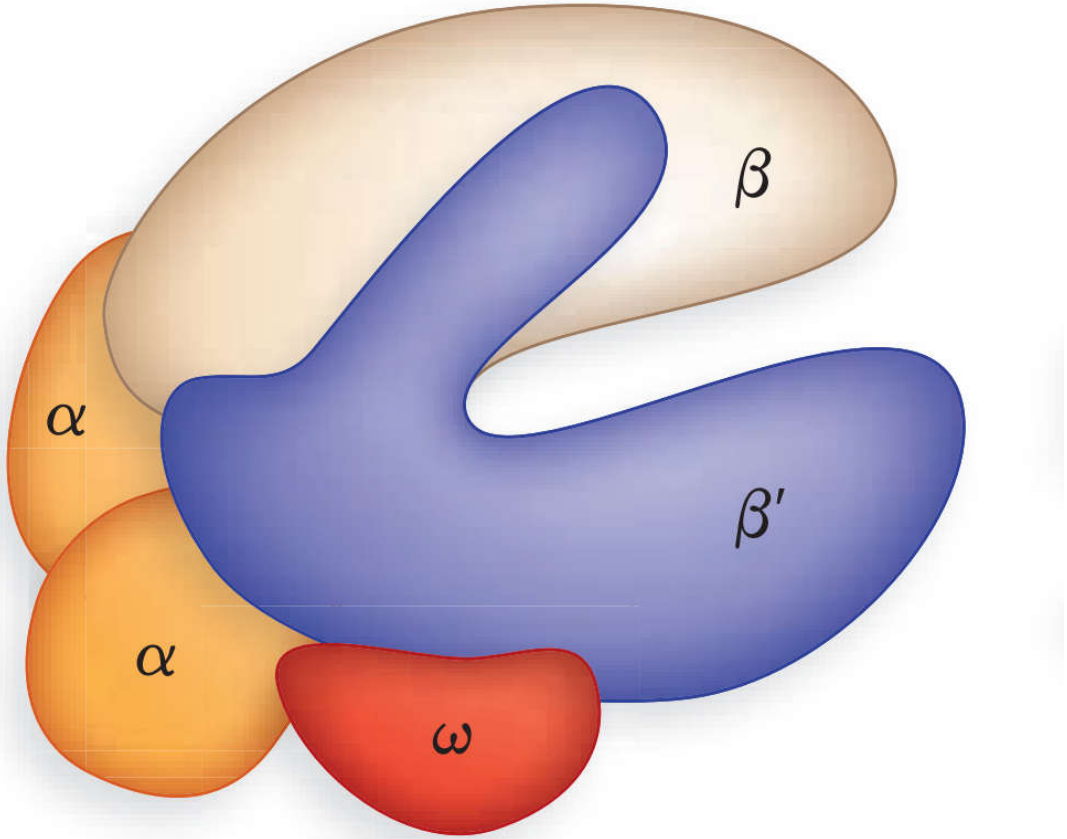
- The holoenzyme consists of five types (五种，六个）of subunits ($\alpha_2\beta\beta'\omega\ \sigma$)and its is used to synthesize all the RNA molecules *in E. coli*.
- The multiple functions include:
  - searches for initiation sites on the DNA molecule and unwinds a short stretch of DNA (**initiation**);
  - selects the **correct** NTP **and** catalyzes the formation of phosphodiester bonds (**elongation**);
  - detects termination signals for RNA synthesis (**termination**).



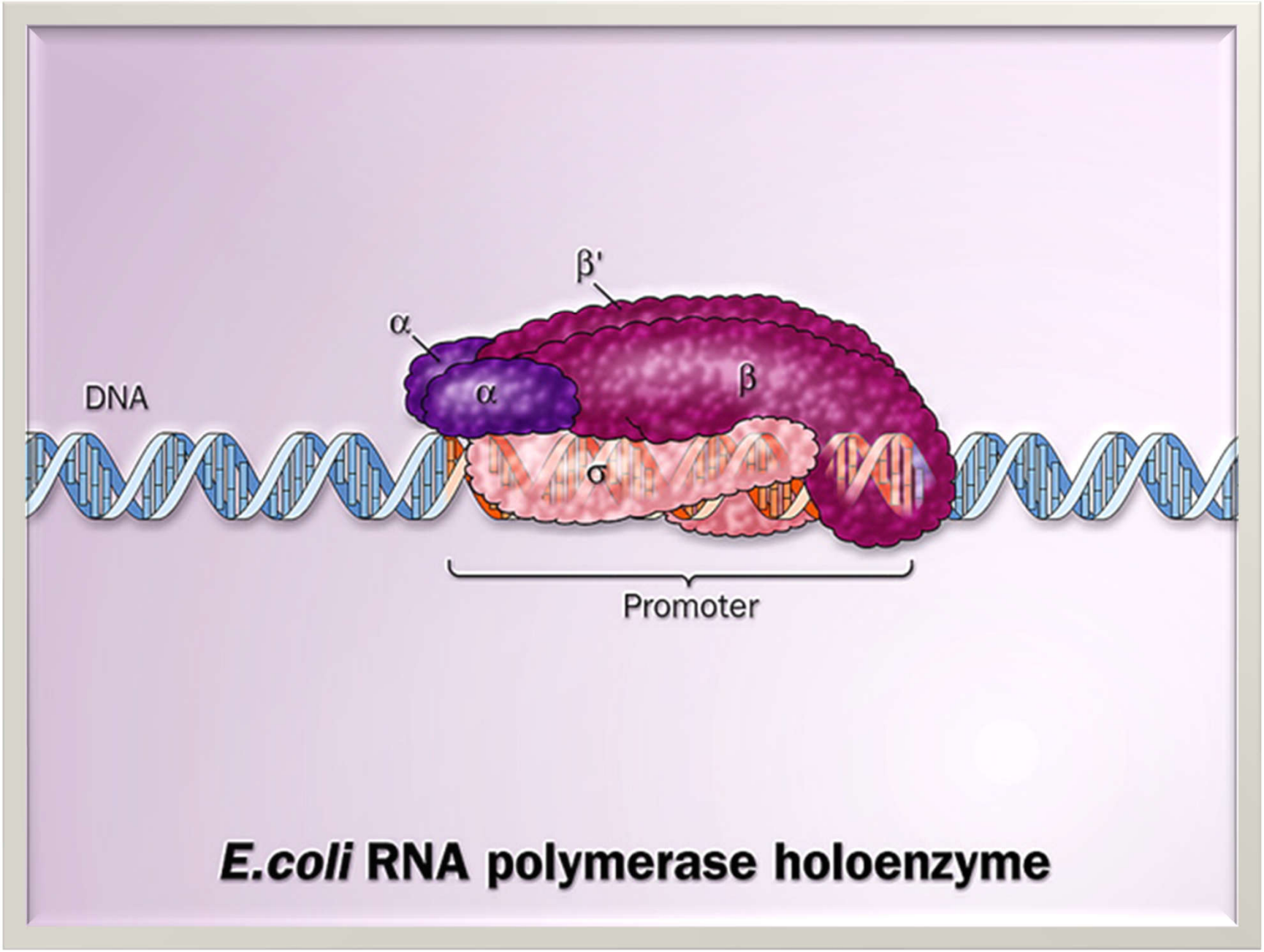


The *E. coli* RNA polymerase holoenzyme consists of six subunits: $\alpha_2\beta\beta'\omega\ \sigma$.

core enzyme



five core subunits $(\alpha_2\beta\beta'\omega;\ M_r\ 390{,}000)$



***E.coli* RNA polymerase holoenzyme**

Subunits of RNA Polymerase from *E. coli:*

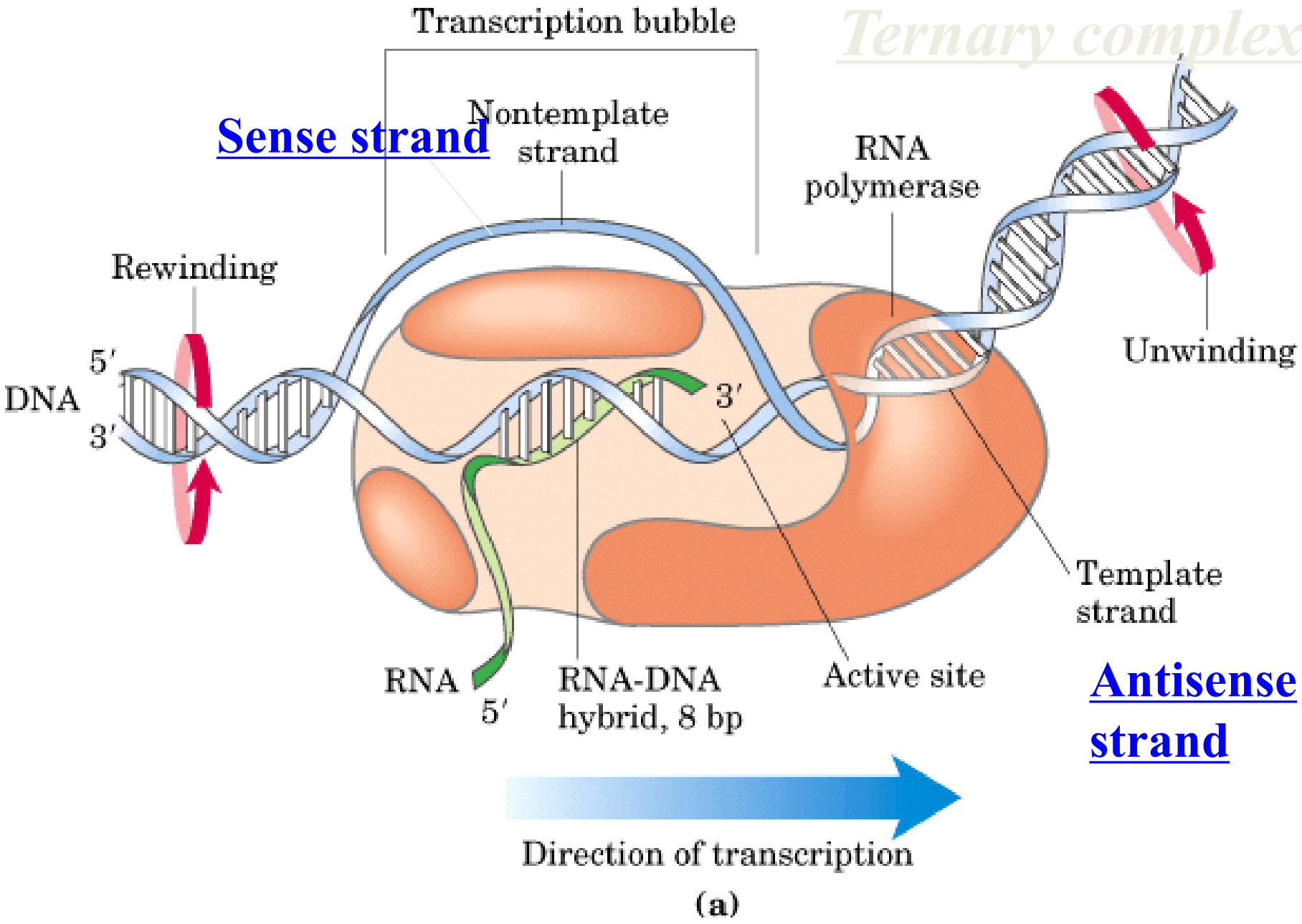
Subunit	Number	Mass (kd)	Gene	Role
α	2	37	*rpoA*	Binds regulatory sequences
β	1	151	*rpoB*	Forms phosphodiester bonds
β'	1	155	*rpoC*	Binds DNA template
σ^{70}	1	70	*rpoD*	Recognizes promoter, initiates synthesis



3. RNA synthesis occurs in a “moving” transcription bubble on the DNA template



Ternary complex



Sense strand

(5′) CGCTATAGCGTTT(3′)	DNA nontemplate (coding) strand
(3′) GCGATATCGCAAA(5′)	DNA template strand
(5′) CGCUAUAGCGUUU(3′)	RNA transcript

Antisense strand



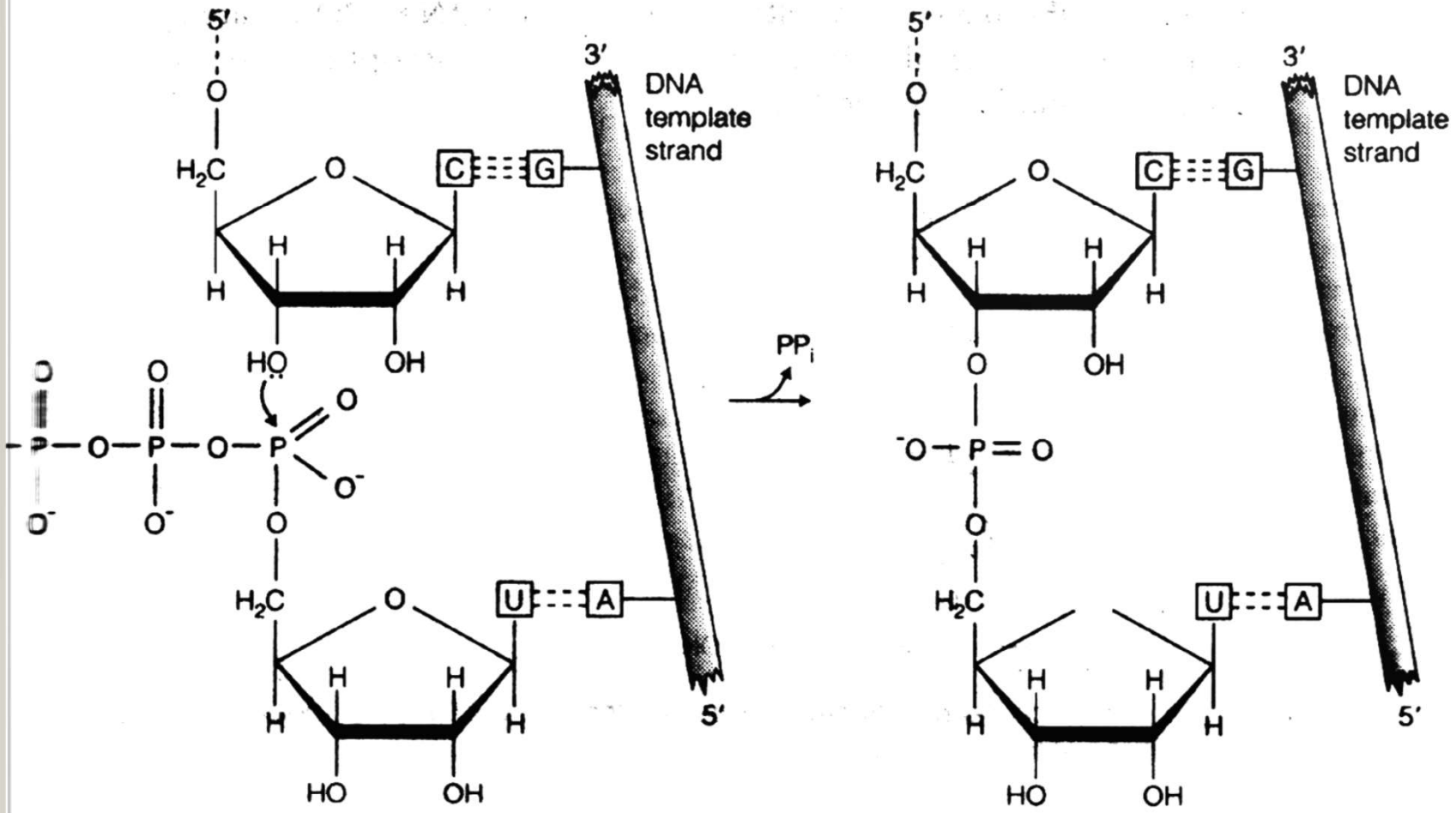


Fig. 2. Transcription by RNA polymerase. In each step the incoming ribonucleotide selected is that which can base-pair with the next base of the DNA template strand. In the diagram, the incoming nucleotide is rUTP to base-pair with A residue of the template DNA. A 3′5′ phosphodiester bond is formed, extending the RNA chain by one nucleotide, and pyrophosphate is released. Overall the RNA molecule grows in a 5′ to 3′ direction.

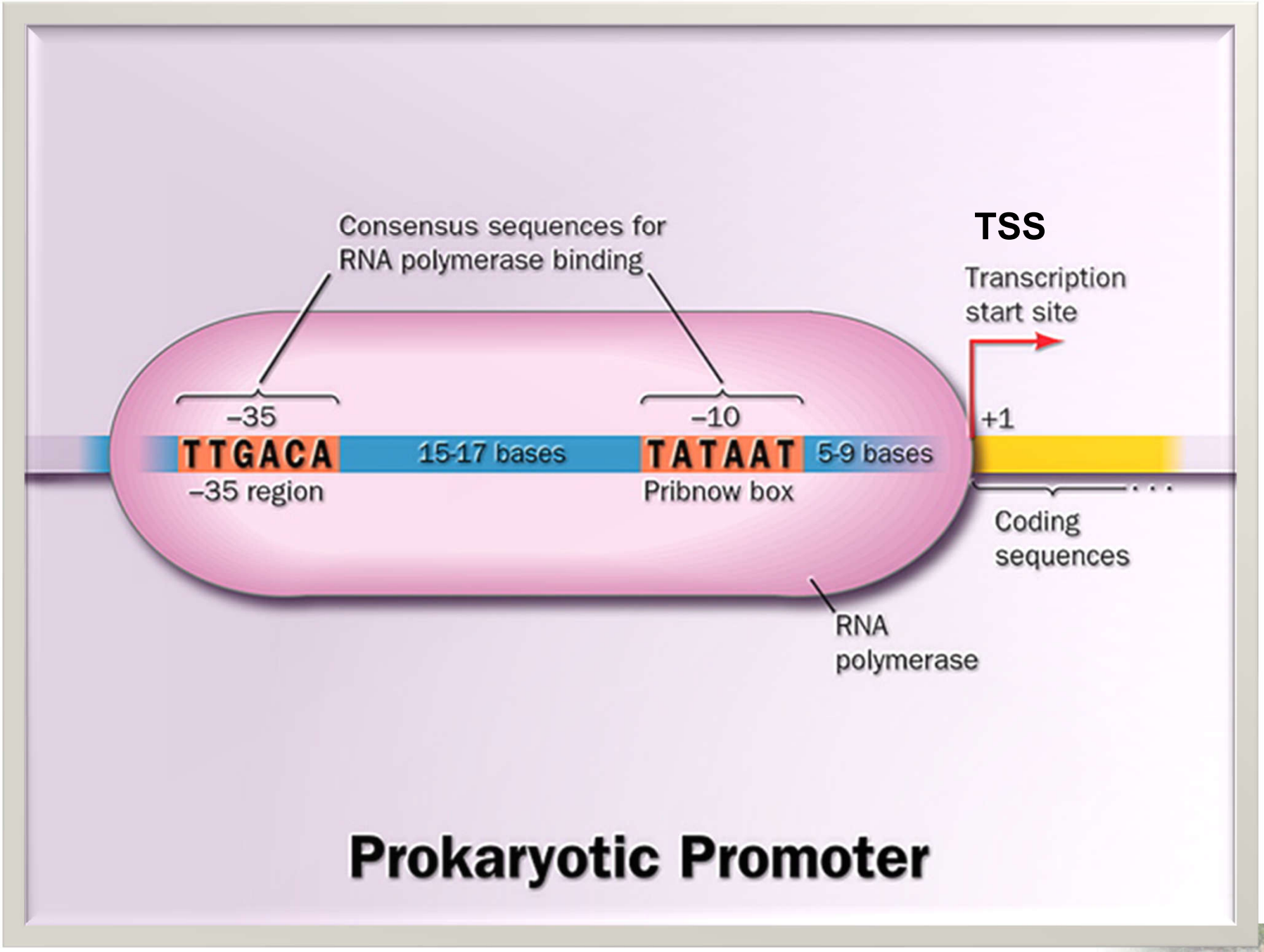
4. RNA polymerase recognizes specific promoter sequences on DNA to initiate transcription

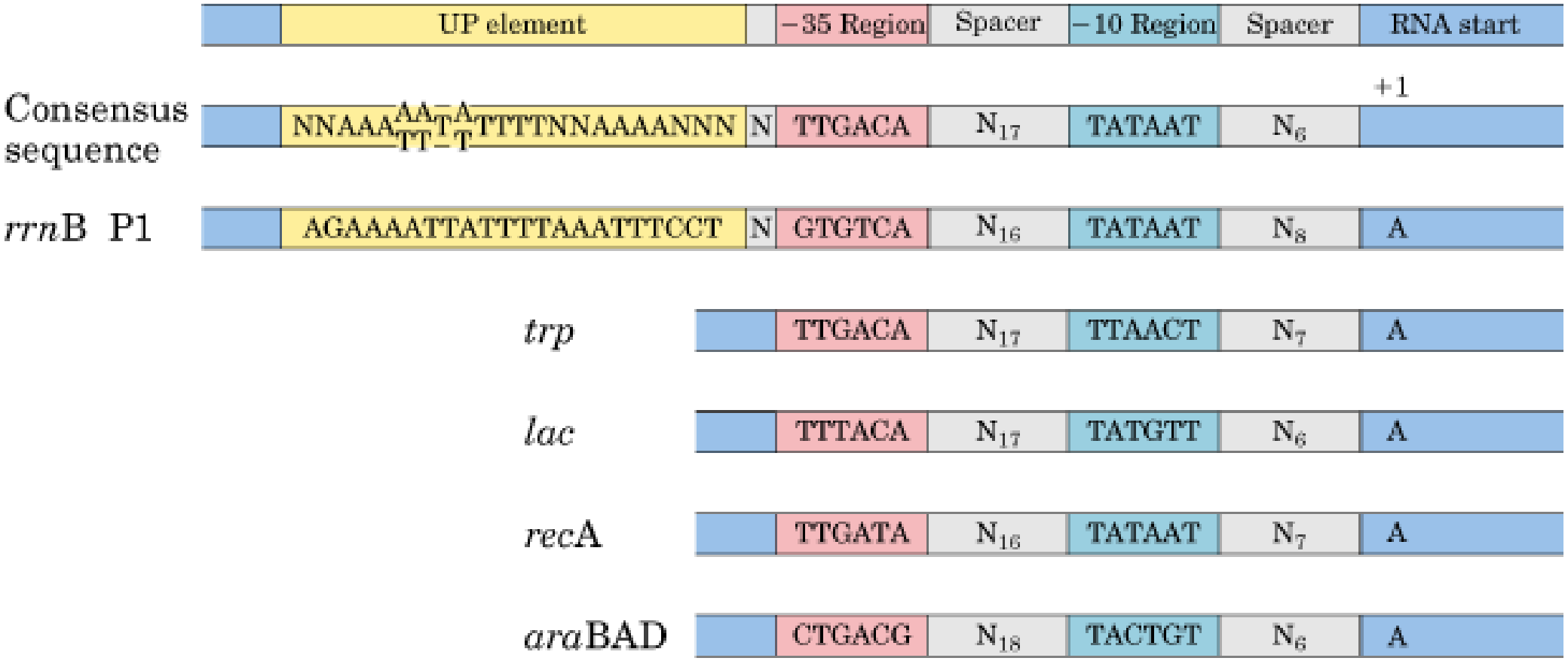
- **An RNA polymerase binds to specific sequences in the DNA called promoters, which direct the transcription of adjacent segments of DNA (genes).**
- **Promoter** sequences are located adjacent to genes.
- Promoters have two **consensus sequences** centered at –10 and –35 positions (the first residue of the RNA is given +1).

TSS
Transcription start site





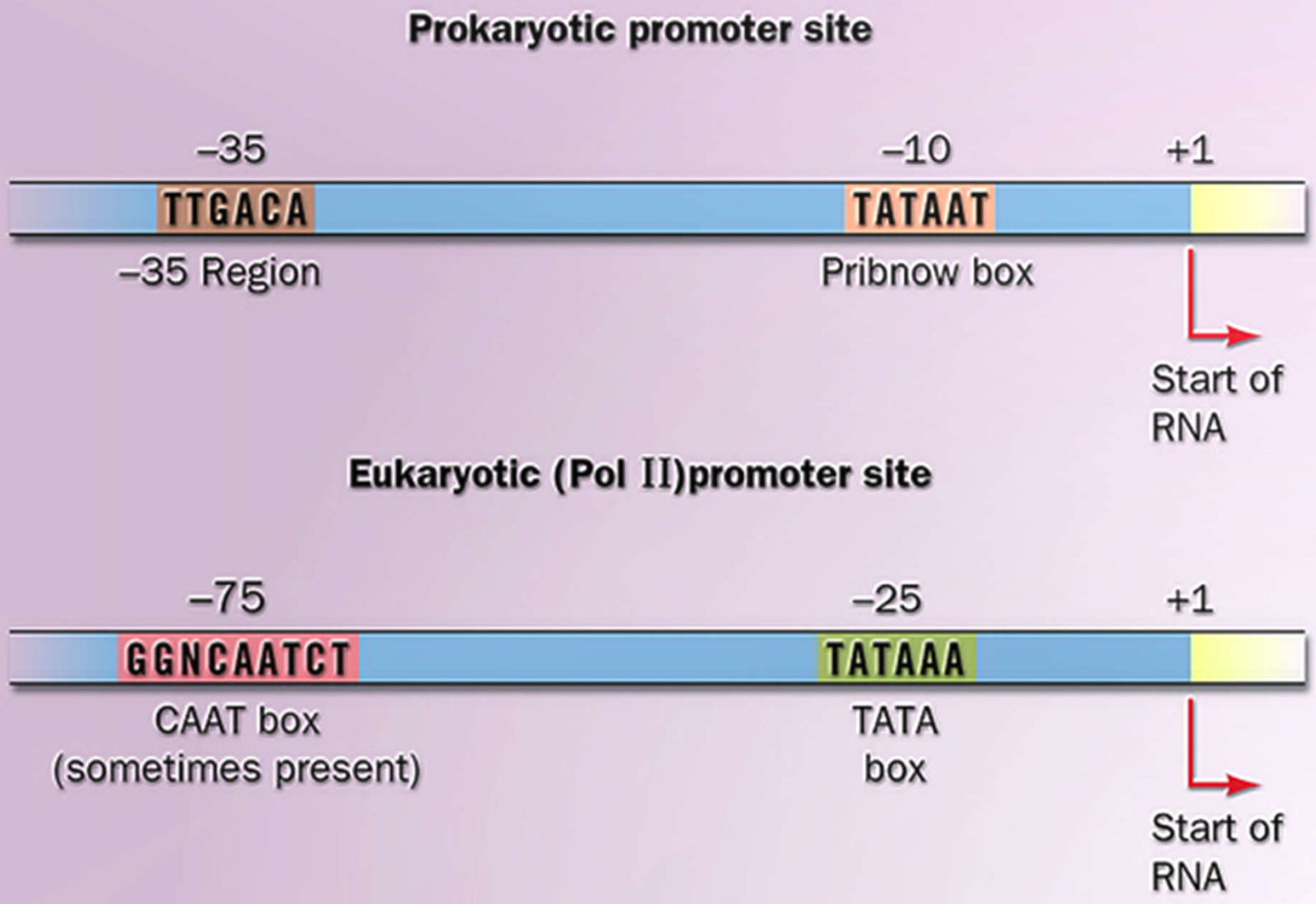

Consensus sequences for
RNA polymerase binding
TSS
Transcription
start site
–35
–10
+1
TTGACA
15-17 bases
TATAAT
5-9 bases
–35 region
Pribnow box
Coding
sequences
RNA
polymerase
Prokaryotic Promoter



the –10 sequence, TATAAT (the Pribnow box)

the –35 consensus region, TTGACA




Prokaryotic promoter site
–35
–10
+1
TTGACA
TATAAT
–35 Region
Pribnow box
Start of RNA
Eukaryotic (Pol II)promoter site
–75
–25
+1
GGNCAATCT
TATAAA
CAAT box (sometimes present)
TATA box
Start of RNA

5. The σ subunits enable the *E.coli* RNA polymerase to recognize specific promoter sites

- The RNA polymerase without the σ subunit (i.e., the $\alpha_2\beta\beta'\omega$, **core enzyme**) is unable to start transcription at a promoter.
- Promoter vary up to 1000-fold in their efficiency of initiation which depends on the exact sequence of the key promoter elements as well as flanking sequences.



Three stages of transcription

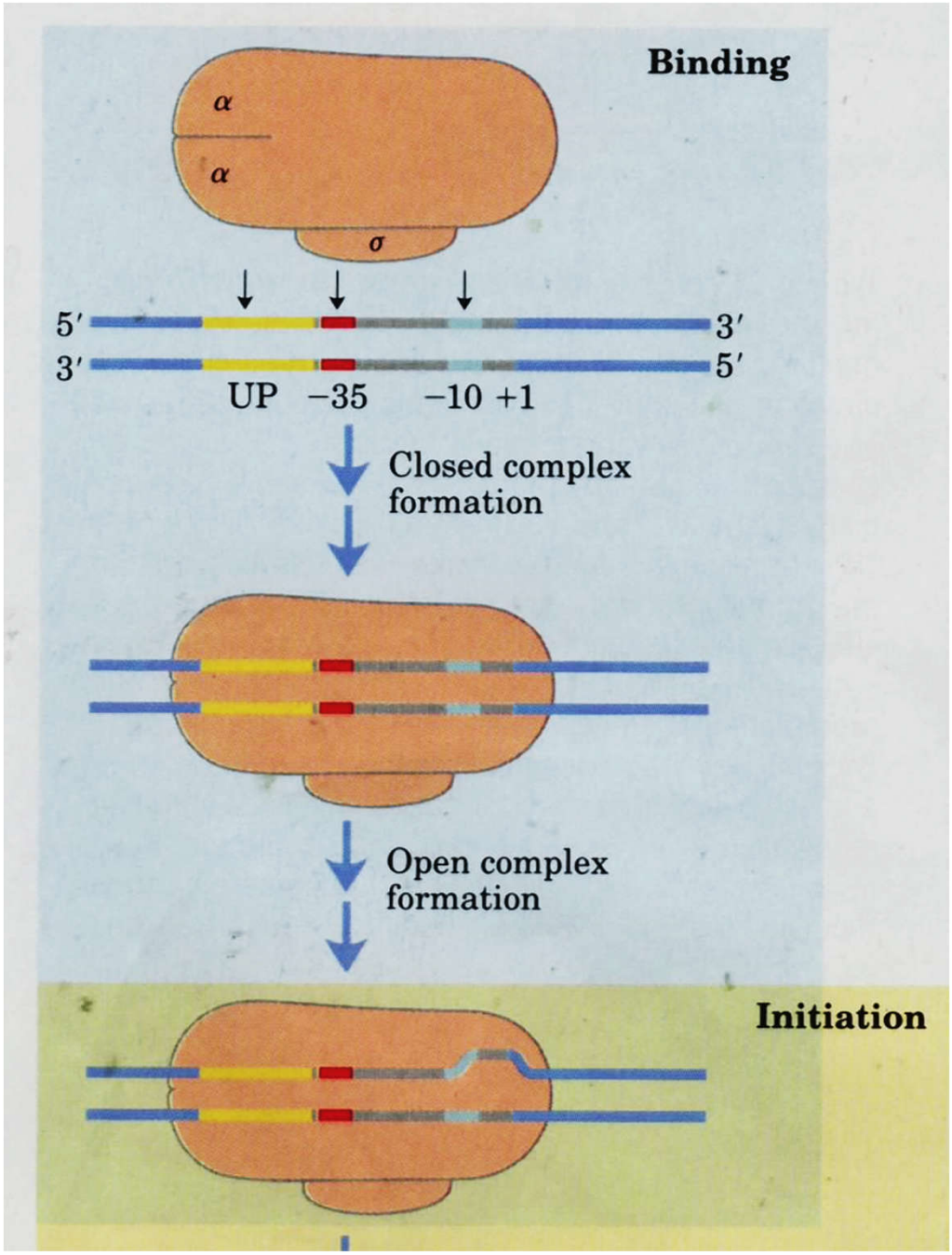
- Initiation
- Elongation
- Termination

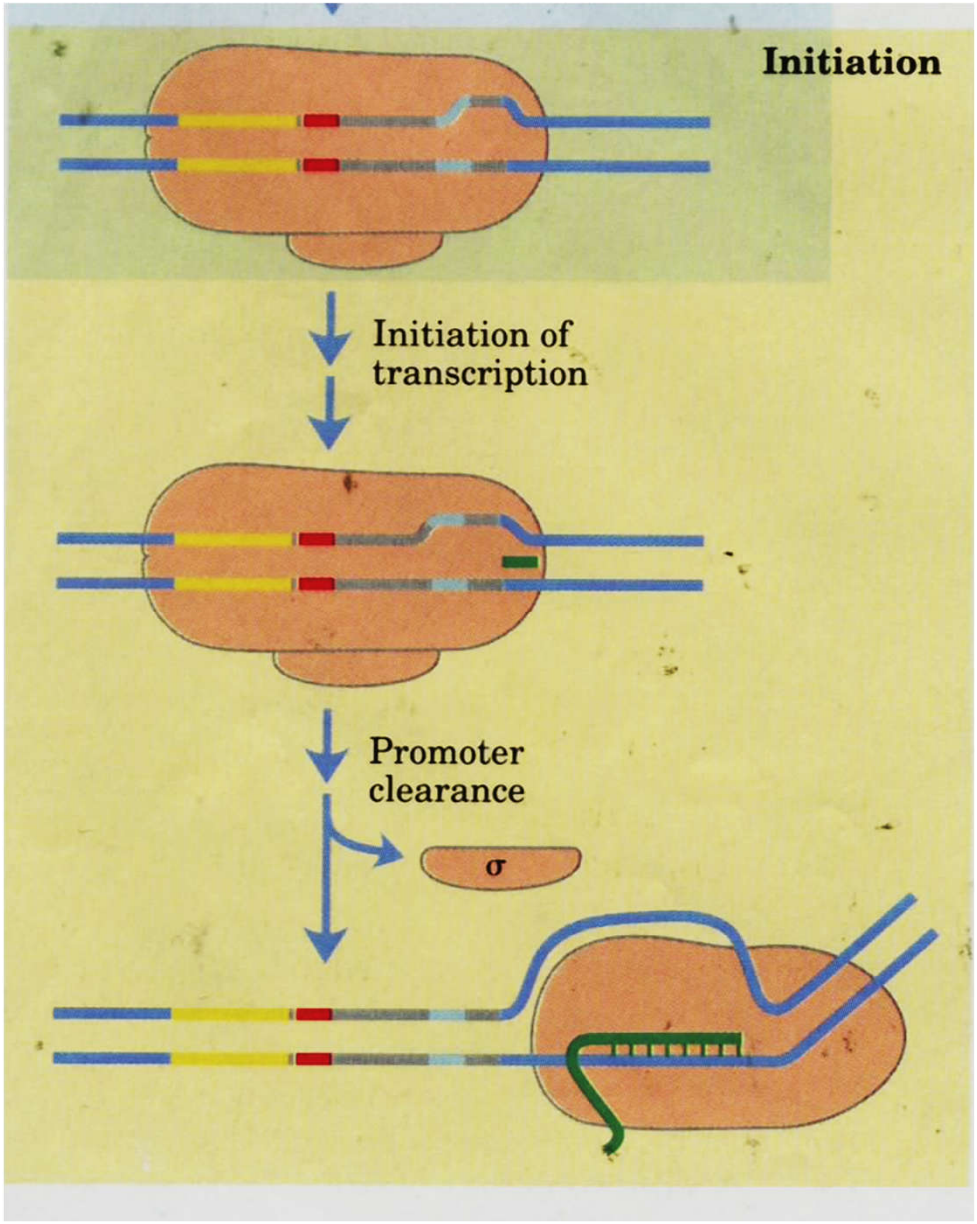


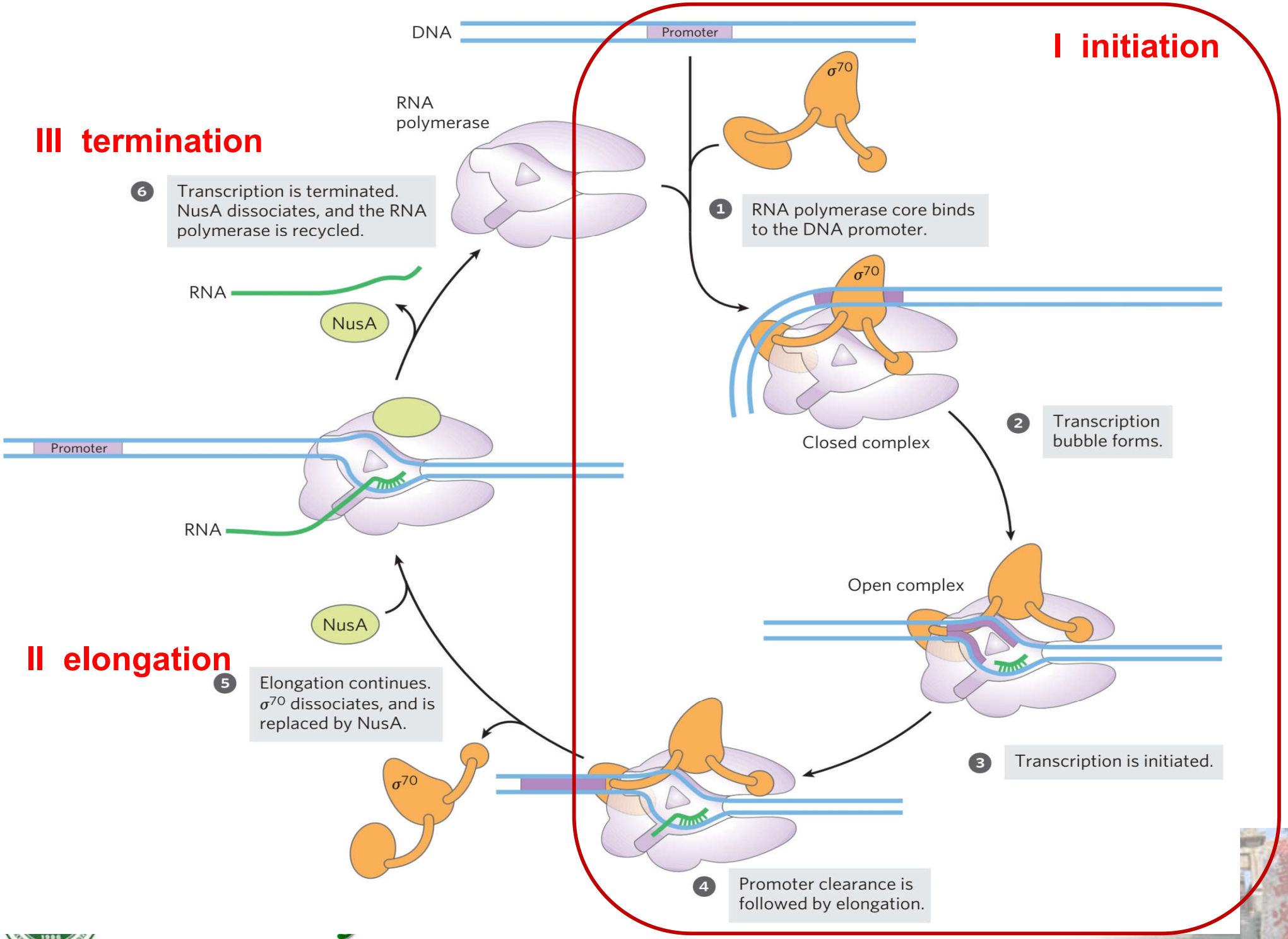
6. RNA polymerase unwinds the template DNA then initiate RNA synthesis -Initiation

- The enzyme slides to a promoter region and forms a more tightly bound "**closed complex**".
- Then the polymerase-promoter complex has to be converted to an "**open complex**", in which a 12-15 bp covering the region from the AT-rich –10 site to +3 site is unwound.
- The essential transition from a "closed" to an "open" complex sets the stage for RNA synthesis, after which the core polymerase moves away from the promoter. **The σ subunit dissociates** at random as the polymerase enters the elongation phase of transcription.




Binding
α
α
σ
5′
3′
3′
5′
UP
−35
−10
+1
Closed complex formation
Open complex formation
Initiation


Initiation
Initiation of transcription
Promoter clearance
σ

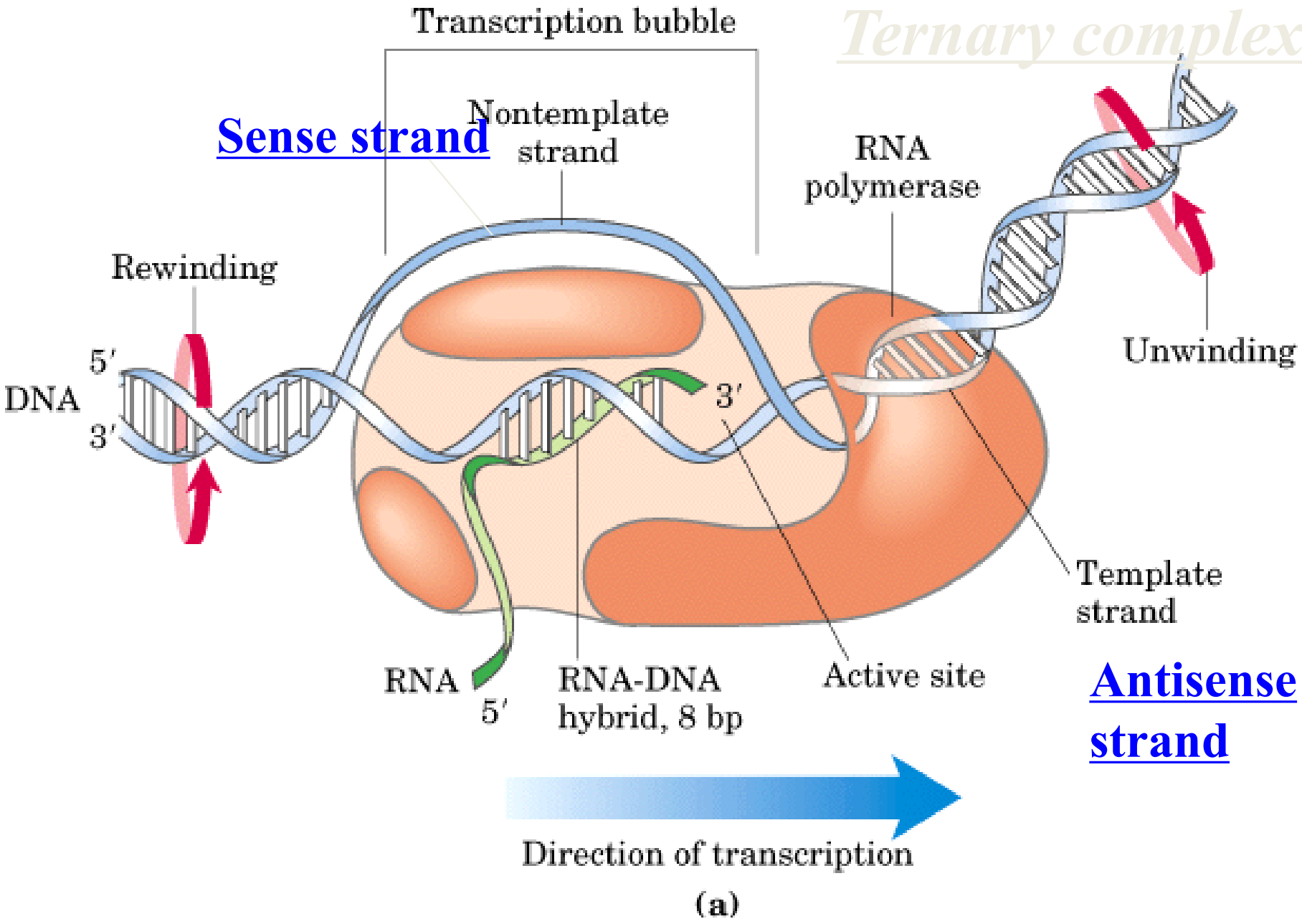

I initiation
II elongation
III termination
DNA
Promoter
RNA polymerase
σ70
1 RNA polymerase core binds to the DNA promoter.
Closed complex
2 Transcription bubble forms.
Open complex
3 Transcription is initiated.
4 Promoter clearance is followed by elongation.
5 Elongation continues. σ70 dissociates, and is replaced by NusA.
NusA
RNA
6 Transcription is terminated. NusA dissociates, and the RNA polymerase is recycled.

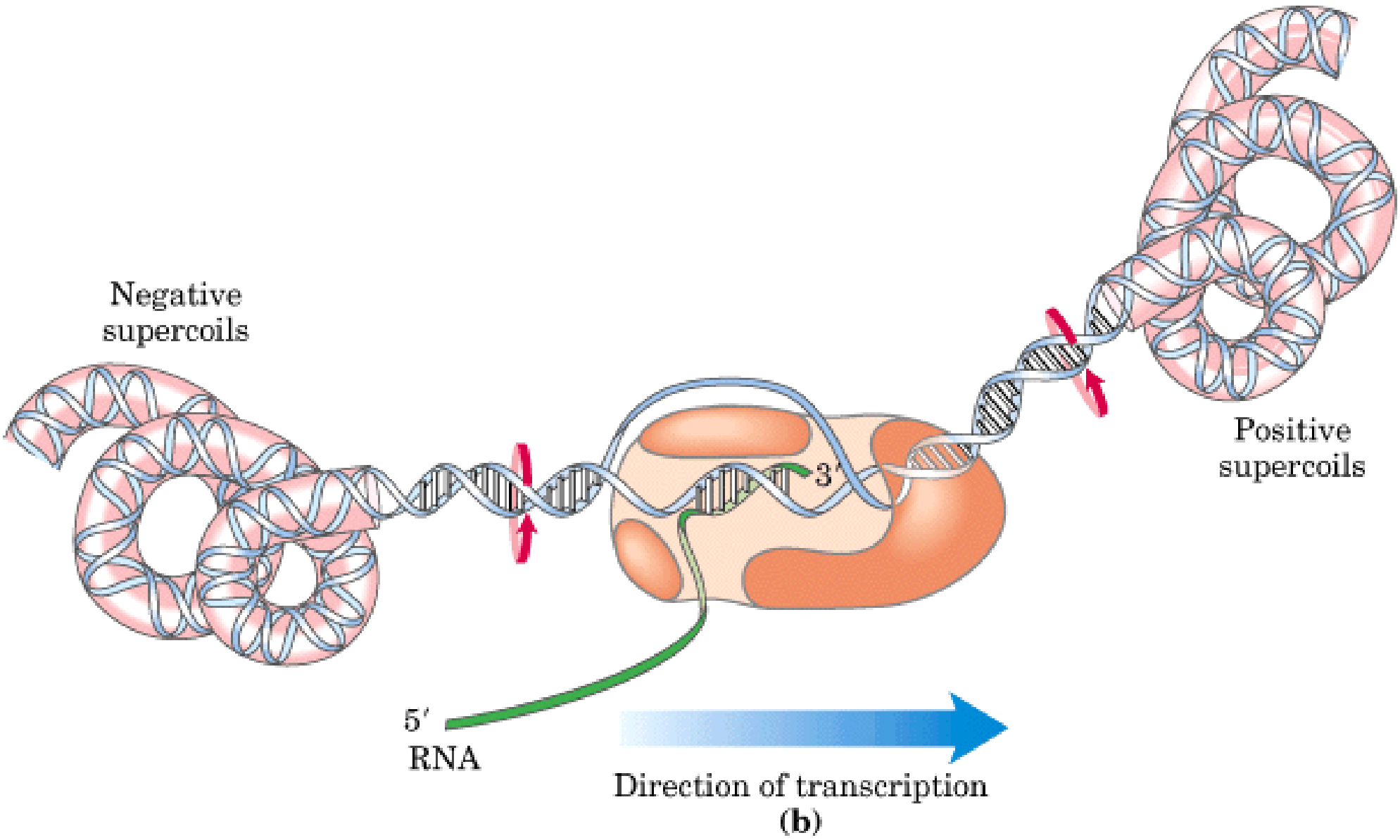
7. After initiation, the factor σ is released to leave the core enzyme which continues elongation

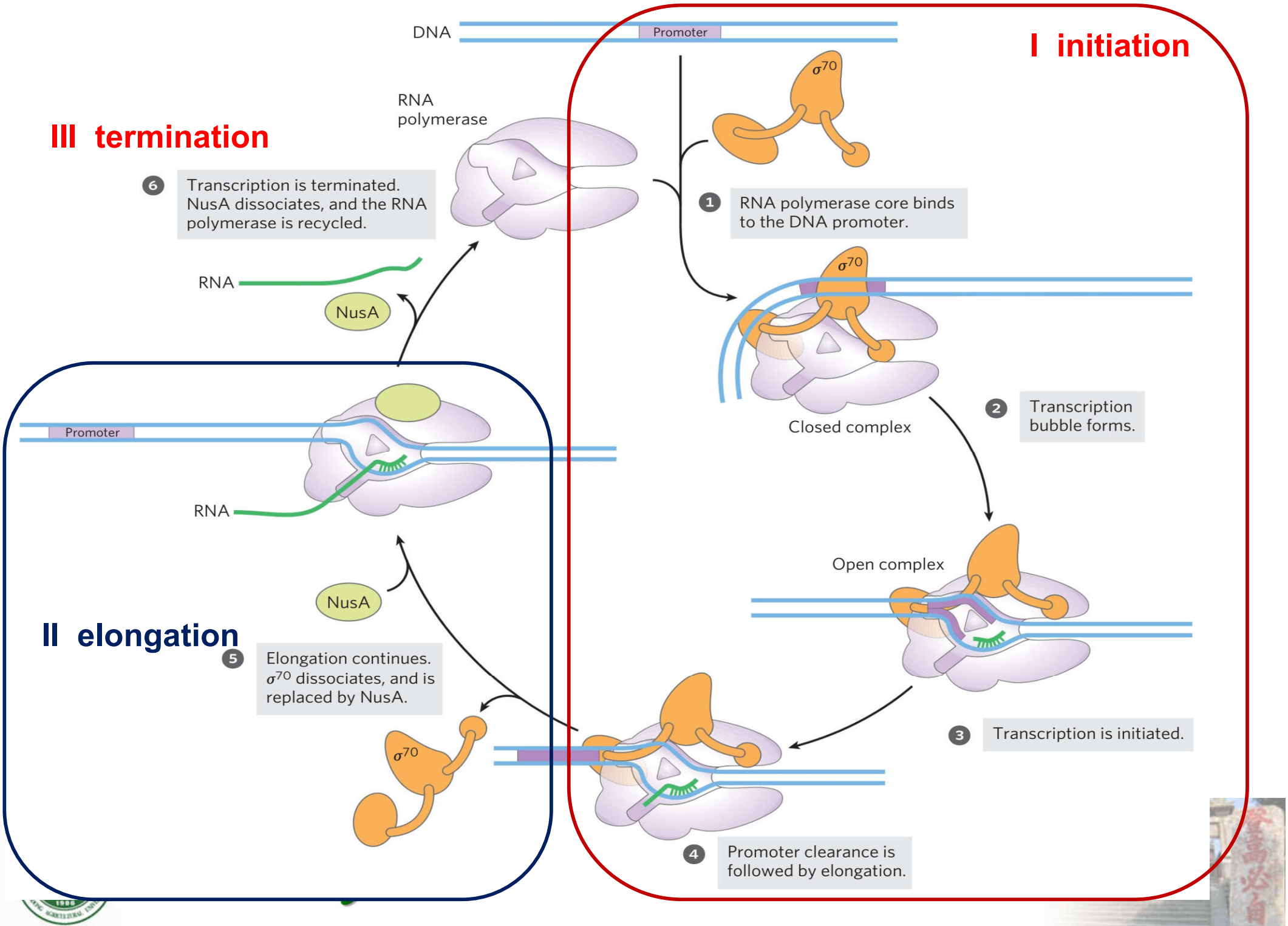
The core enzyme ($\alpha_2\beta\beta'\omega$) continues elongation of the RNA transcript. The RNA synthesizes RNA in the 5'-3' direction, using the four ribonucleoside 5' triphosphates (ATP, GTP, CTP, UTP) as precursors.



Ternary complex




Negative
supercoils
3′
Positive
supercoils
5′
RNA
Direction of transcription
(b)

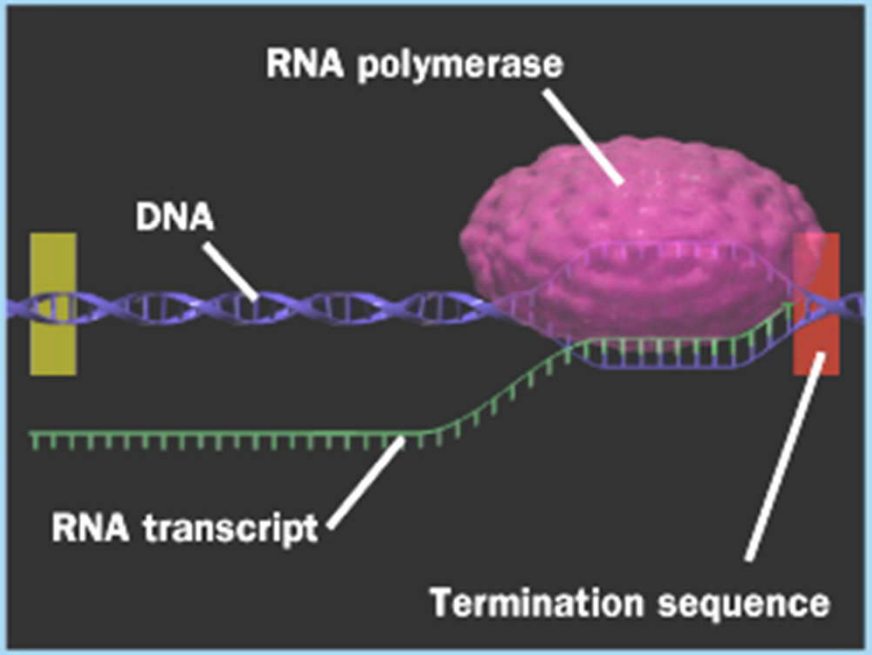

I initiation
II elongation
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DNA
Promoter
RNA polymerase
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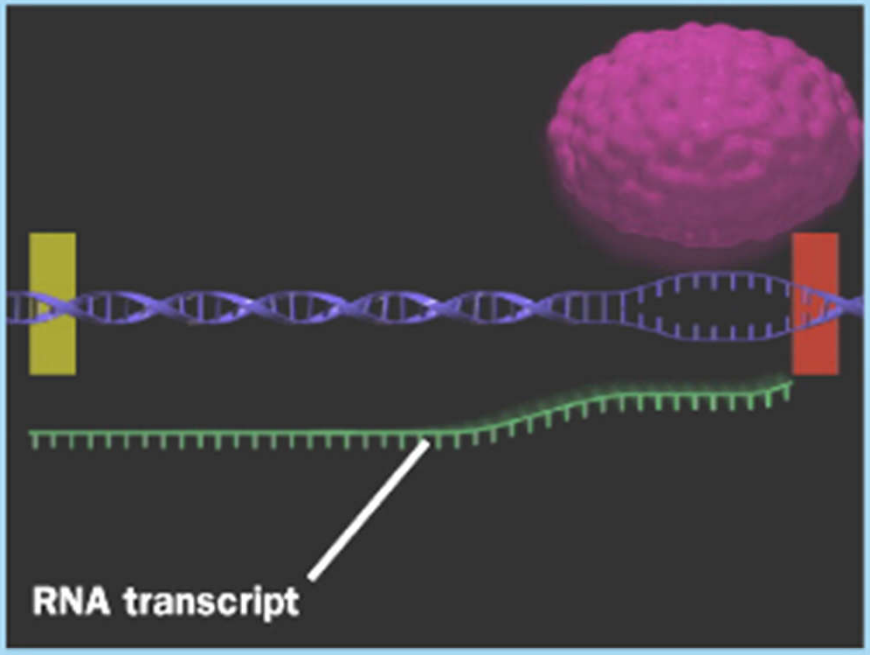
8. *E.coli* RNA polymerase stops synthesizing RNA at specific terminator DNA sequences -Termination

- A common termination signal is a hairpin structure formed by a palindromic GC-rich region, followed by an AT-rich sequence.
- Other signals are also used which require the assistance of ρ protein for effective termination.



TERMINATION of TRANSCRIPTION





When RNA polymerase reaches the termination sequence on the DNA template strand...

...the RNA transcript and RNA polymerase are released.

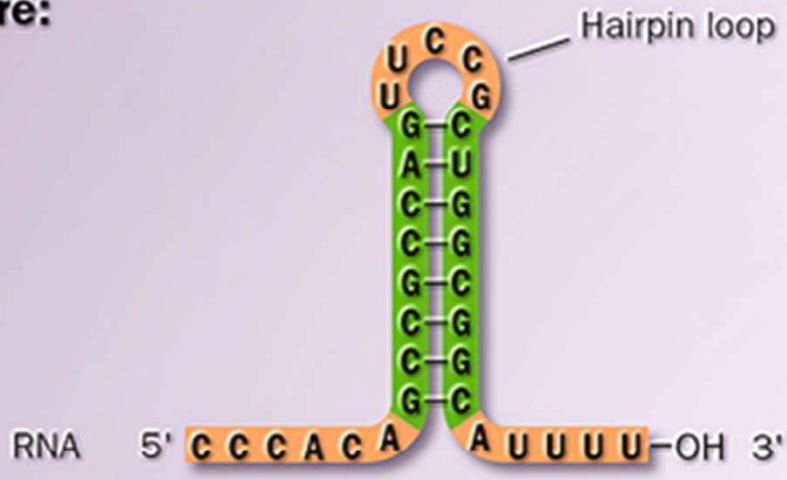
Palindrome sequences
回文序列

Primary structure:

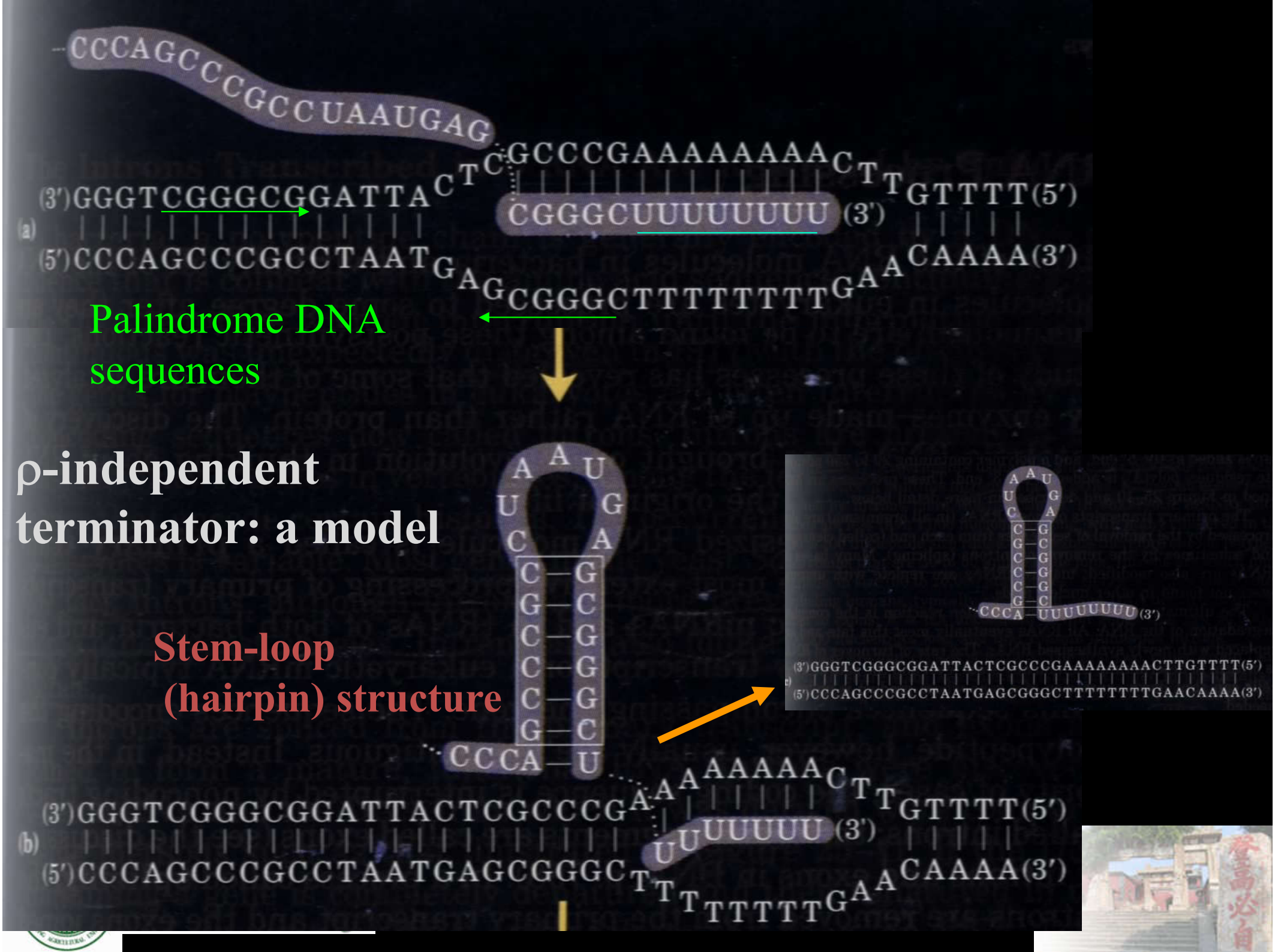
RNA 5' C C C A C A G C C G C C A G U U C C G C U G G C G G C A U U U U–OH 3'

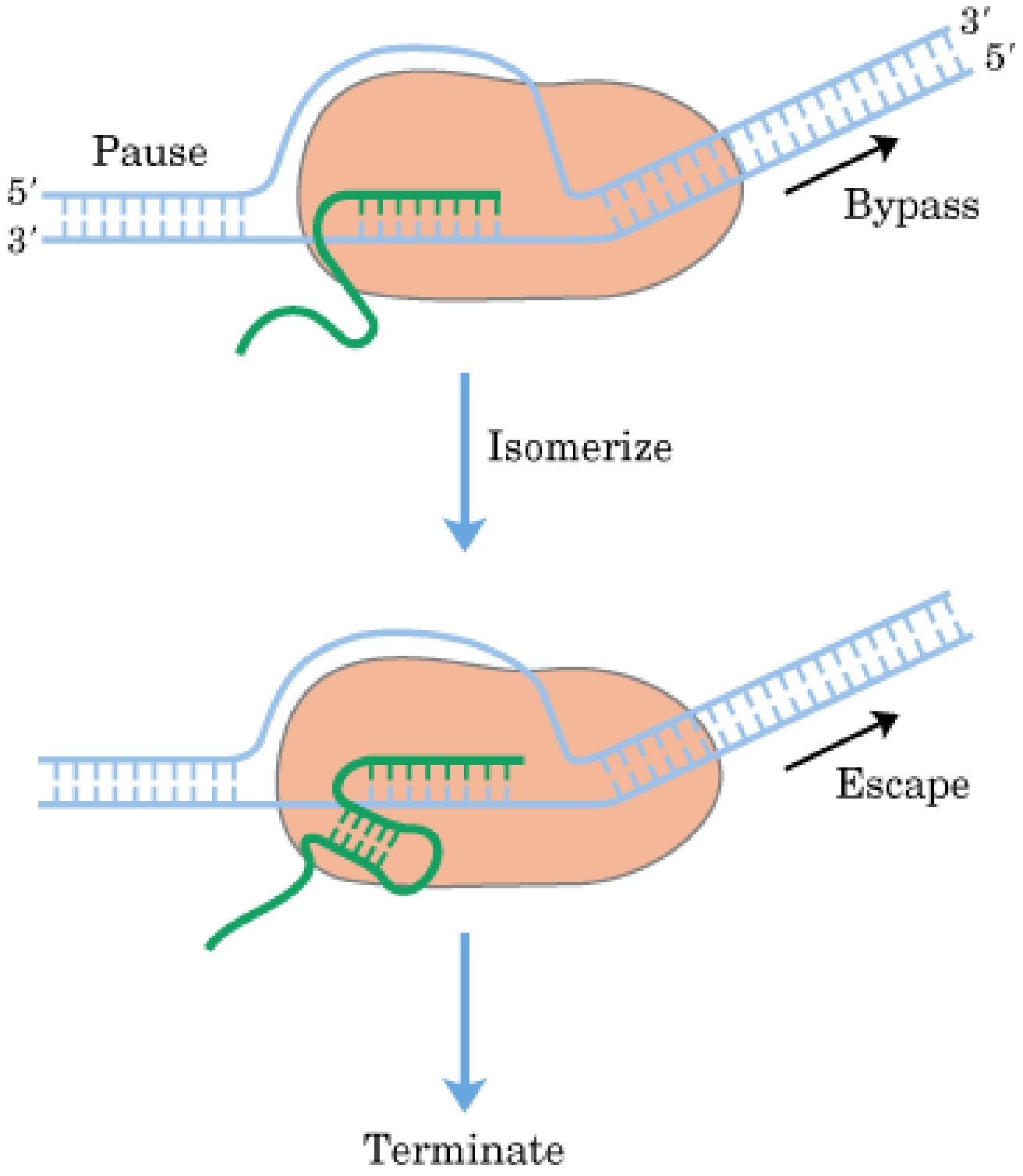
3′–C G G C G G U C G A C C G C C G –5′

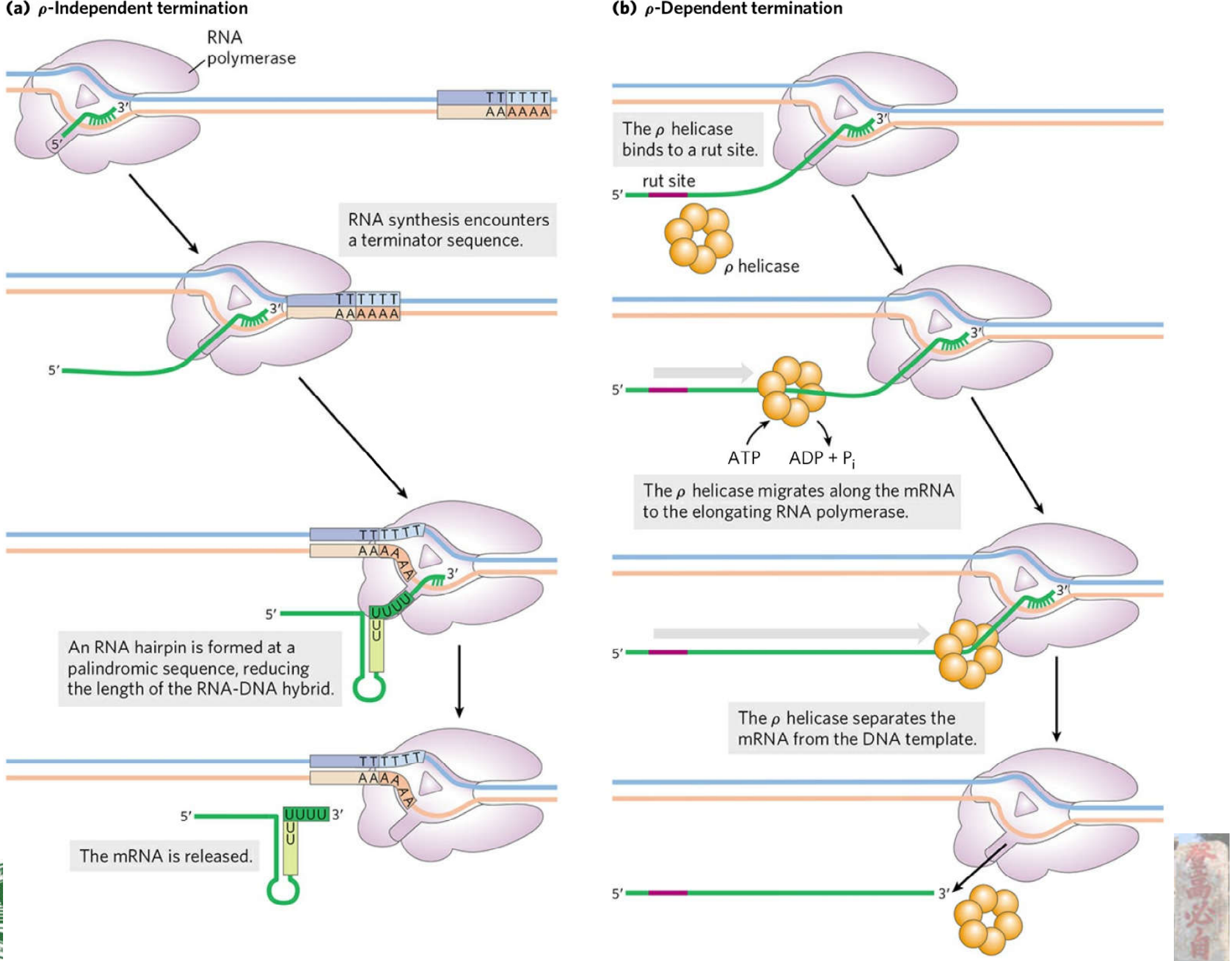
Secondary structure:

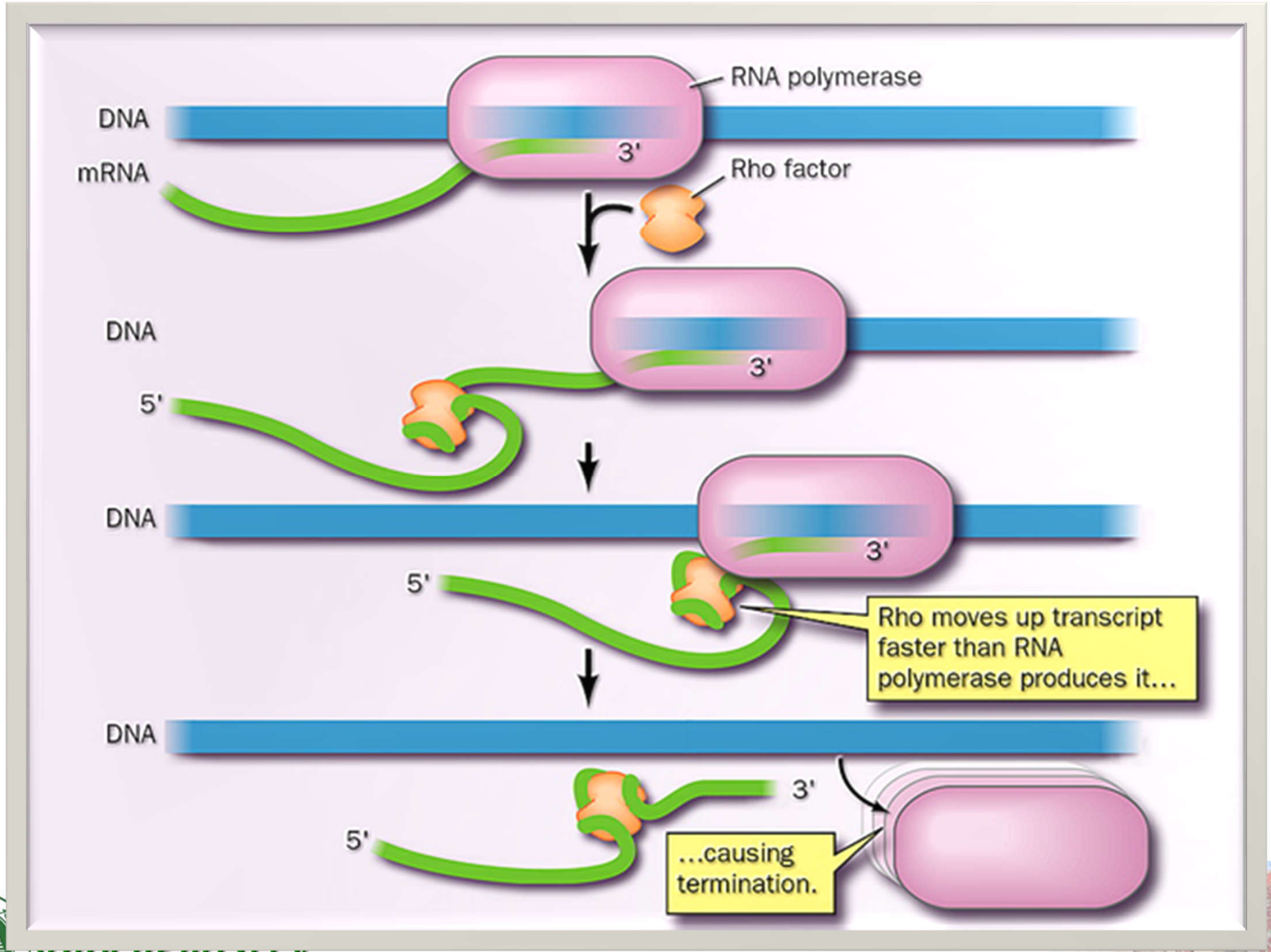


Termination Sequence


CCCAGCCCGCCTAATGAG
GCCCCGAAAAAAAAACTTGTTTT
(3')GGGTCGGGCGGATTA
CGGGGCUUUUUUUU (3')
(5')CCCAGCCCGCCTAAT
GAGCGGGCTTTTTTTTTGAACAAAA(3')
Palindrome DNA sequences
ρ-independent terminator: a model
Stem-loop (hairpin) structure
CCCA
UUUUUUUU (3')
(3')GGGTCGGGCGGATTACTCGCCCGAAAAAAAACTTGTTTT(5')
(5')CCCAGCCCGCCTAATGAGCGGGCTTTTTTTTTGAACAAAA(3')
(3')GGGTCGGGCGGATTACTCGCCCG
UUUUUU (3')
(5')CCCAGCCCGCCTAATGAGCGGGC
TTTTTTTTTGAACAAAA(3')


3′
5′
Pause
5′
3′
Bypass
Isomerize
Escape
Terminate


(a) ρ-Independent termination
RNA polymerase
3′
5′
TTTTTT
AAAAAA
RNA synthesis encounters a terminator sequence.
An RNA hairpin is formed at a palindromic sequence, reducing the length of the RNA-DNA hybrid.
UUUU
CC
The mRNA is released.
(b) ρ-Dependent termination
The ρ helicase binds to a rut site.
rut site
ρ helicase
ATP
ADP + Pi
The ρ helicase migrates along the mRNA to the elongating RNA polymerase.
The ρ helicase separates the mRNA from the DNA template.


RNA polymerase
DNA
mRNA
3'
Rho factor
DNA
3'
5'
DNA
3'
5'
Rho moves up transcript faster than RNA polymerase produces it...
DNA
3'
5'
...causing termination.

Model for an ρ-dependent terminator

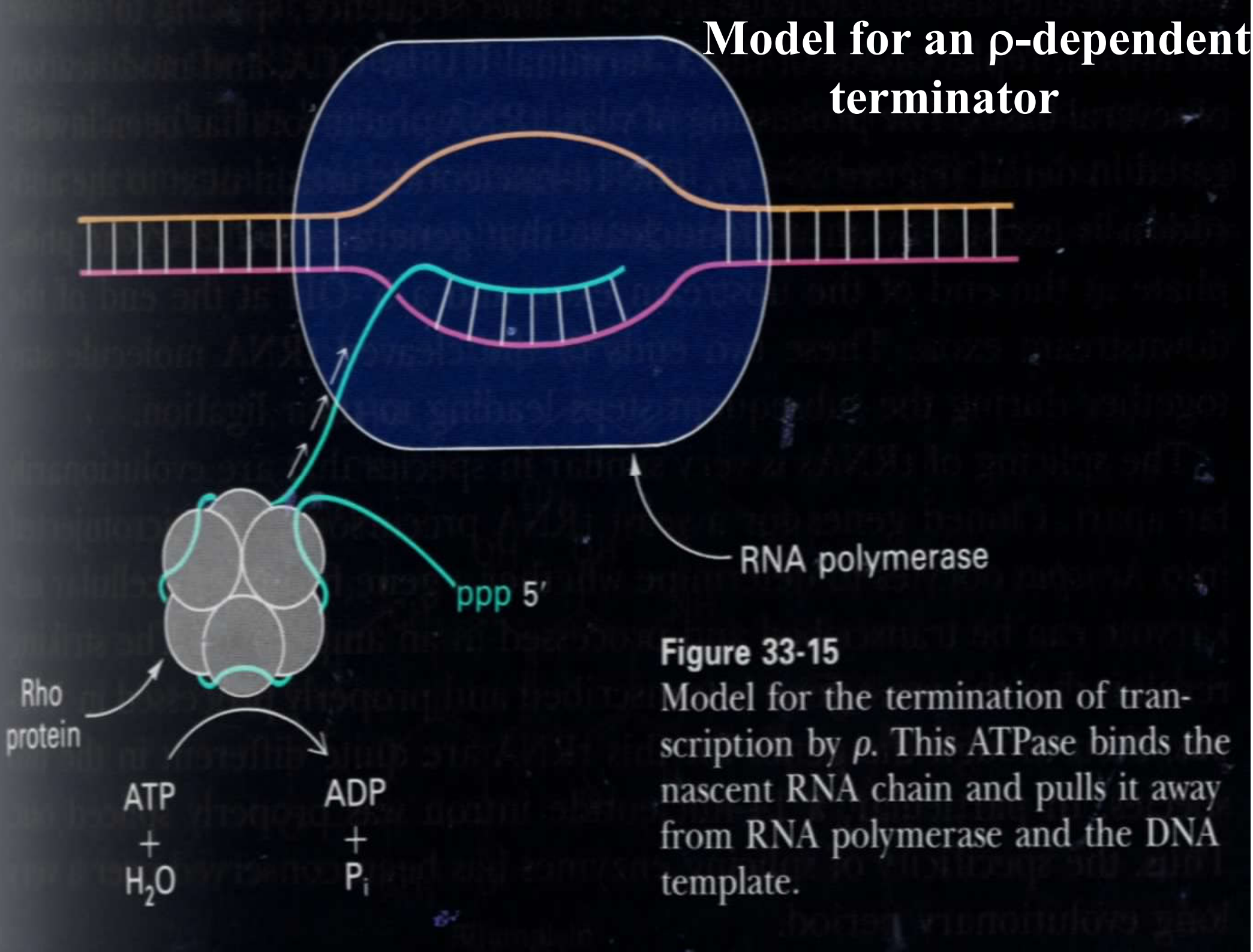
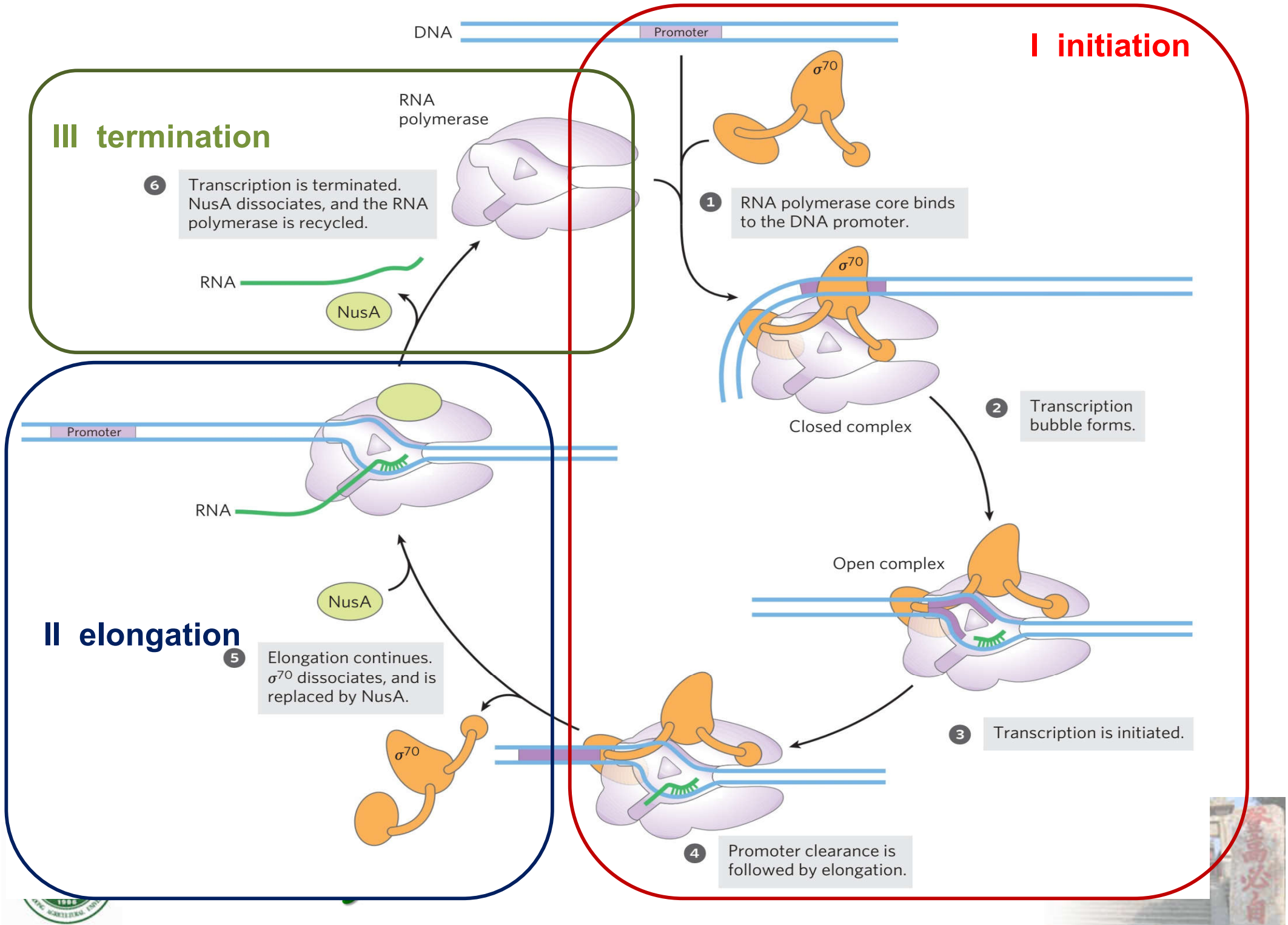


Figure 33-15
Model for the termination of transcription by *ρ*. This ATPase binds the nascent RNA chain and pulls it away from RNA polymerase and the DNA template.

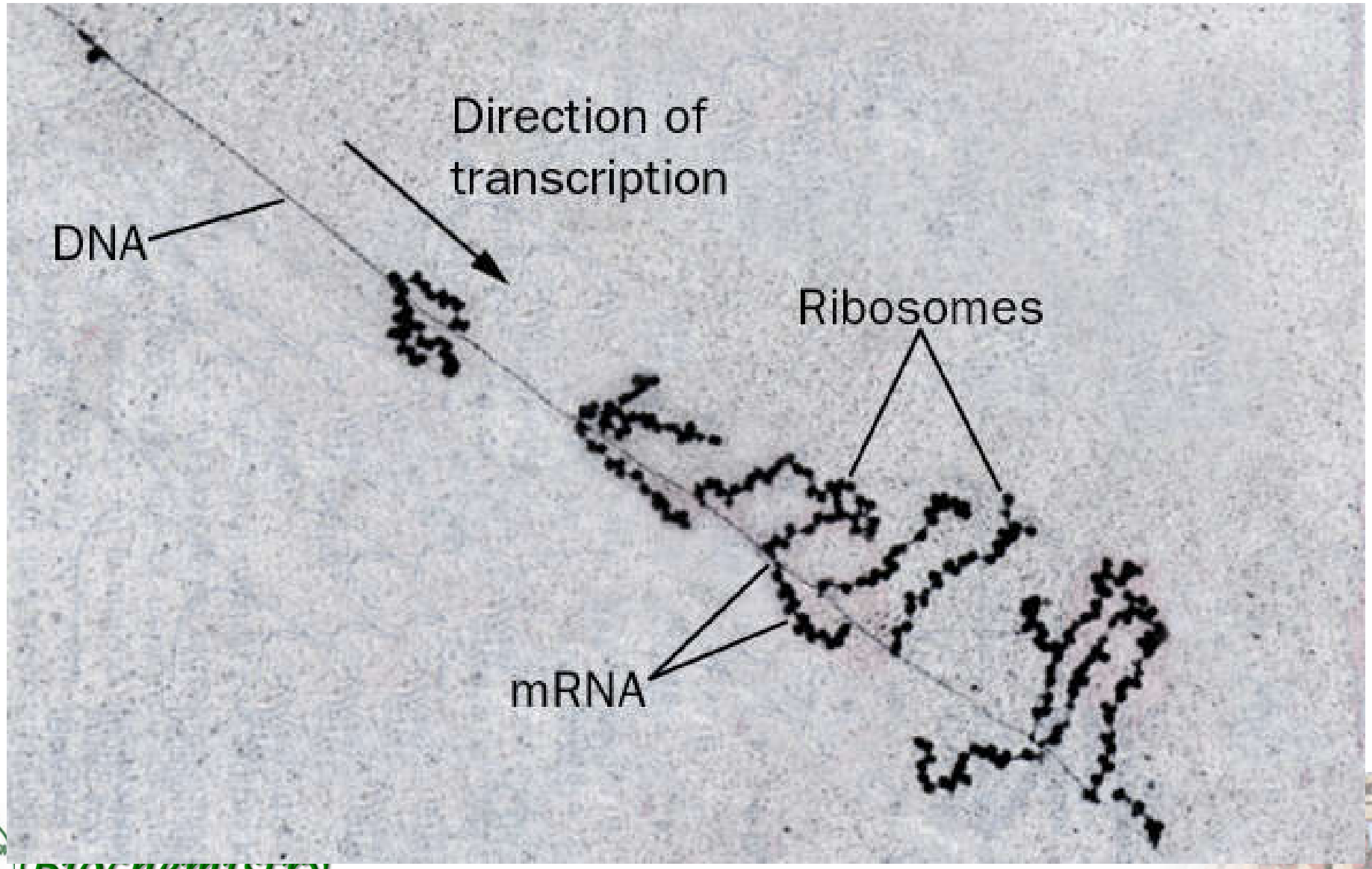

I initiation
DNA
Promoter
σ70
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Closed complex
2 Transcription bubble forms.
Open complex
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II elongation
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NusA
Promoter
RNA
III termination
6 Transcription is terminated. NusA dissociates, and the RNA polymerase is recycled.
RNA polymerase
RNA
NusA

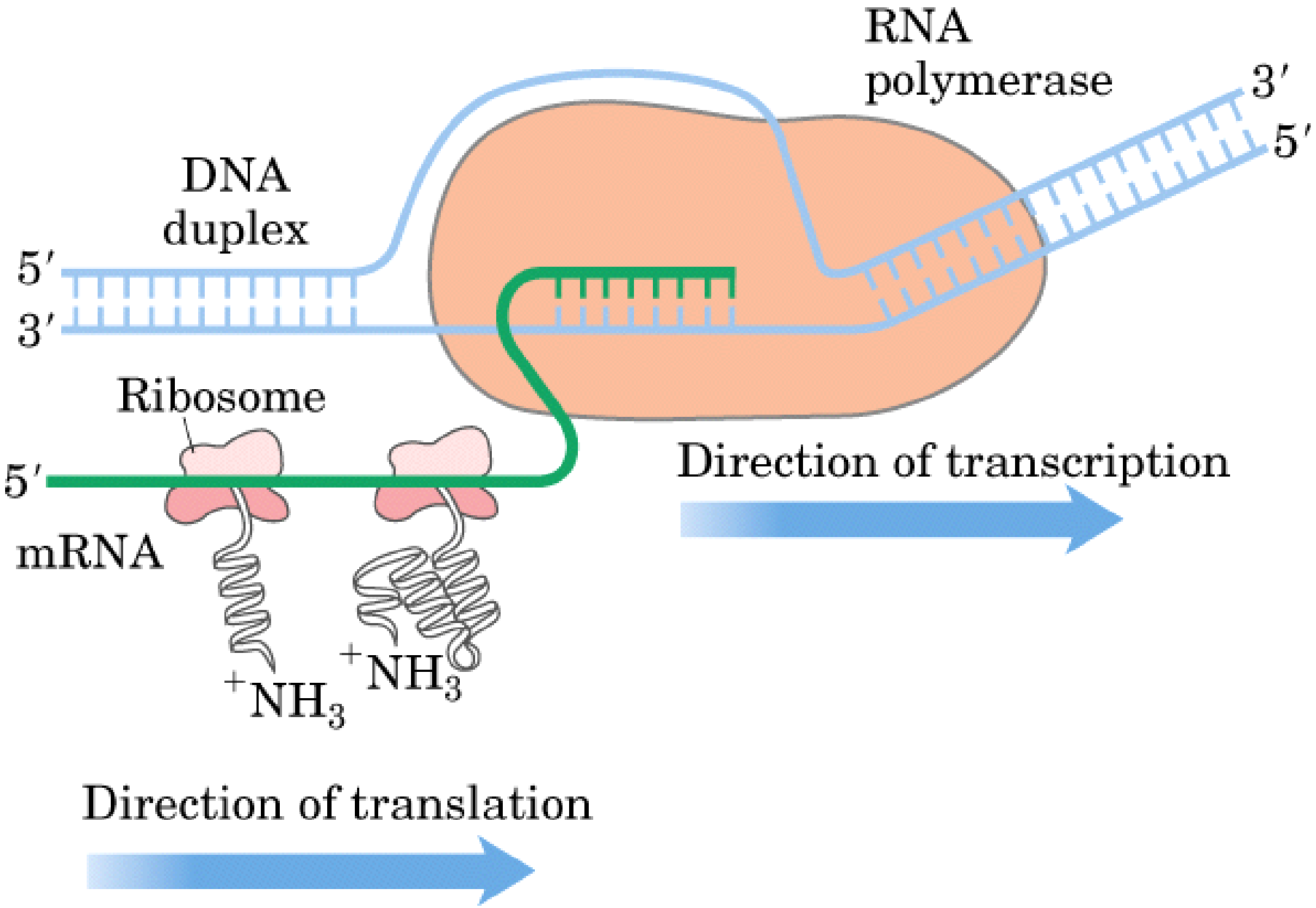
9. RNA processing

- In prokaryotes, RNA transcribed from protein-coding genes (messenger RNA, mRNA), requires little or no modification prior to translation
- However, mRNA in eukaryotes, ribosomal RNA (rRNA) and transfer RNA (tRNA) in both procaryotes and eukaryotes are synthesized as precursor molecules that do require post-transcriptional processing（转录后加工） (G8, G9, G10)



This E. coli gene is being simultaneously transcribed and translated




RNA
polymerase
3′
5′
DNA
duplex
5′
3′
Ribosome
5′
mRNA
+NH3
+NH3
Direction of transcription
Direction of translation

顺反子（cistron）：

即结构基因，决定一条多肽链合成的功能单位。

单顺反子（monocistron）：

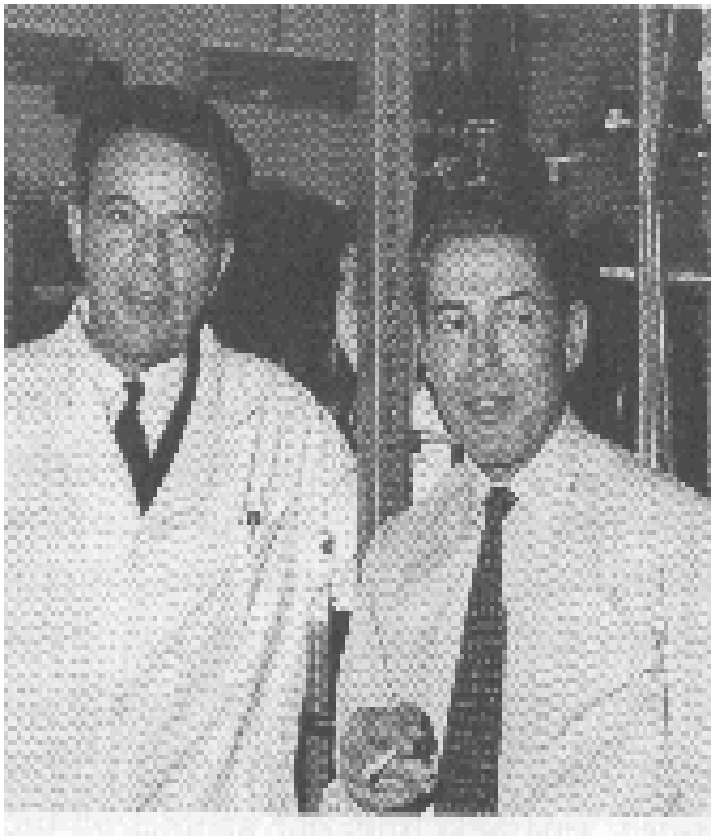
真核基因转录产物为单顺反子，即一个基因编码一条多肽链或RNA链，每个基因转录有各自的调节元件。

多顺反子（polycistron）：

在原核细胞中，通常是几种不同的mRNA连在一起，相互之间由一段短的不编码蛋白质的间隔序列所隔开，这种mRNA叫做多顺反子mRNA。



G3 The *Lac* Operon



操纵子学说

1、操纵子模型的提出

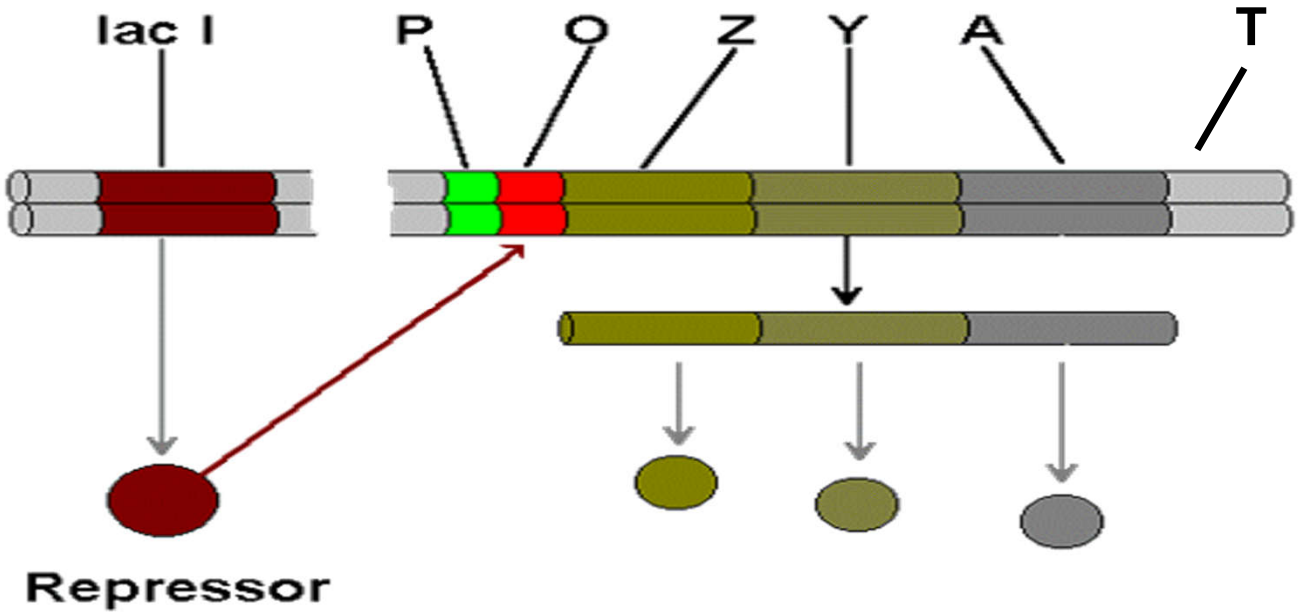
1961年，Monod和Jacob提出乳糖操纵子模型

获1965年诺贝尔生理学和医学奖



2、操纵子（operon）的定义

操纵子：是基因表达的协调单位，由启动子、操纵基因及其所控制的一组功能上相关的结构基因所组成。操纵基因受调节基因产物的控制。



■操纵子模型（lac operon）

操纵子的基本组成元件包括：

启 动 子（promoter，P）

操纵基因（operator，O）

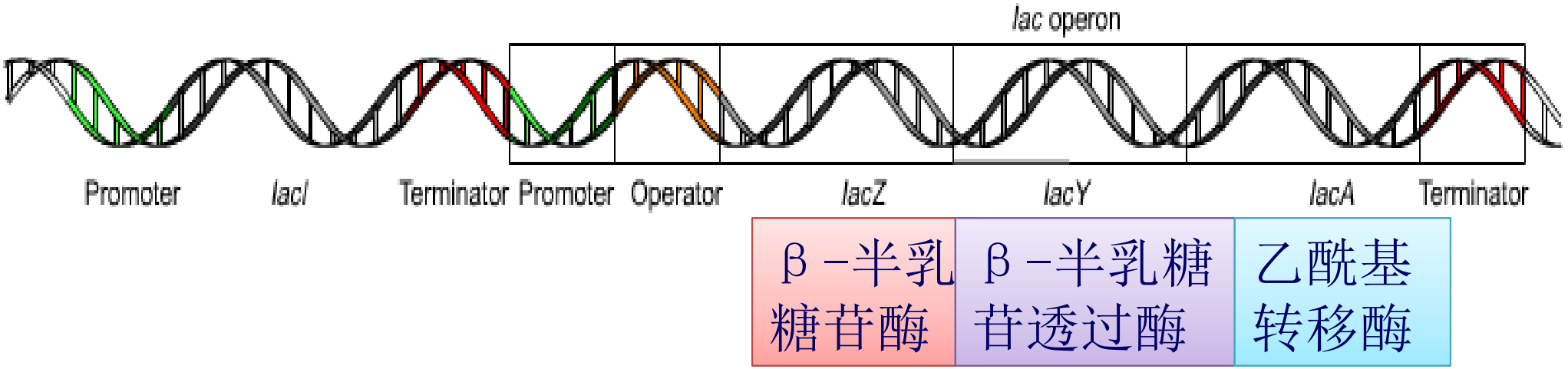
结构基因（structural gene(s)）

终 止 子 （terminator）

调节基因（regulatory gene）

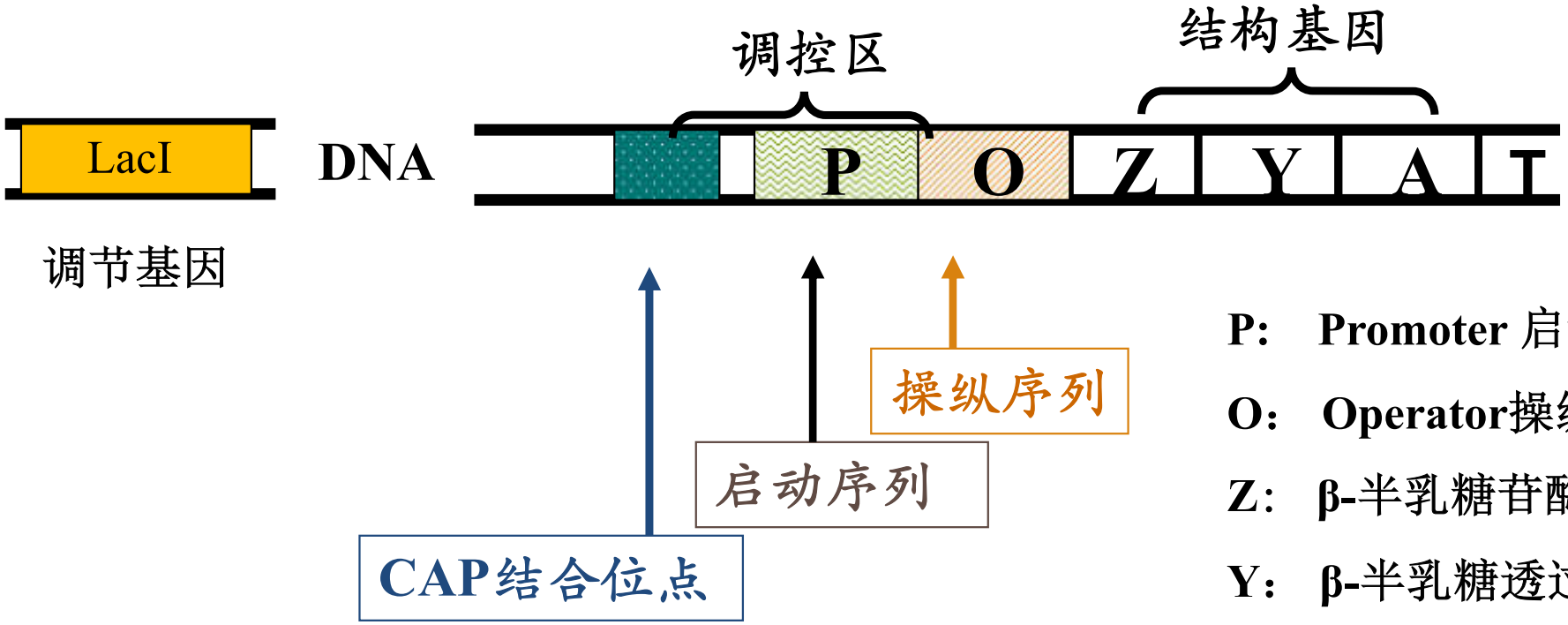


Lac operon (乳糖操纵子)



β-galactosidase (*Z*),
galactoside permease (*Y*),
thiogalactoside transacetylase (*A*)

乳糖操纵子-正、负调控模型



P: Promoter 启动子

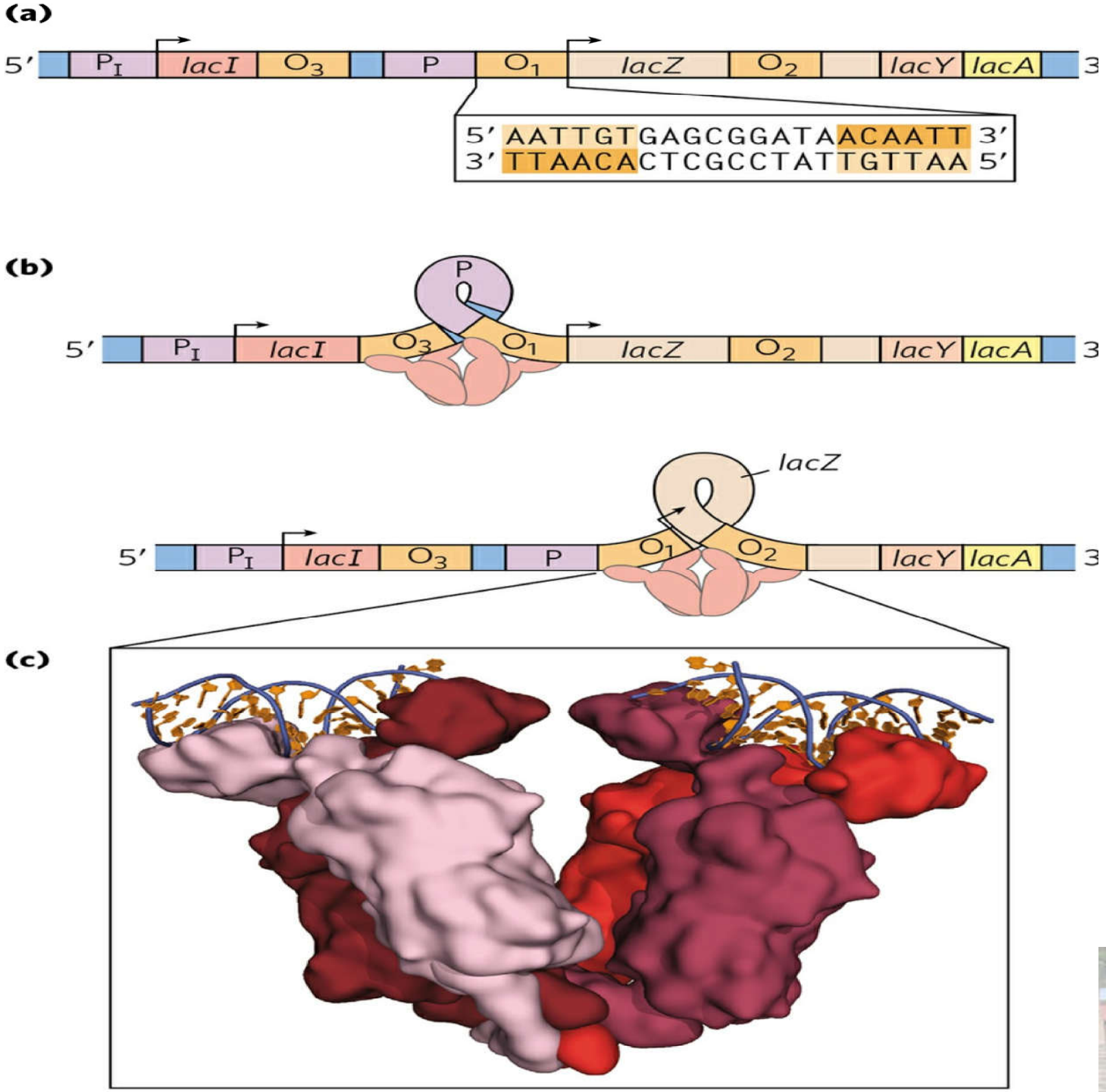
O: Operator操纵基因

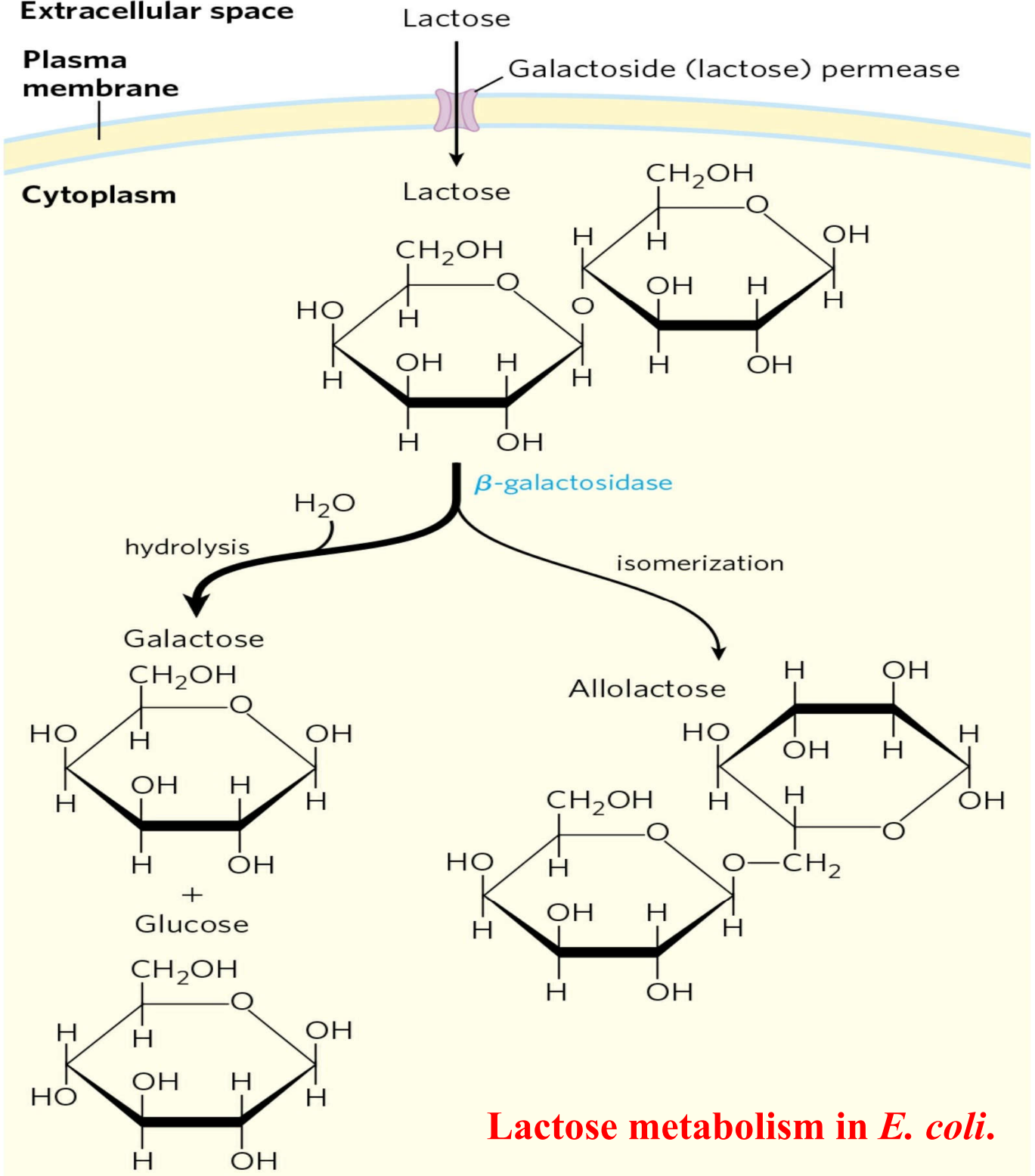
Z: β-半乳糖苷酶

Y: β-半乳糖透过酶

A: 乙酰基转移酶

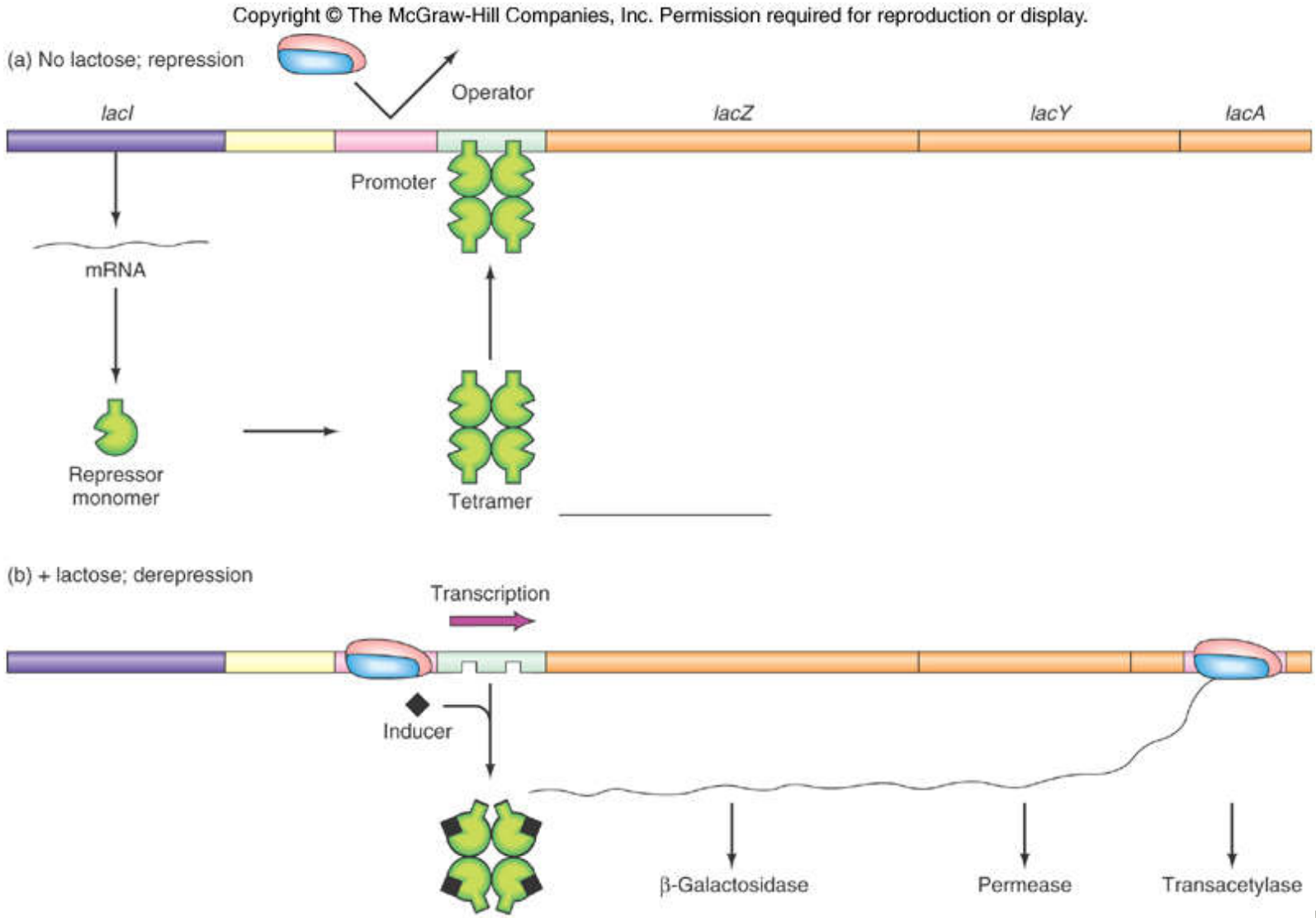
T: Terminator终止子

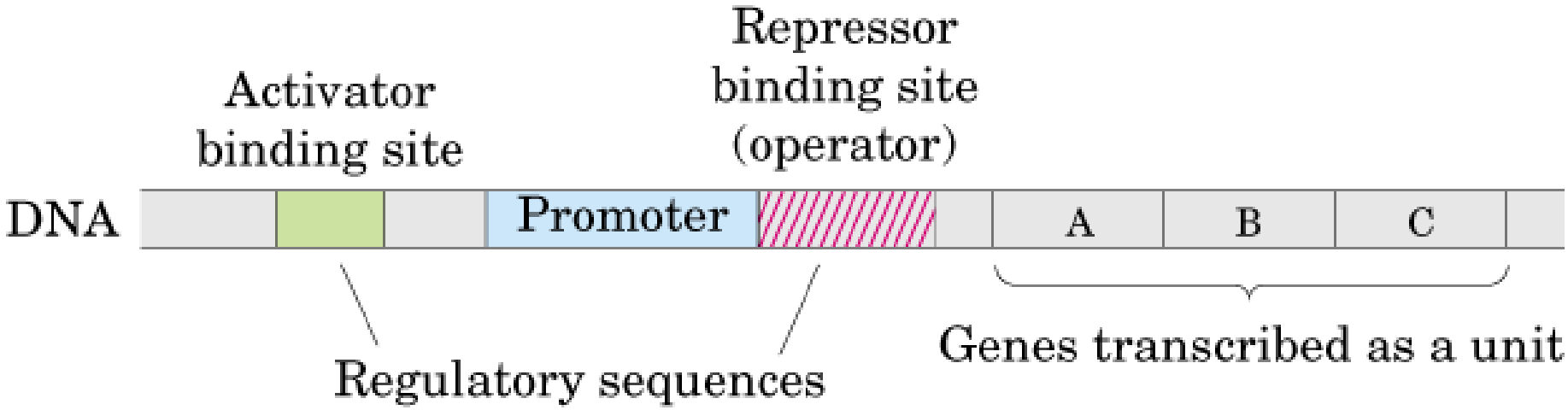

(a)
5′
P_I
lacI
O_3
P
O_1
lacZ
O_2
lacY
lacA
3
5′ AATTGTGAGCGGATAACAATT 3′
3′ TTAACACTCGCCTATTGTTAA 5′
(b)
P
5′
P_I
lacI
O_3
O_1
lacZ
O_2
lacY
lacA
3
lacZ
5′
P_I
lacI
O_3
P
O_1
O_2
lacY
lacA
3
(c)



Lactose metabolism in *E. coli*.

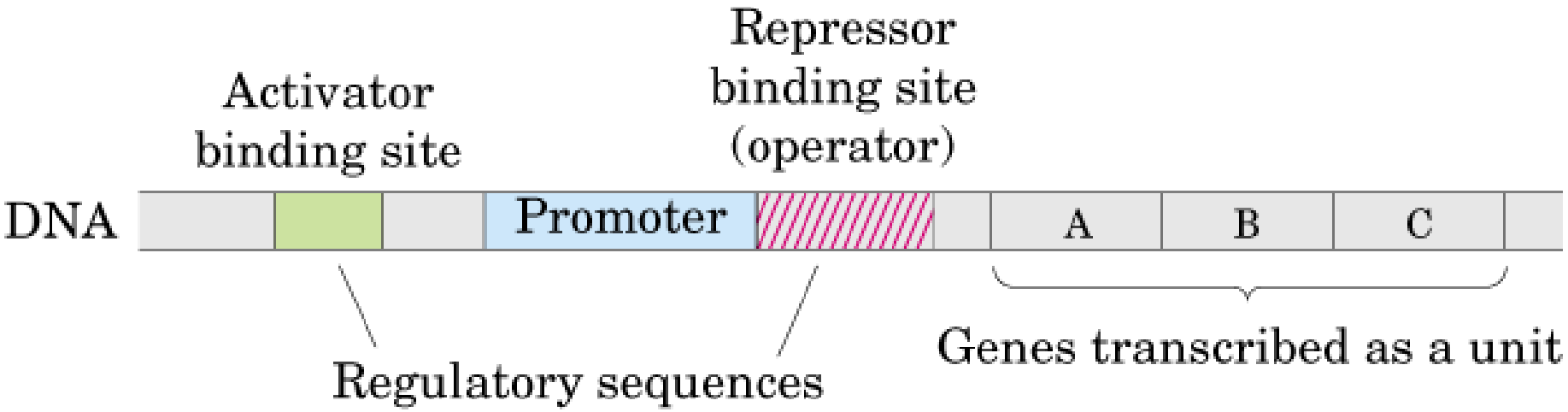
Negative Control of the *lac* Operon




Activator
binding site
Repressor
binding site
(operator)
DNA
Promoter
A
B
C
Regulatory sequences
Genes transcribed as a unit

cAMP与代谢物激活蛋白-正调控模型

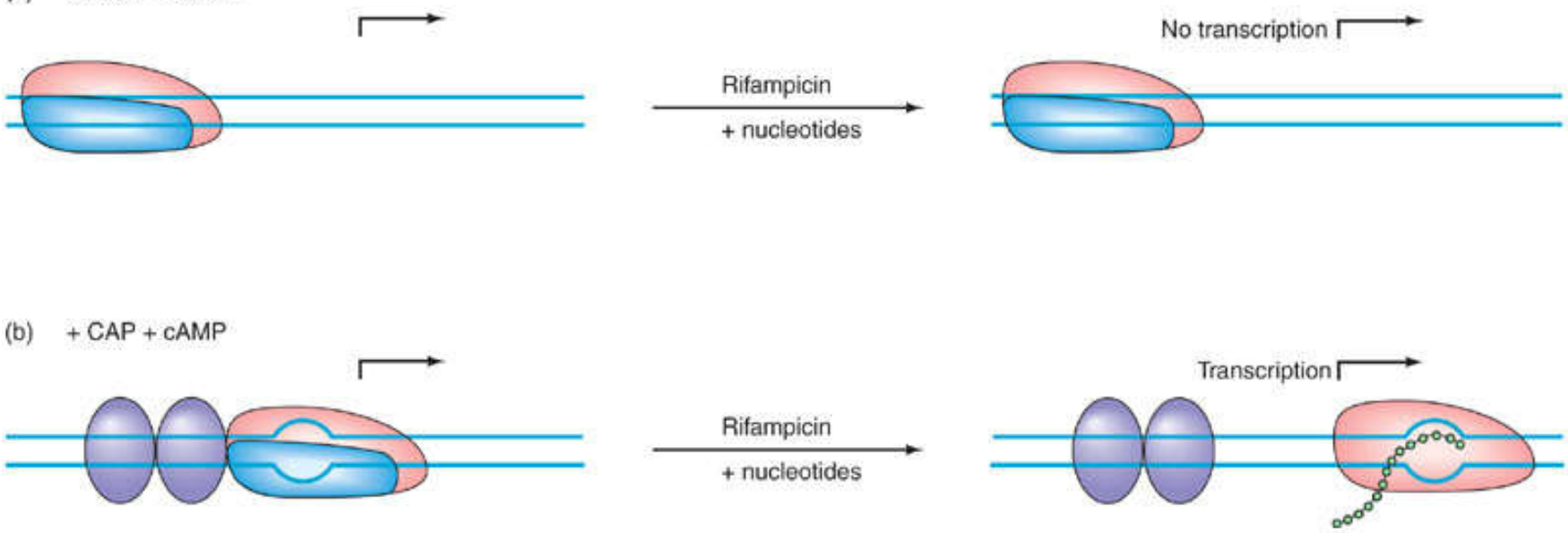
代谢物激活蛋白（CAP/CRP）(catabolite activator protein/cAMP receptor protein)



CAP Plus cAMP Action

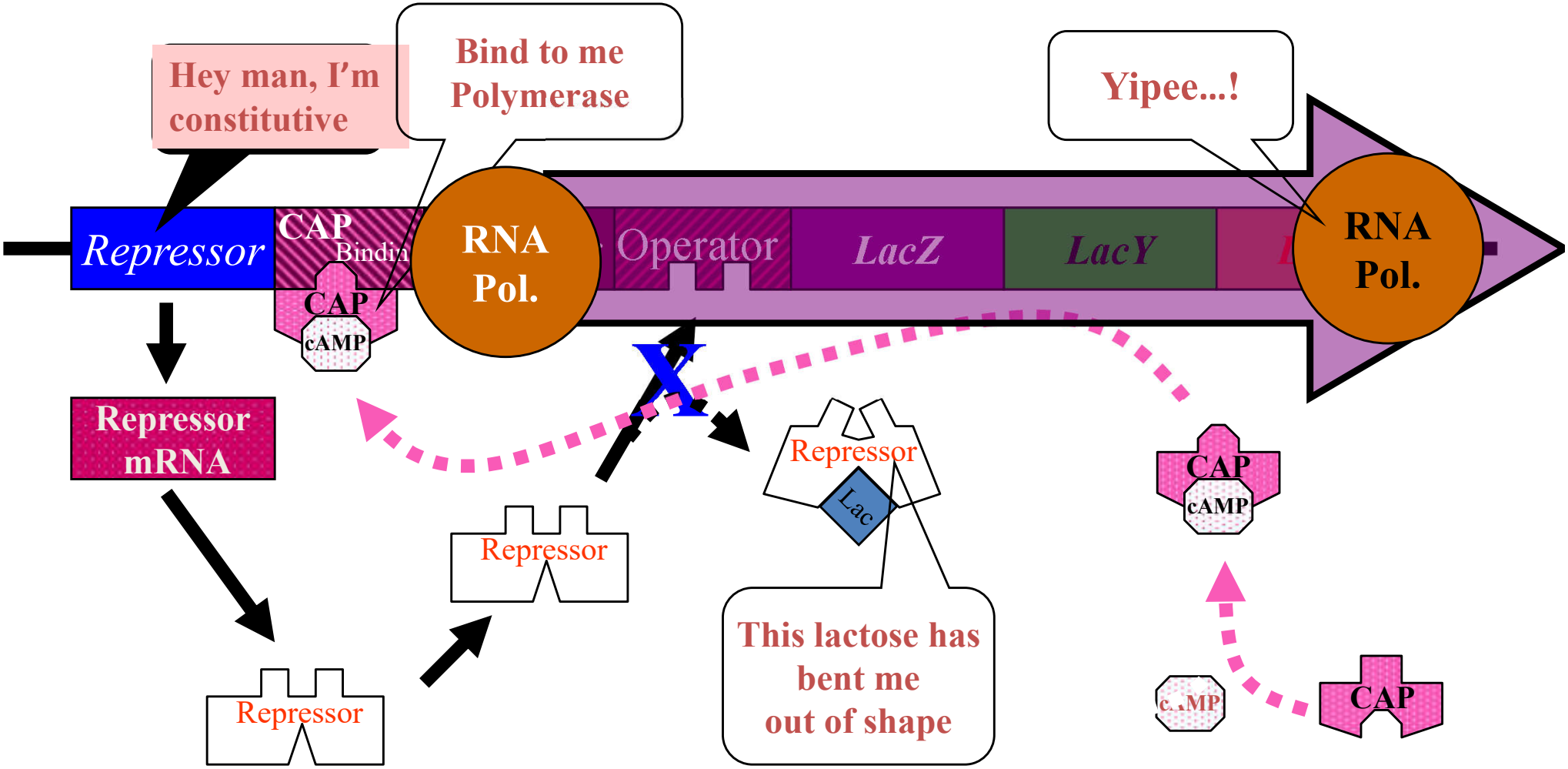
- The open promoter complex does not form even if RNA polymerase has bound the DNA unless the CAP-cAMP complex is also bound

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The *Lac* Operon:

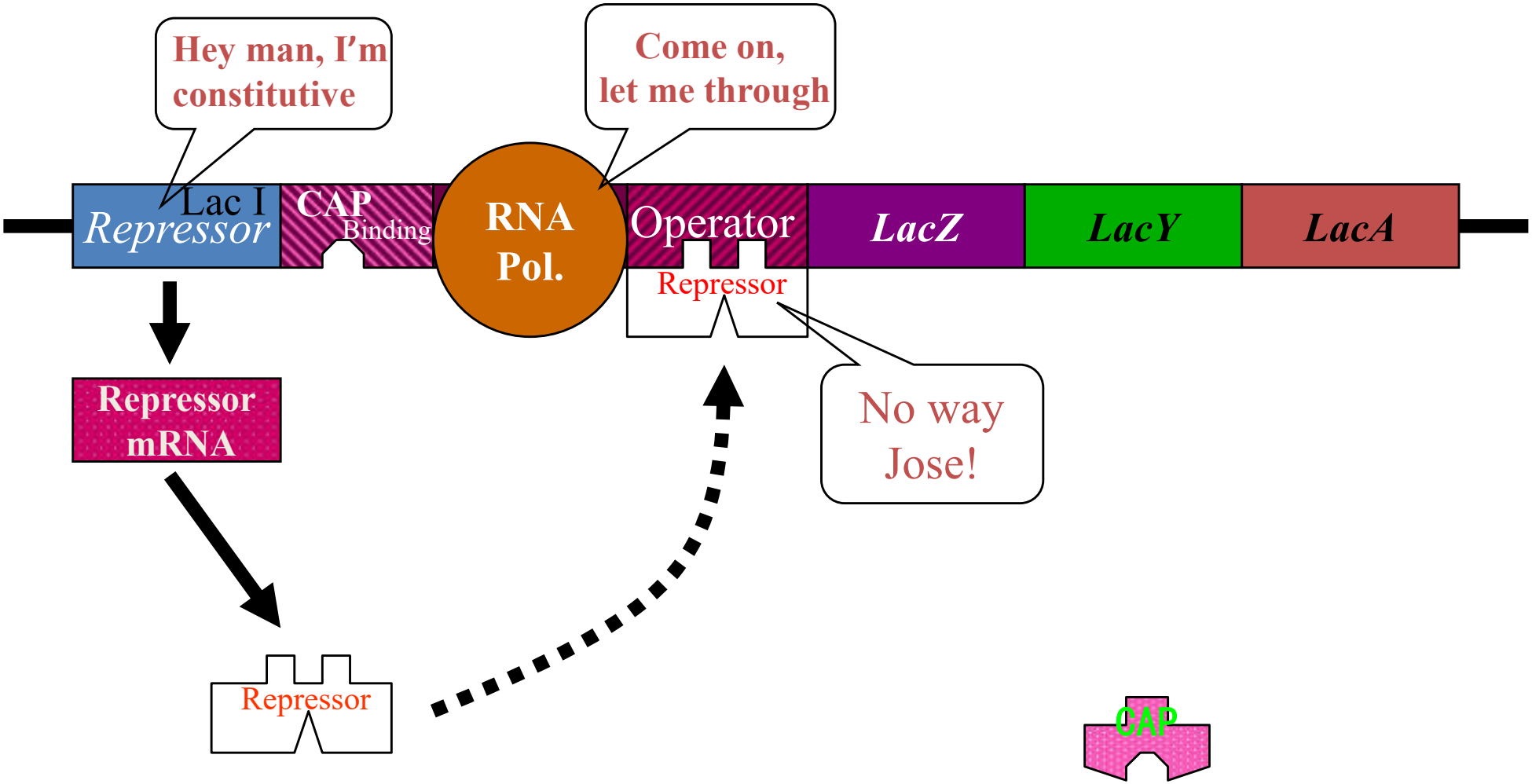
When Lactose Is Present But Not Glucose



Biochemistry

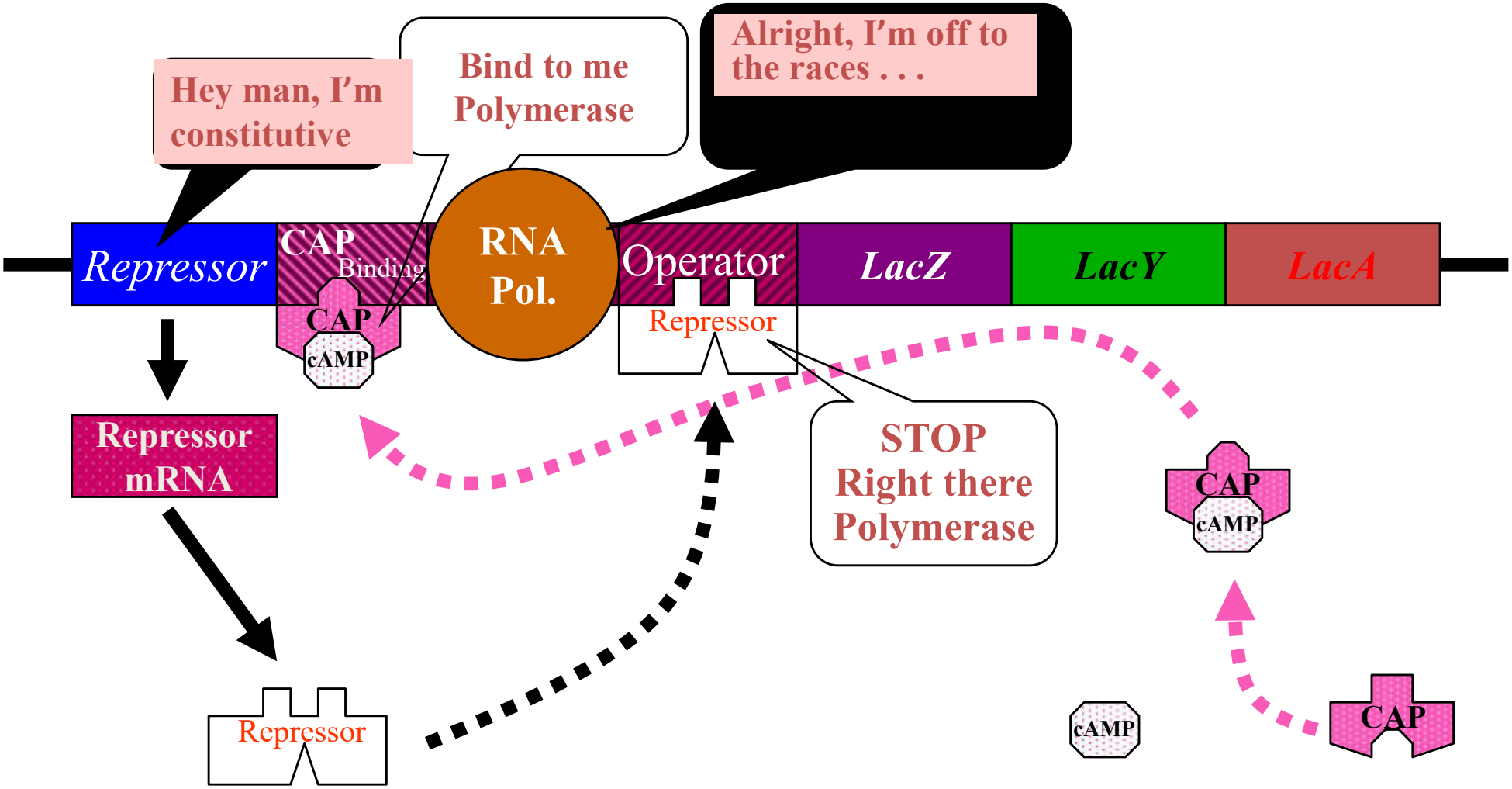
The *Lac* Operon:

When Glucose Is Present But Not Lactose

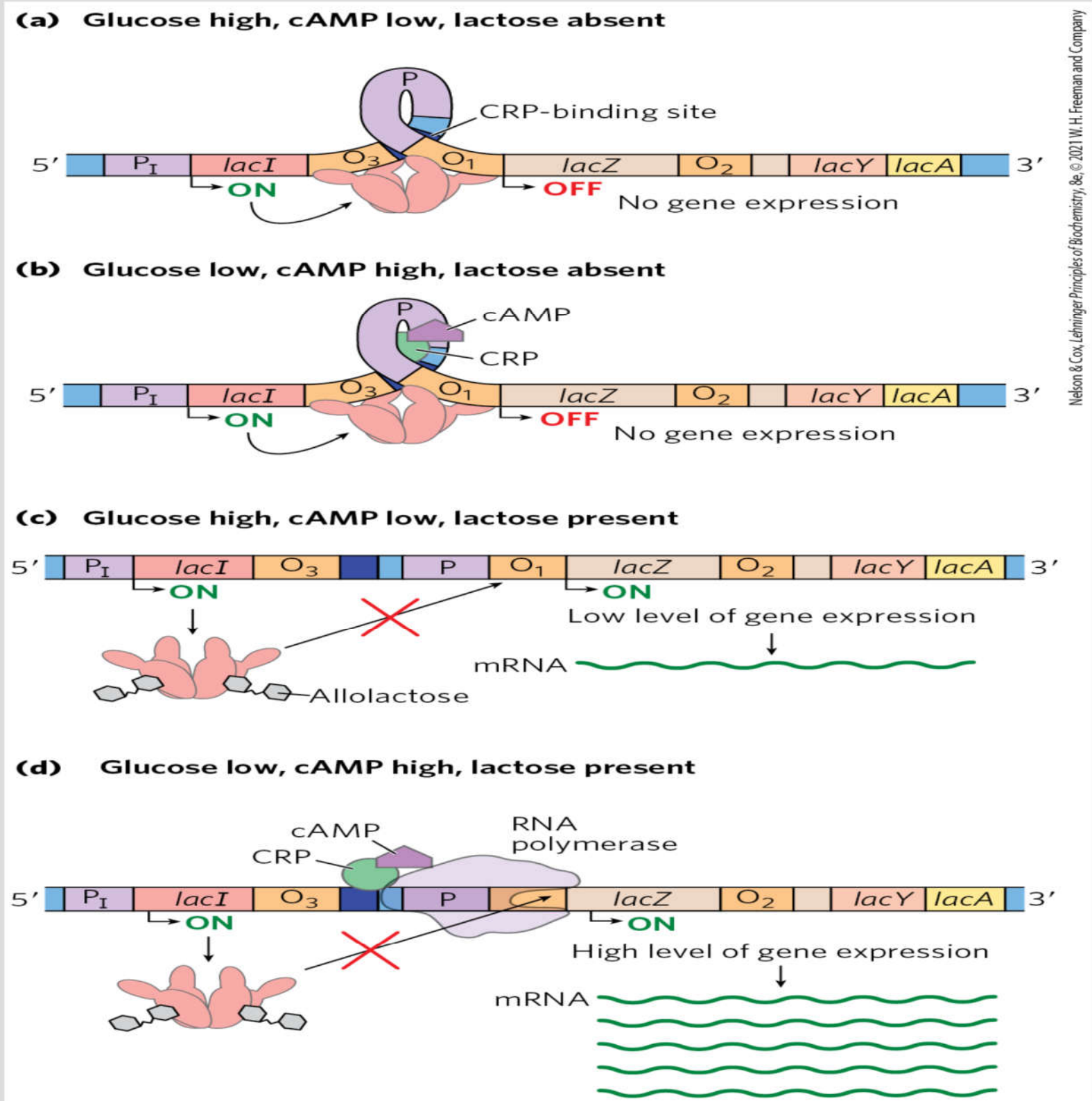


The *Lac* Operon:

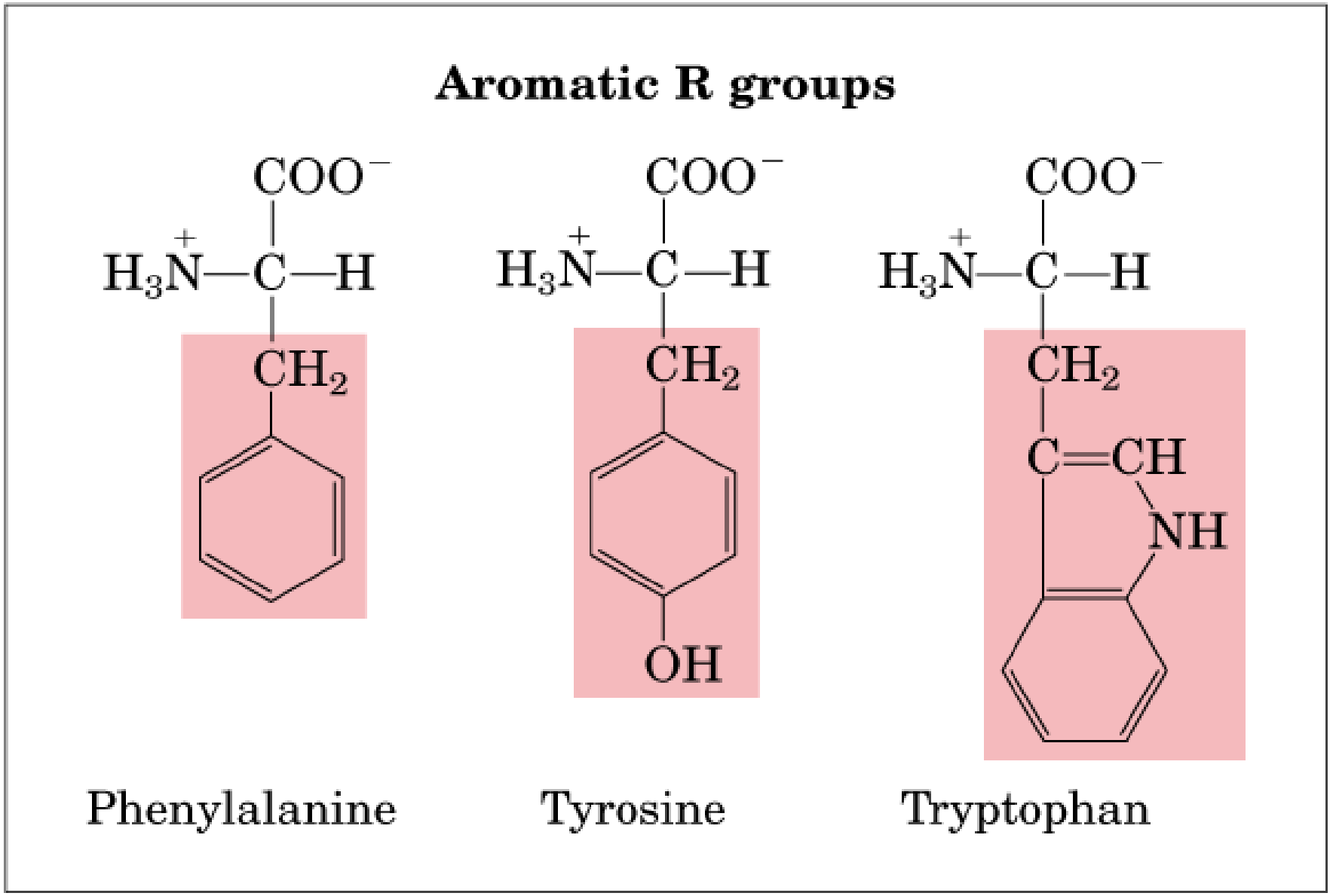
When Neither Lactose Nor Glucose Is Present

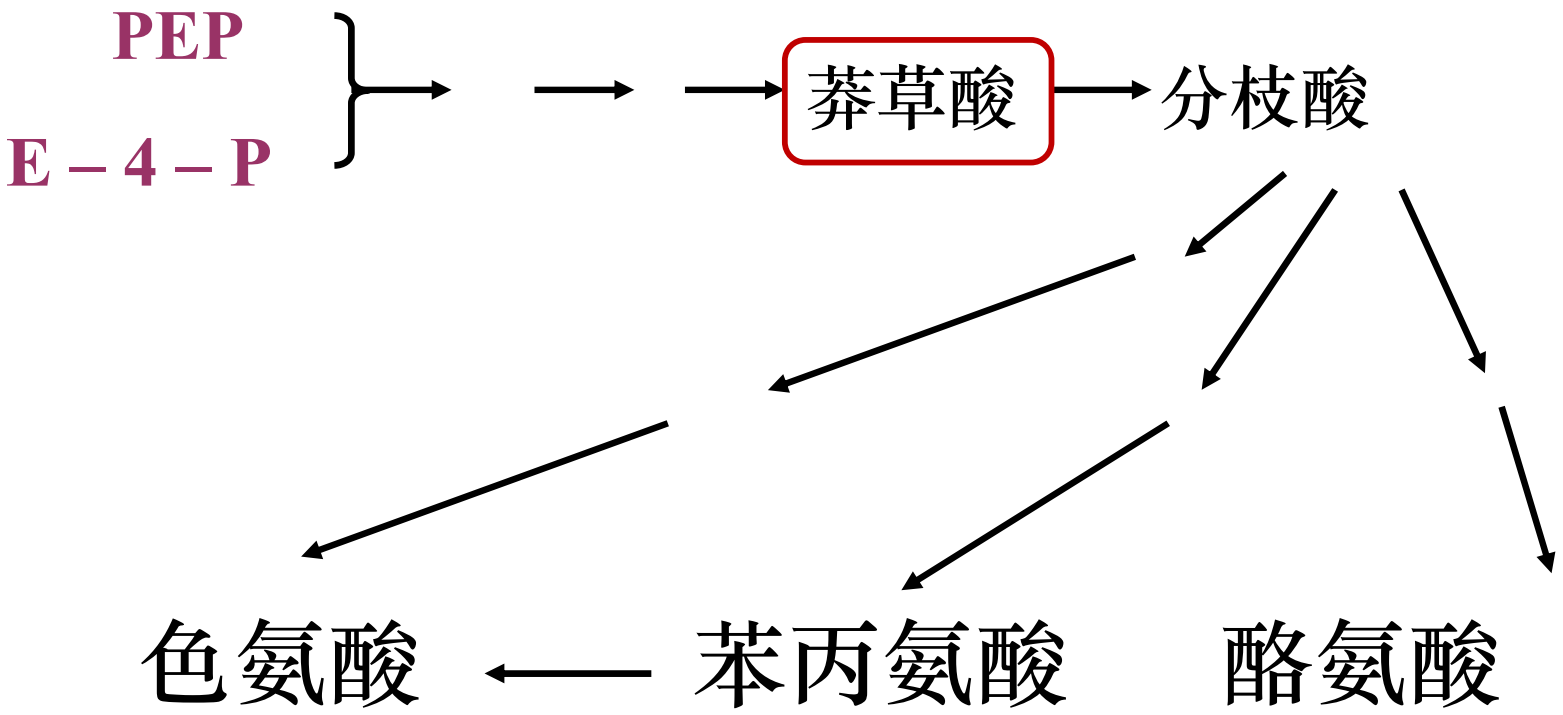


Biochemistry

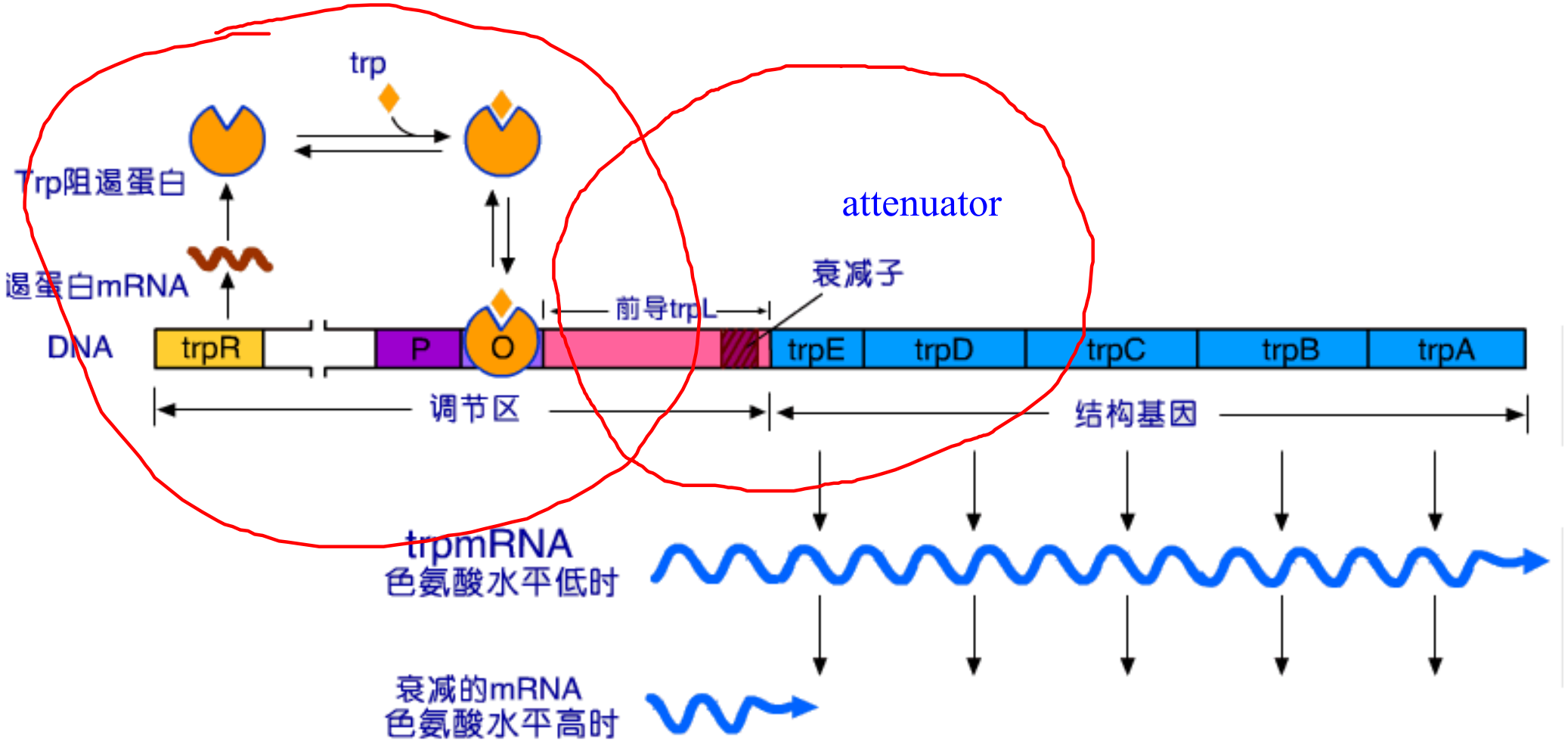

(a) Glucose high, cAMP low, lactose absent
P
CRP-binding site
5′
P_I
lacI
O_3
O_1
lacZ
O_2
lacY
lacA
3′
ON
OFF
No gene expression
(b) Glucose low, cAMP high, lactose absent
P
cAMP
CRP
5′
P_I
lacI
O_3
O_1
lacZ
O_2
lacY
lacA
3′
ON
OFF
No gene expression
(c) Glucose high, cAMP low, lactose present
5′
P_I
lacI
O_3
P
O_1
lacZ
O_2
lacY
lacA
3′
ON
ON
Low level of gene expression
mRNA
Allolactose
(d) Glucose low, cAMP high, lactose present
cAMP
CRP
RNA polymerase
5′
P_I
lacI
O_3
P
lacZ
O_2
lacY
lacA
3′
ON
ON
High level of gene expression
mRNA
Nelson & Cox, Lehninger Principles of Biochemistry, 8e, © 2021 W. H. Freeman and Company

Phe, F, 苯丙氨酸; Tyr, Y, 酪氨酸; Trp, W, 色氨酸



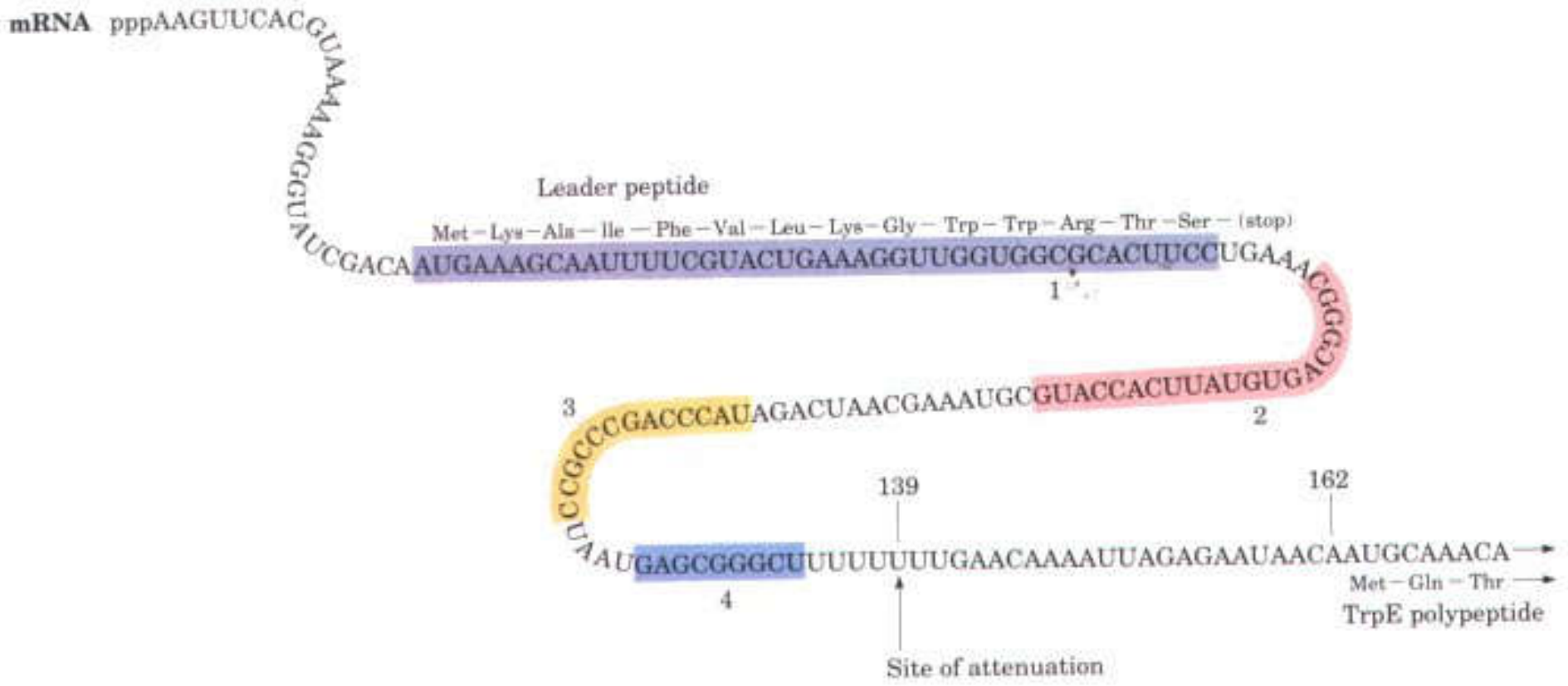

PEP
E – 4 – P
莽草酸
分枝酸
色氨酸
苯丙氨酸
酪氨酸

3. The tryptophan operon



色氨酸操纵子的结构与阻遏蛋白的负性调控

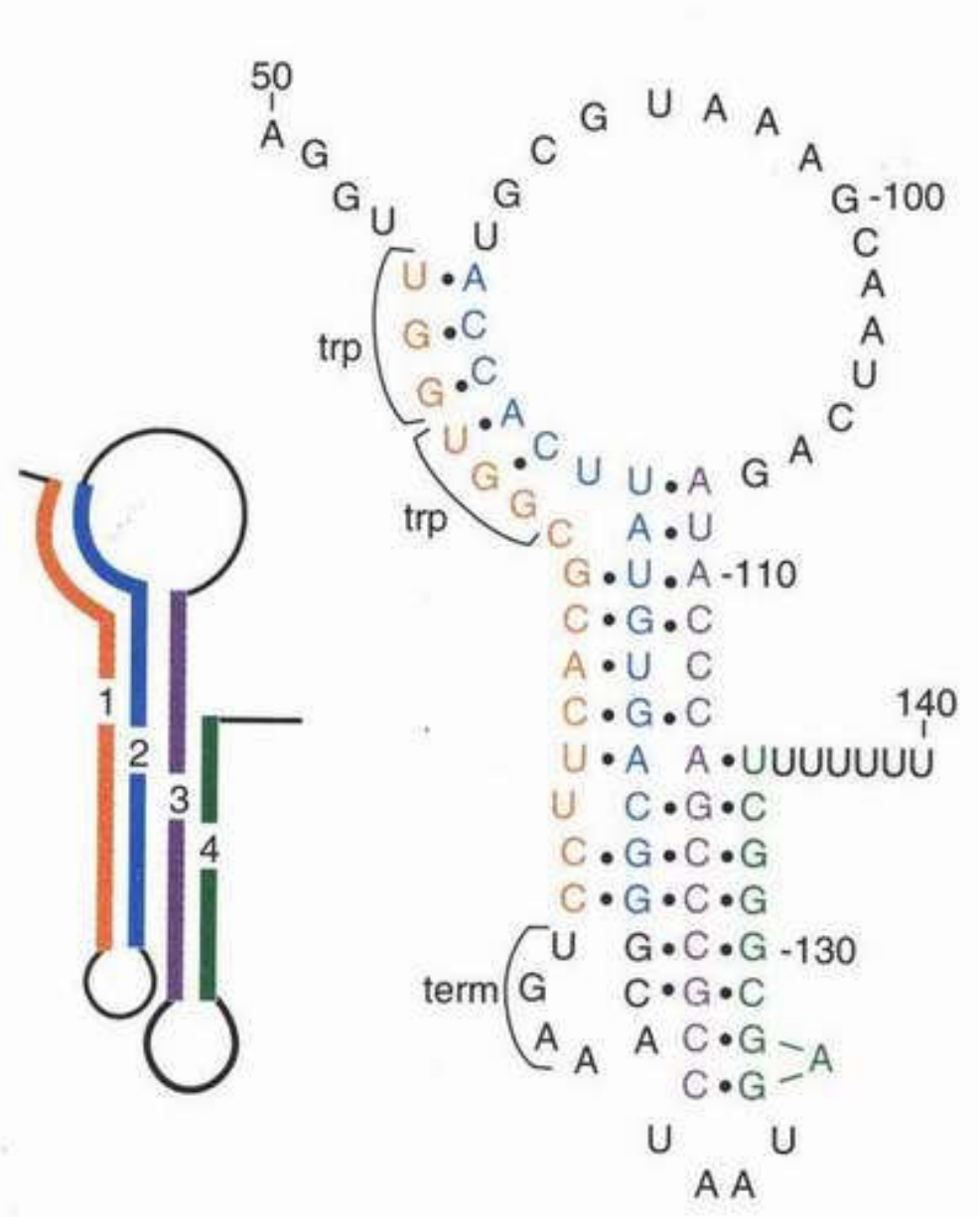
Leader peptide of trp operon



Attenuator (衰减子） (internal terminator), like a trancriptional terminator contains a palindromic GC-rich （回文结构）and a poly (U).



Leader peptide of trp operon



转录、翻译偶联

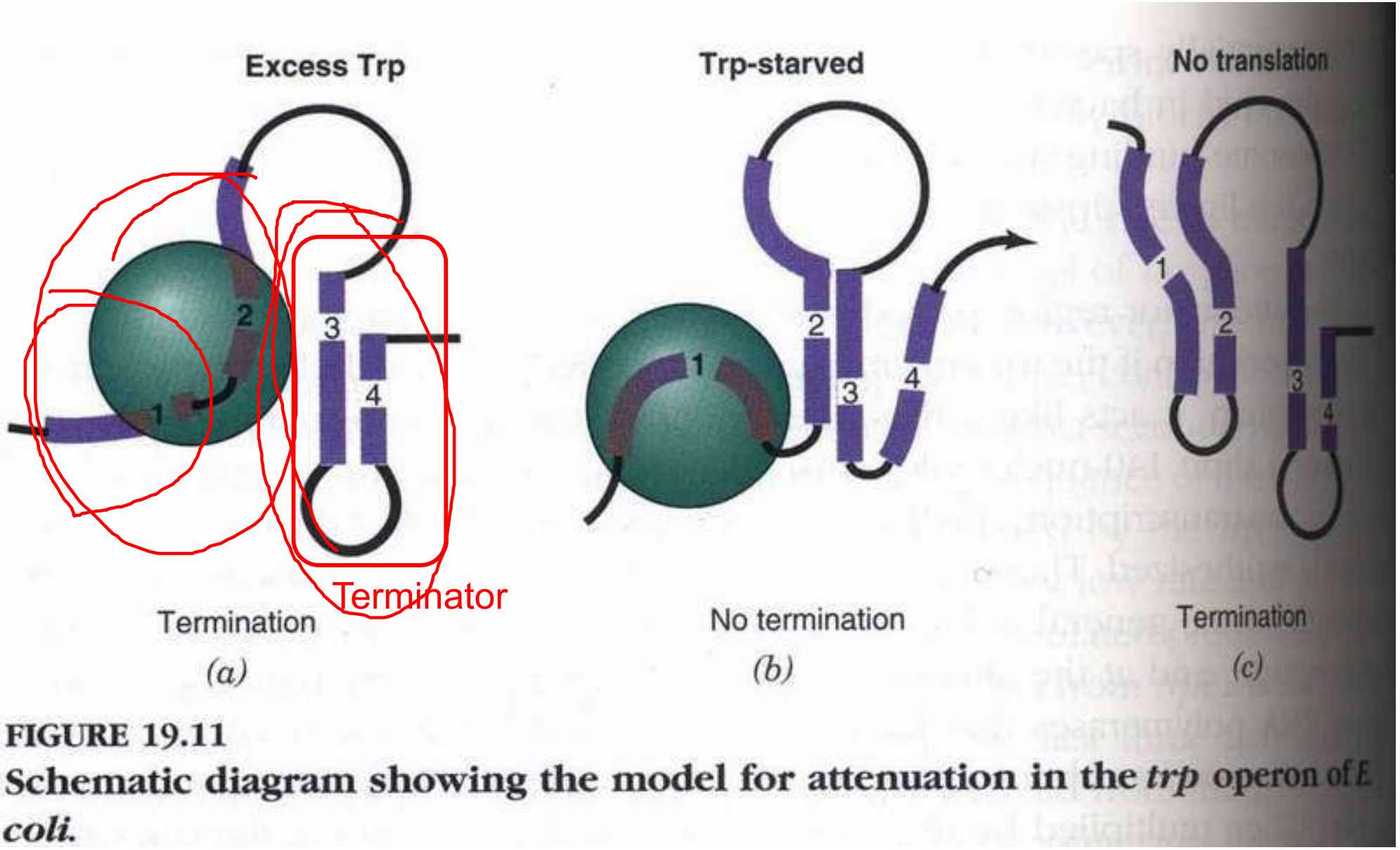
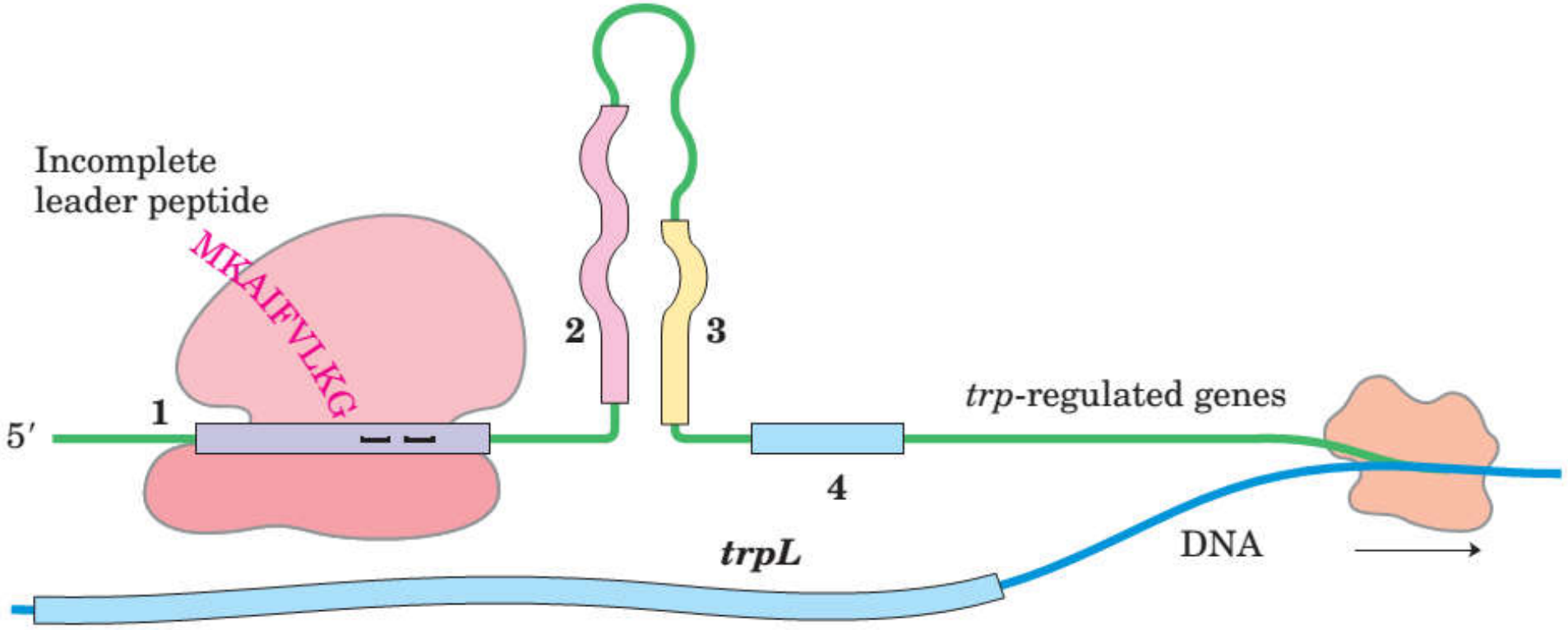
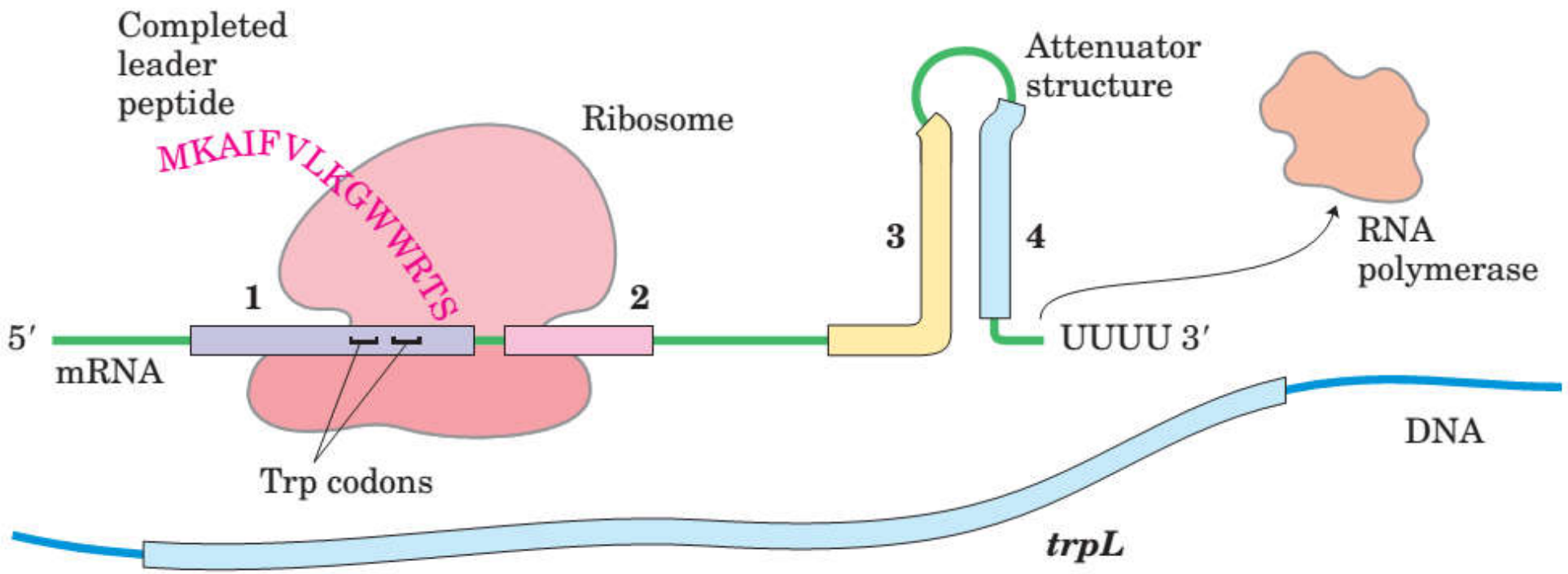


FIGURE 19.11
Schematic diagram showing the model for attenuation in the *trp* operon of *E. coli*.

When Trp levels are low, the ribosome pauses at the trp codons in sequence 1. Formation of the paired structure between sequence 2 and 3 prevents attenuation, because sequence 3 is no longer available to form the attenuator structure with sequence 4.



When Trp levels are high, the ribosome quickly translates sequence 1 and blocks sequence 2 before sequence 3 is transcribed. Continued transcription leads to the terminator-like structure formed by sequence 3 and 4



G5 Transcription in Eukaryotes: an Overview

G6 Transcription of Protein-Coding Genes in Eukaryotes

Biochemistry



1. Three kinds of RNA polymerases (I, II, and III) have been revealed for making RNAs in the nucleus of eukaryotic cells

- Each is responsible for the transcription of a certain groups of genes: Ribosomal RNA（rRNA）, Messenger RNA （mRNA） or Transfer RNA （tRNA） genes.
- The enzymes are often identified by examining their sensitivity towards α-amanitin （鹅膏蕈碱） (from a toxic mushroom).





Amanita Phalloides



Amanita bisporigera

真核生物**RNA**聚合酶Ⅰ,Ⅱ,Ⅲ对α-鹅膏蕈素的敏感程度不同。

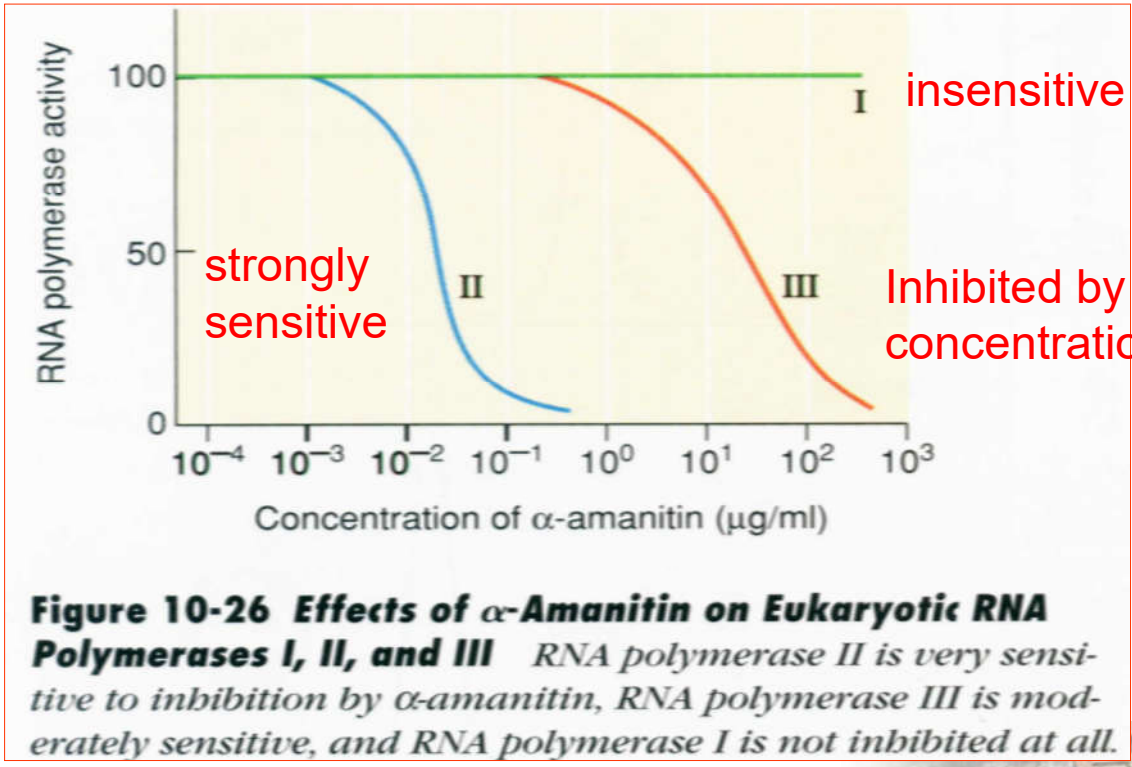


Figure 10-26 ***Effects of α-Amanitin on Eukaryotic RNA Polymerases I, II, and III*** *RNA polymerase II is very sensitive to inhibition by α-amanitin, RNA polymerase III is moderately sensitive, and RNA polymerase I is not inhibited at all.*

α-鹅膏蕈素对真核生物RNA聚合酶Ⅰ,Ⅱ,Ⅲ的影响

Eukaryotic RNA Polymerases

Type	Localization	Cellular transcripts
I	Nucleolus	18S, 5.8S, and 28S ribosomal RNAs (rRNA)
II	Nucleoplasm	mRNA precursors and small nuclear RNAs (snRNA)
III	Nucleoplasm	tRNAs and 5S rRNAs



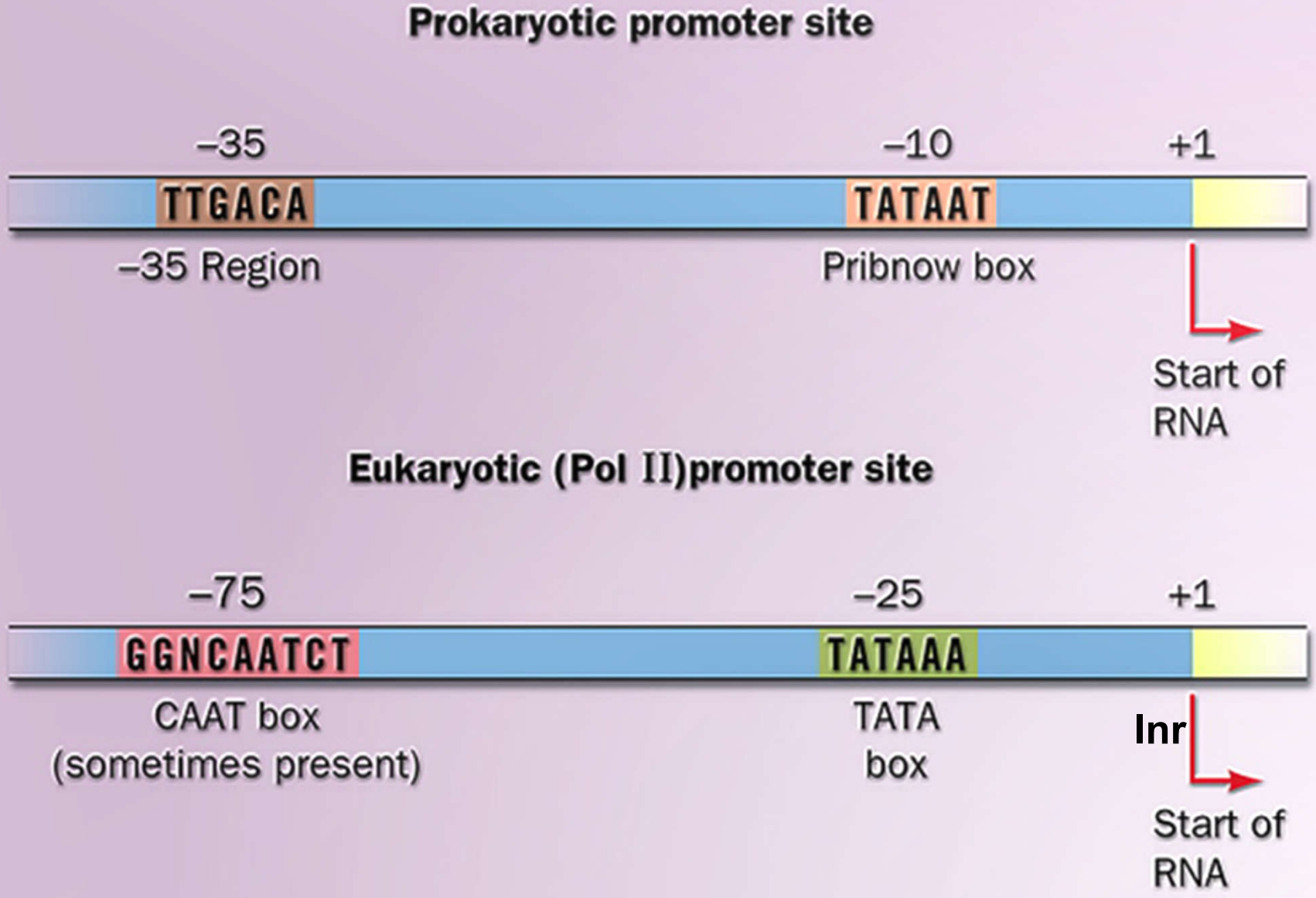
Each of the three RNA polymerases contains 12 or more subunits, some of which are similar to those of *E. coli* RNA polymerase. However, four to seven subunits in each enzyme are unique to that enzyme.



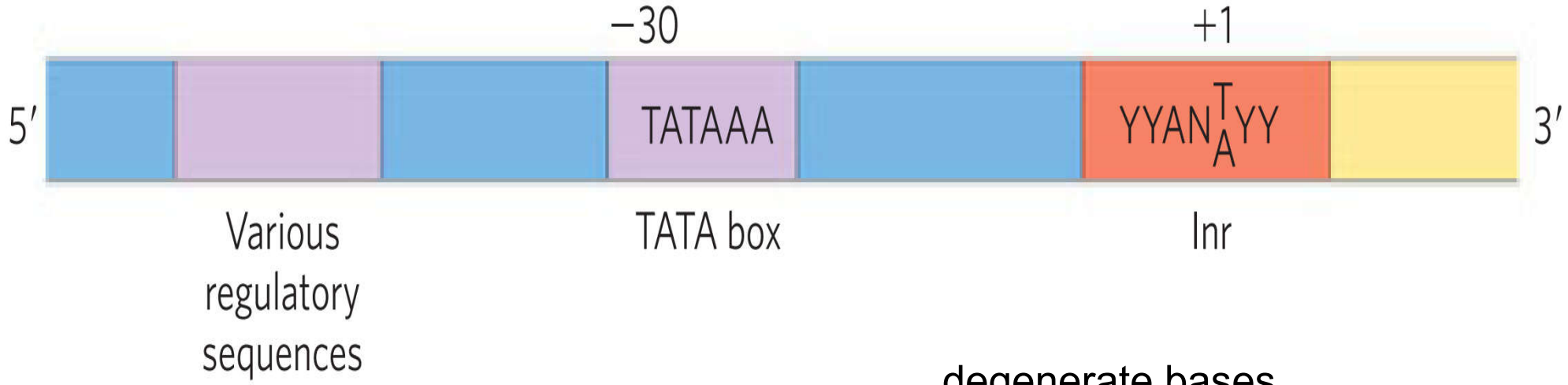
2. RNA polymerase II (Pol II) binds to promoters of thousands of protein-coding genes

- **Many Pol II promoters contain a TATAAA sequence (called a TATA box) at -25 (-30) position and an initiator sequence (Inr) at +1 position.**
- **The preinitiation complex (including Pol II) is believed to assemble at the TATA box, with DNA unwound at the Inr sequence.**
- **However, many Pol II promoters lack a TATA or Inr or both sequences!**




Prokaryotic promoter site
–35
–10
+1
TTGACA
TATAAT
–35 Region
Pribnow box
Start of RNA
Eukaryotic (Pol II)promoter site
–75
–25
+1
GGNCAATCT
TATAAA
CAAT box (sometimes present)
TATA box
Inr
Start of RNA

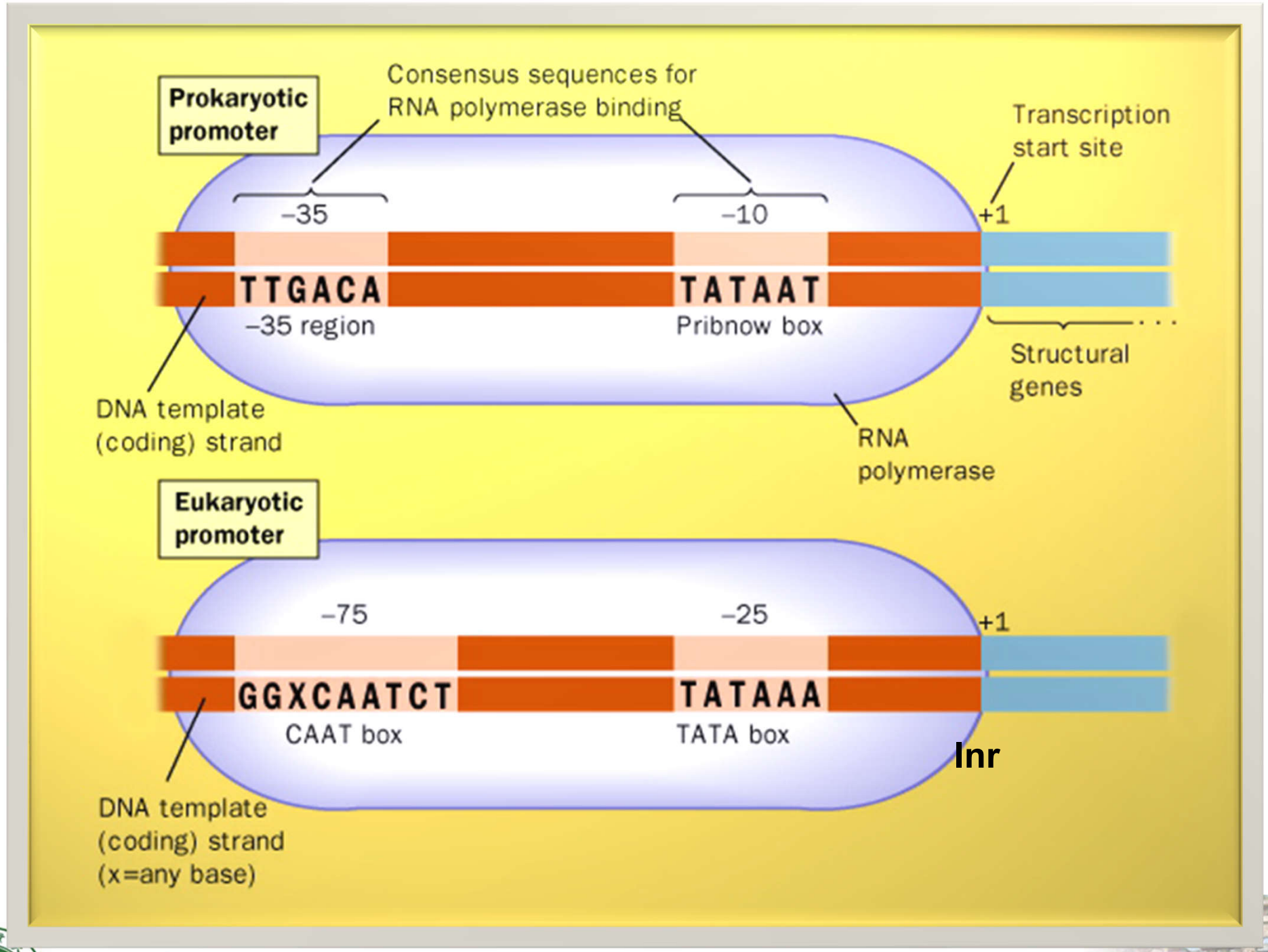
General features of promoters for protein-coding genes in higher eukaryotes



degenerate bases

简并碱基符号对应表

简并碱基		正常碱基
R		A/G
Y		C/T
N		A/G/C/T

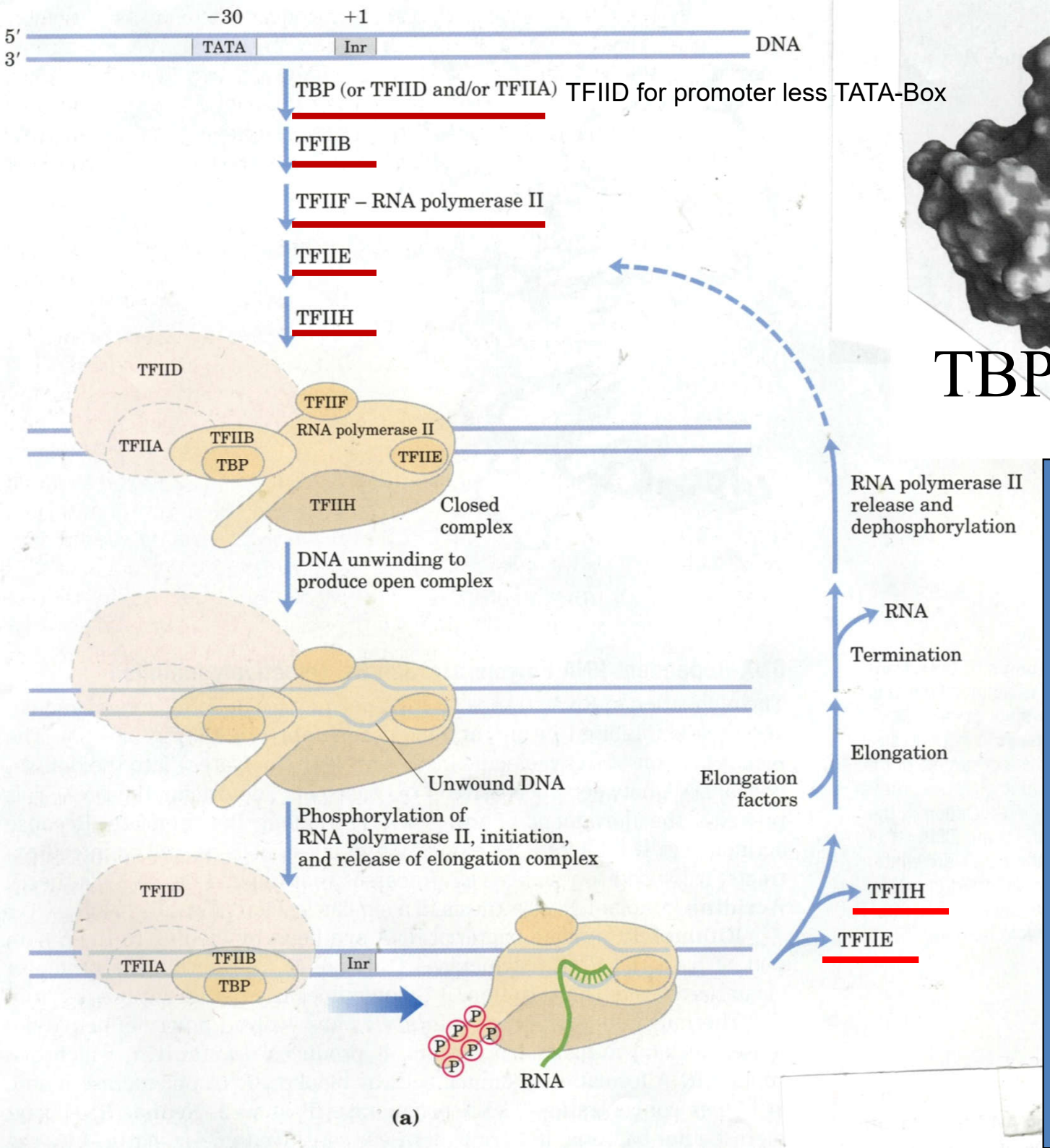

Prokaryotic promoter
Consensus sequences for RNA polymerase binding
Transcription start site
−35
−10
+1
TTGACA
TATAAT
−35 region
Pribnow box
Structural genes
DNA template (coding) strand
RNA polymerase
Eukaryotic promoter
−75
−25
+1
GGXCAATCT
TATAAA
CAAT box
TATA box
Inr
DNA template (coding) strand (x=any base)

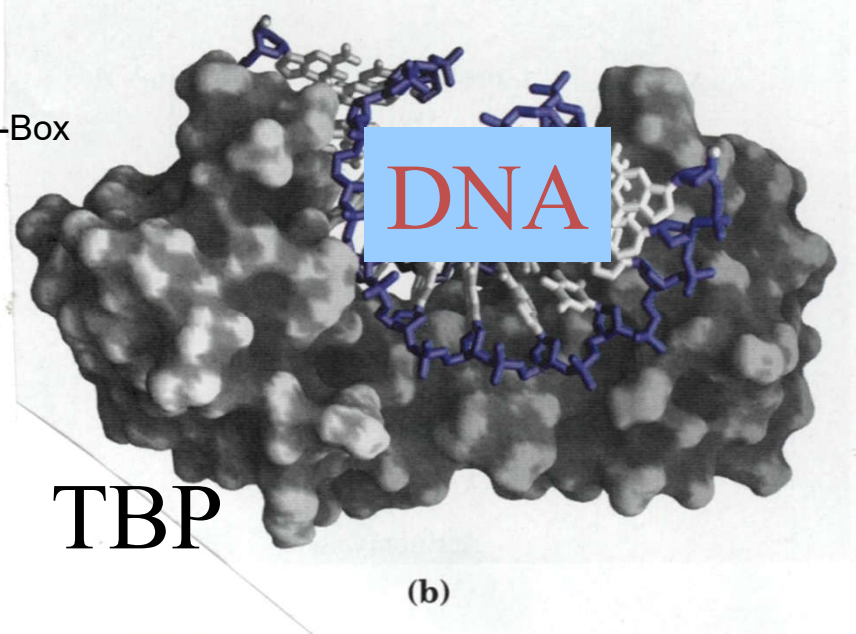
3. Pol II is helped by an arrays of protein factors (called general transcription factors(TF)) to form an active transcription complex at a promoter

- **Firstly, the TATA-binding protein (TBP) binds to the TATA box, then TFIIB, TFIIF-Pol II, TFIIE, and TFIIH (TFII, Transcription Factor for RNA polymerase II) will be added in order forming the closed complex at the promoter.**

- **TFIIH then acts as a helicase to unwind the DNA duplex at the Inr site, forming the open complex.**







A proposed model for Pol II-catalyzed mRNA synthesis

- **A kinase activity of TFIIH will phosphoryate the C-terminal domain of Pol II, which will initiate RNA synthesis and release the elongation complex.**
- **TFIIE and TFIIH will be released after the elongation complex moves forward for a short distance.**



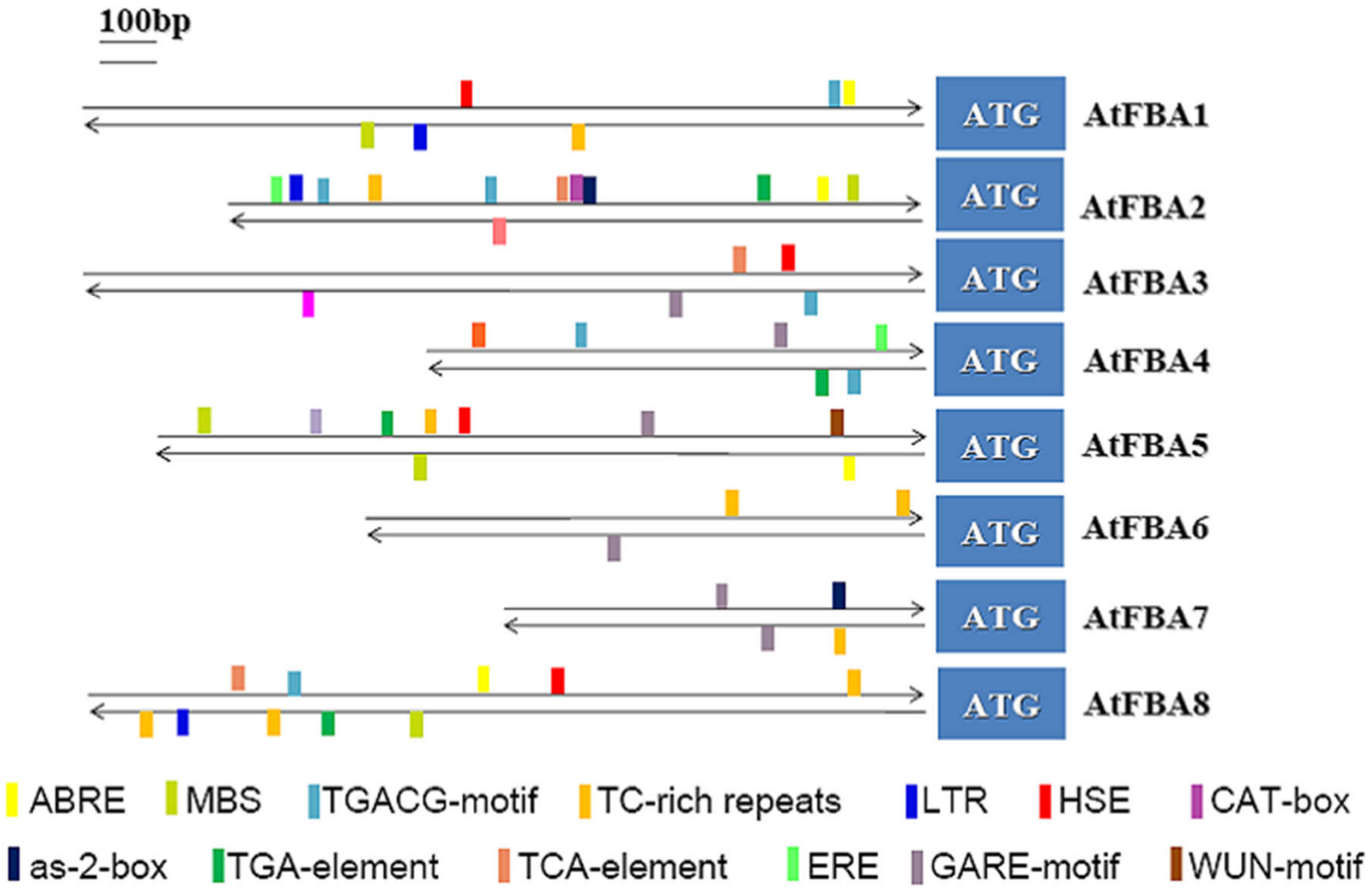
- Transcriptional regulation in a eukaryotic cell is mediated by transcription factors (other than general transcription factors) which recognize and bind to short regulatory DNA sequences (control elements) associated with the gene. These sequences are also called cis-acting elements (cis-elements) (顺式作用元件）. And the transcription factors are also known as trans-acting factors (反式作用因子）.



- The transcription factors regulate specific gene transcription by interacting with the proteins of the transcription initiation complex and either **increase or decrease** the rate of transcription of the target gene.



Many regulatory elements in a promoter



顺式作用元件(cis-action elements)

Each cis-acting element has a specific sequence composition

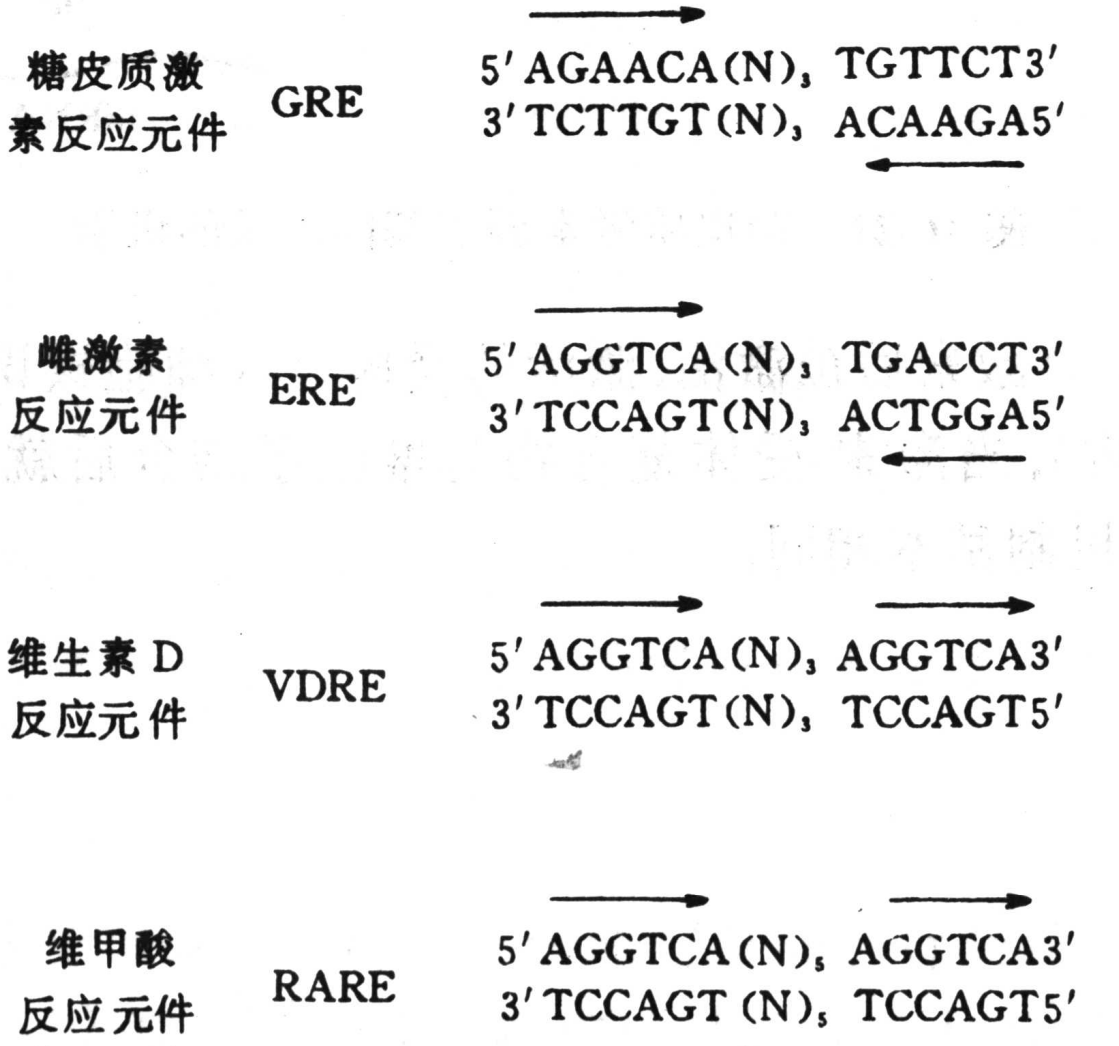
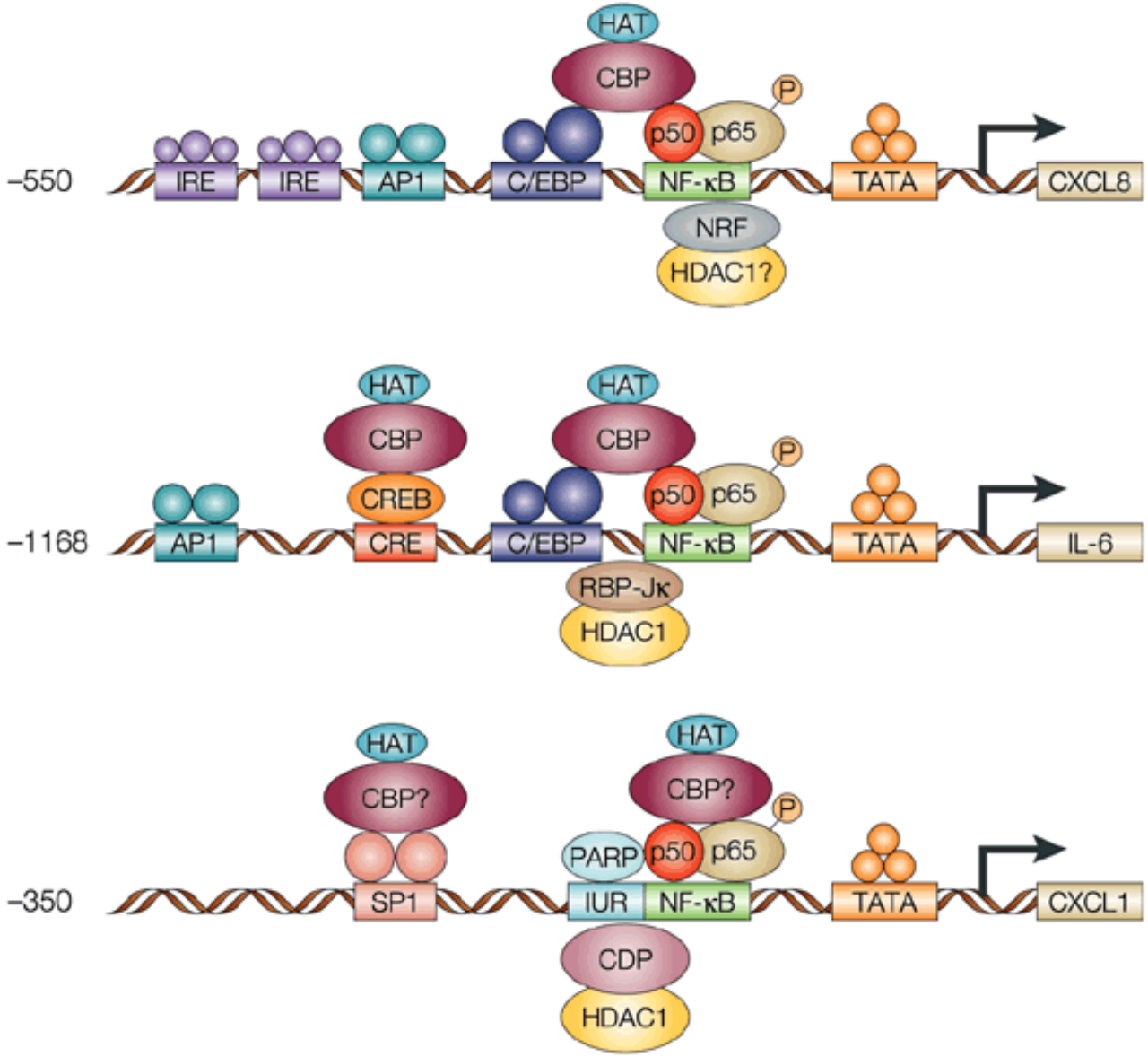


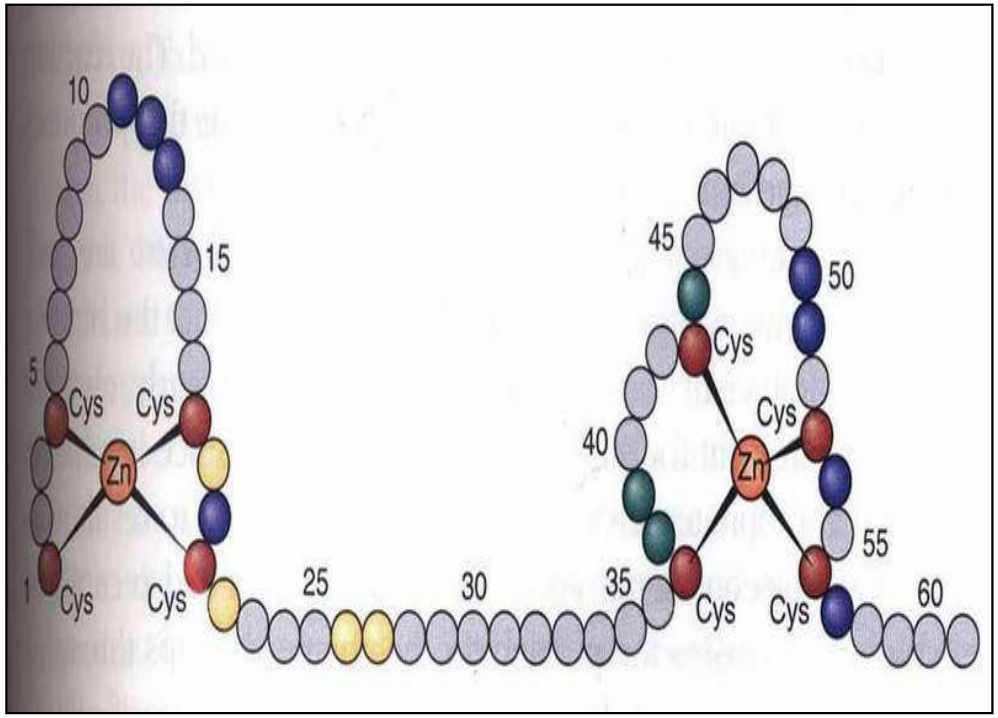
图 16-20 几种脂溶性细胞外分子的 DNA 反应元件

TFs contain a DNA-binding domain (BD) and an activation domain (AD)

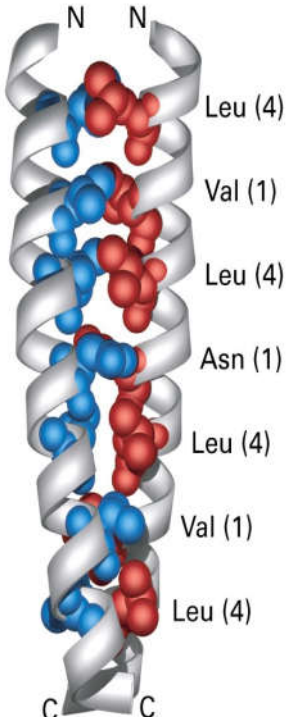


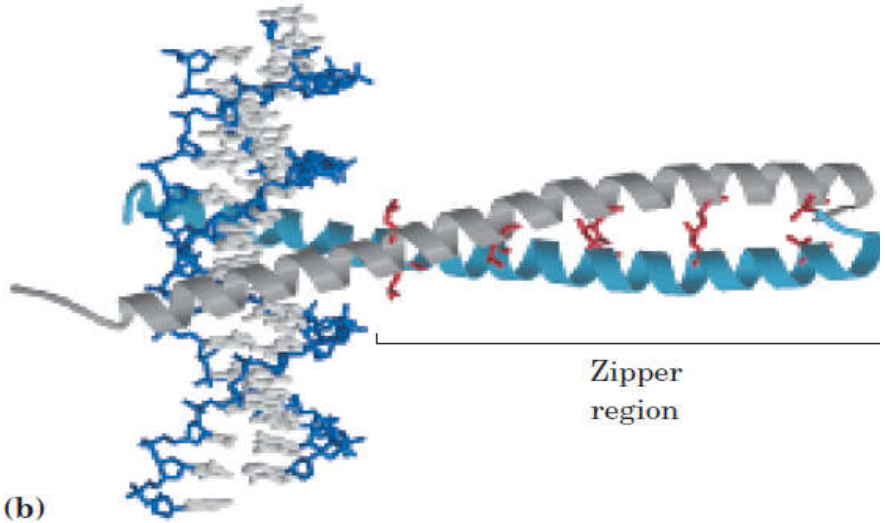
DNA-binding domains

- The BD of most of the TFs can forms a specific structure.
- zinc finger (锌指）



- Basic leucine zipper

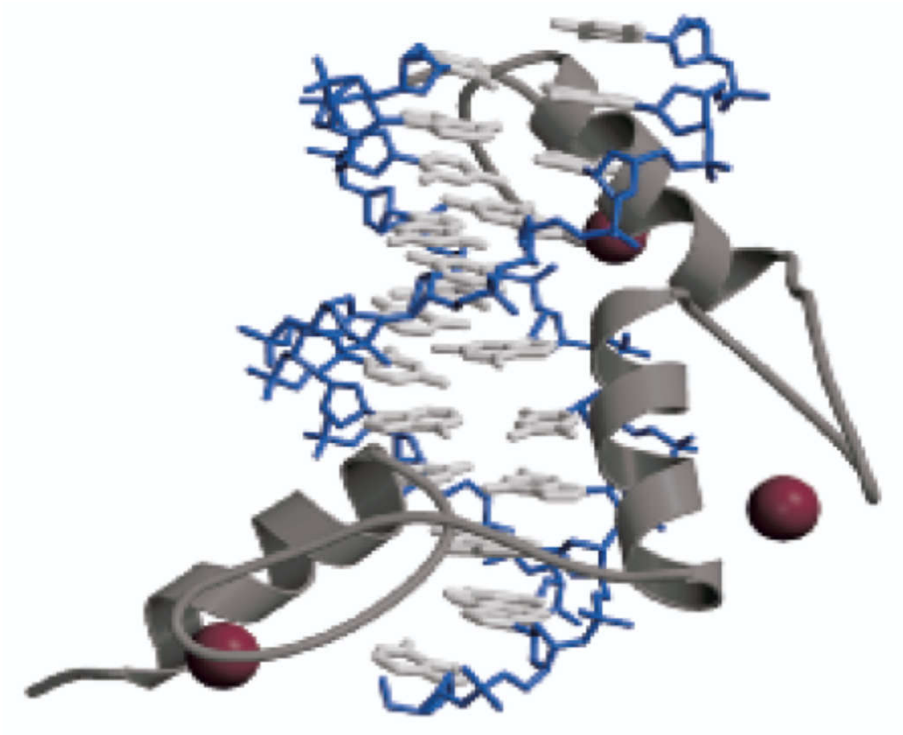




- 一段肽链中每隔7个AA即有一个Leu，α螺旋
  亲水面：亲水AA组成
  疏水面：Leu组成
  （亮氨酸拉链）
- 可形成二聚体发挥作用
  （同二聚体/异二聚体）
- DNA的结合域：
  拉链区的氨基端约30AA的碱性区
  （富含Lys和Arg）



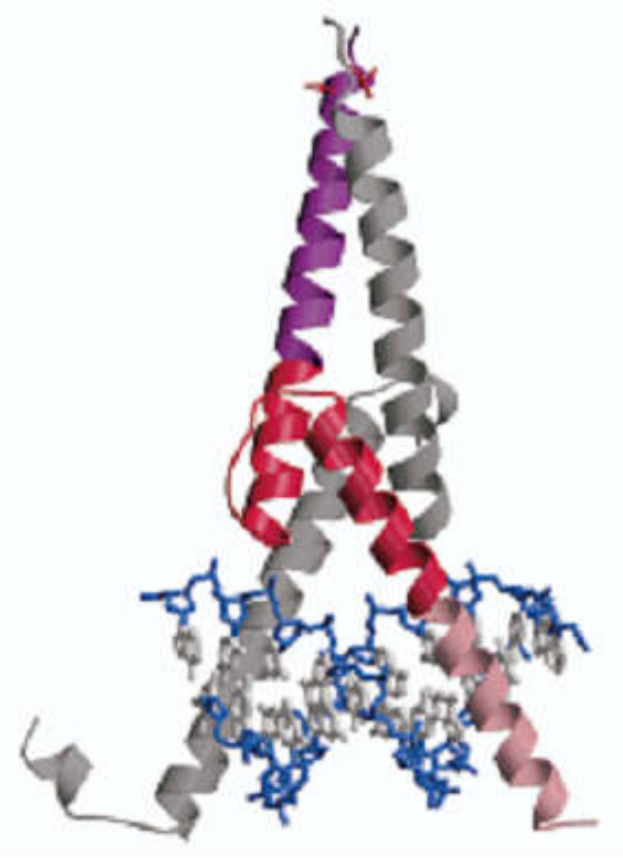
Helix-turn-helix



Homeodomain



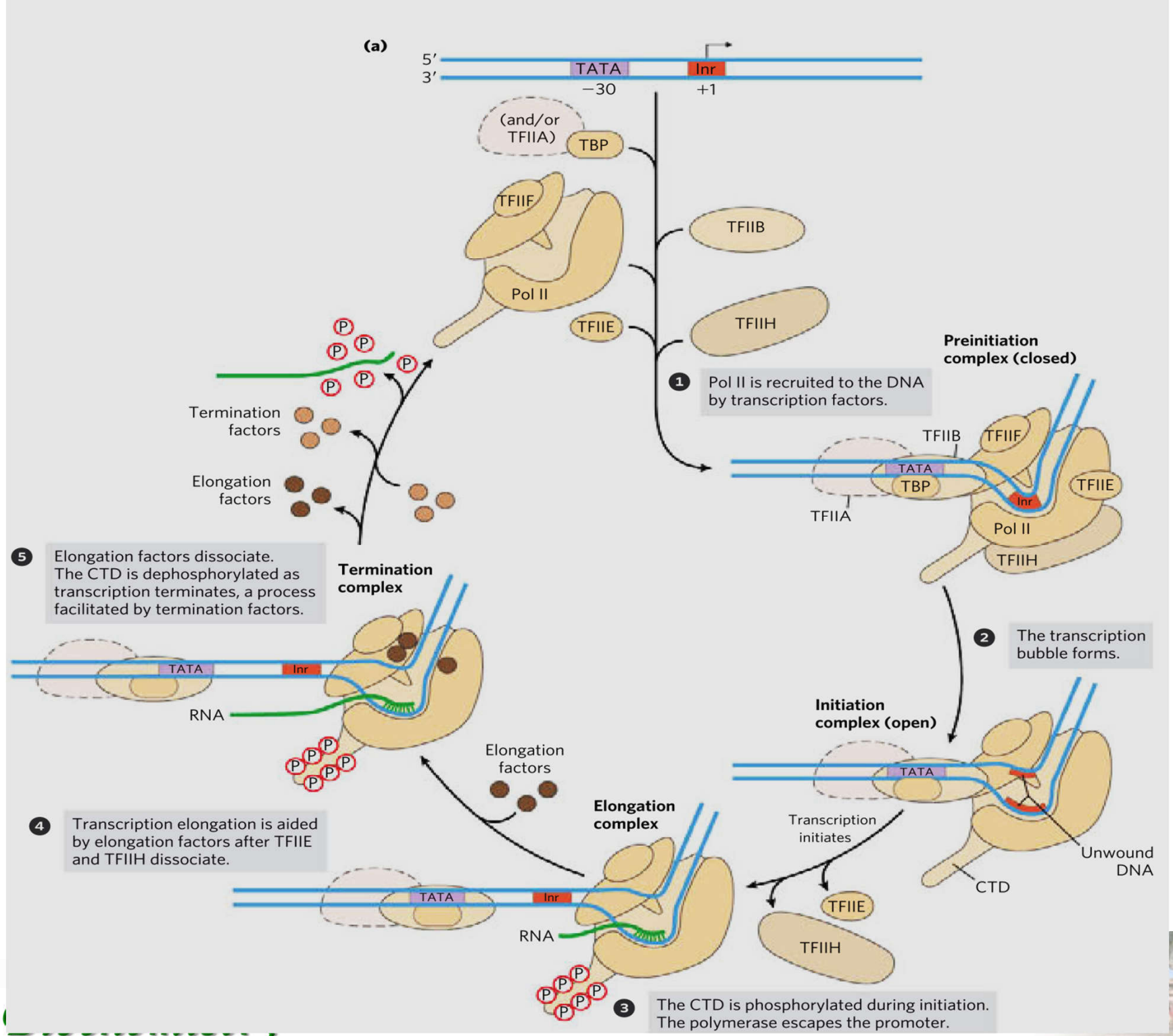
Helix~loop~helix



- **Transcription in Eukaryotes : imitation, elongation, termination**

- **Elongation factors will then join the elongation complex.**
- **Unlike RNA polymerase in prokaryotes, RNA polymerase II does not terminate transcription at specific sites but rather transcription stops at varying distances downstream of the gene with help from termination factors.**




(a)
5′
3′
TATA
Inr
−30
+1
(and/or TFIIA)
TBP
TFIIF
TFIIB
Pol II
TFIIE
TFIIH
Preinitiation complex (closed)
1 Pol II is recruited to the DNA by transcription factors.
TFIIB
TFIIF
TATA
TBP
TFIIE
Inr
TFIIA
Pol II
TFIIH
2 The transcription bubble forms.
Initiation complex (open)
TATA
Transcription initiates
Unwound DNA
CTD
TFIIE
TFIIH
3 The CTD is phosphorylated during initiation. The polymerase escapes the promoter.
Elongation complex
TATA
Inr
RNA
4 Transcription elongation is aided by elongation factors after TFIIE and TFIIH dissociate.
Elongation factors
Termination complex
TATA
Inr
RNA
5 Elongation factors dissociate. The CTD is dephosphorylated as transcription terminates, a process facilitated by termination factors.
Termination factors
Elongation factors
P

- In prokaryotes, RNA transcribed from protein-coding genes (messenger RNA, mRNA), requires little or no modification prior to translation
- However, mRNA in eukaryotes, ribosomal RNA (rRNA) and transfer RNA (tRNA) in both eukaryotes and prokaryotes are synthesized as precursor molelules that do require post-transcriptional processing) (G7, G8, G9)

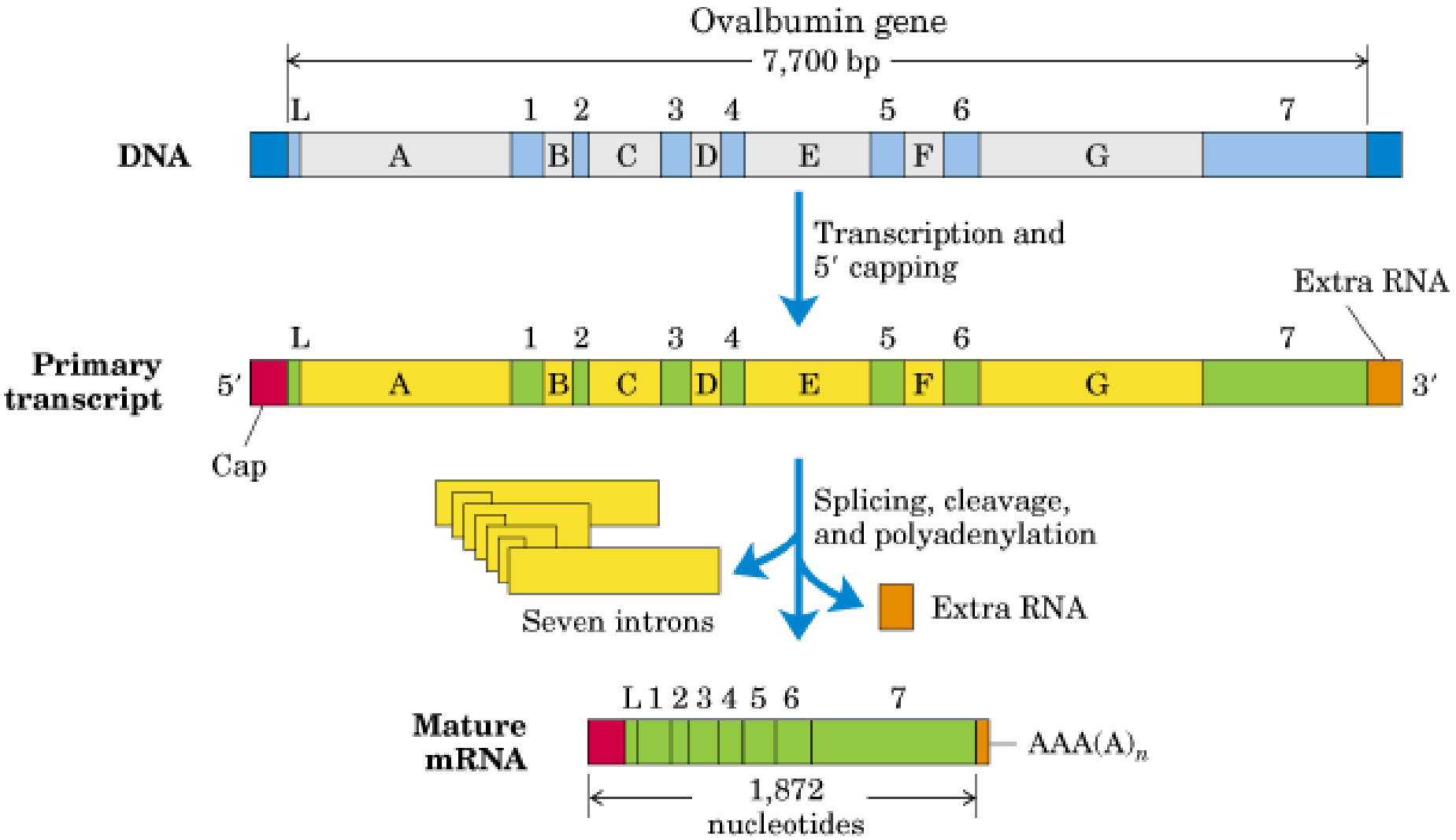


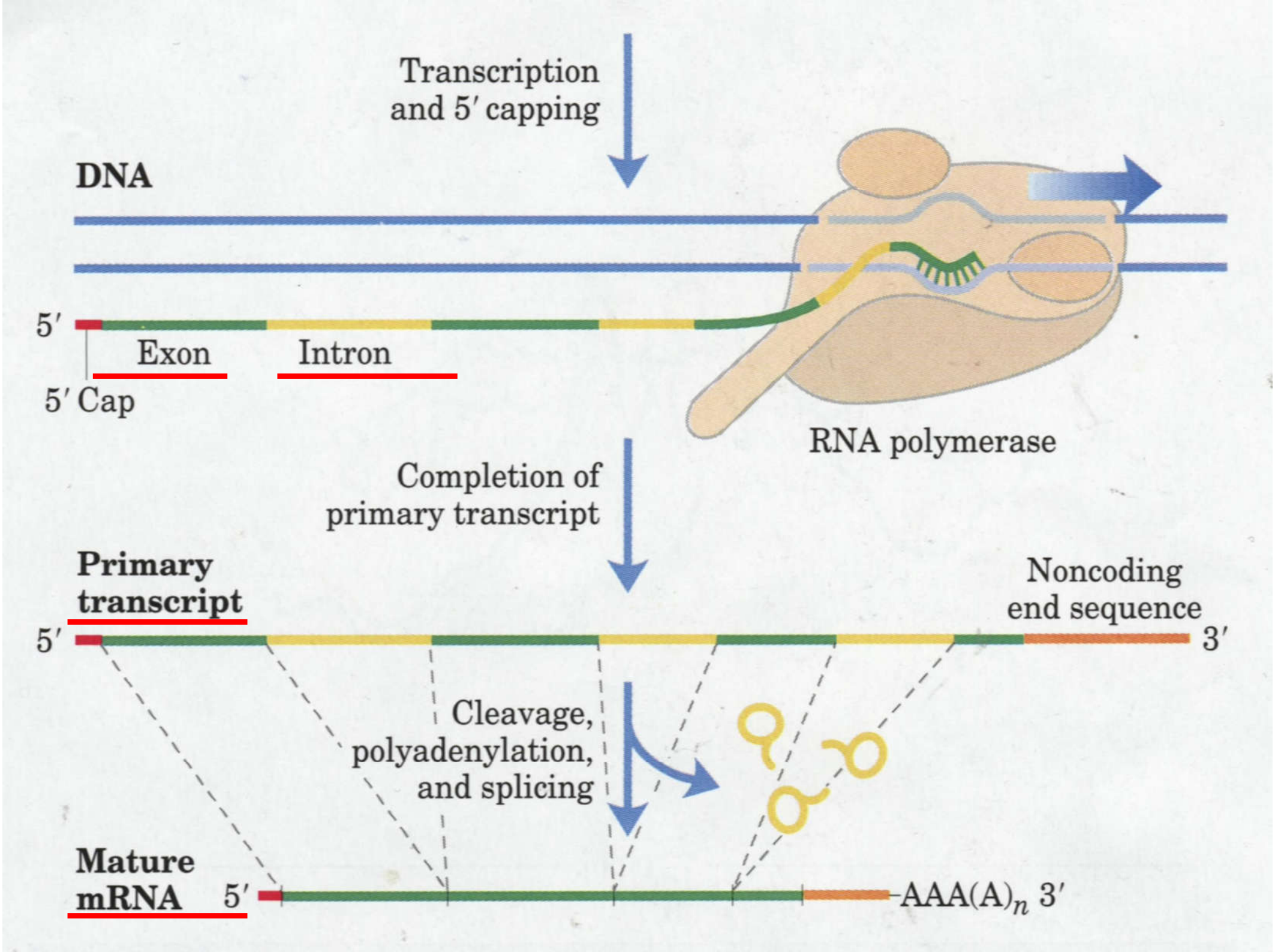
G7 Processing of eukaryotic pre-mRNA

- The primary transcripts of eukaryotic mRNAs are often capped at the 5` end, spliced（剪接） in the middle (introns removed and exons linked), polyadenylated at the 3` end.



Most protein-coding genes in eukaryotes are discontinuous




Transcription
and 5′ capping
DNA
5′
Exon
Intron
5′ Cap
RNA polymerase
Completion of
primary transcript
Primary
transcript
Noncoding
end sequence
5′
3′
Cleavage,
polyadenylation,
and splicing
Mature
mRNA
5′
AAA(A)n 3′

The 5’ cap is added before synthesis of the primary transcript is completed.

A noncoding sequence following the last exon is shown in orange.

Splicing can occur either before or after the cleavage and polyadenylation steps.

All of the processes shown here take place within the nucleus.



All RNA molecules in eukaryotes are processed to some degree after they are synthesized.

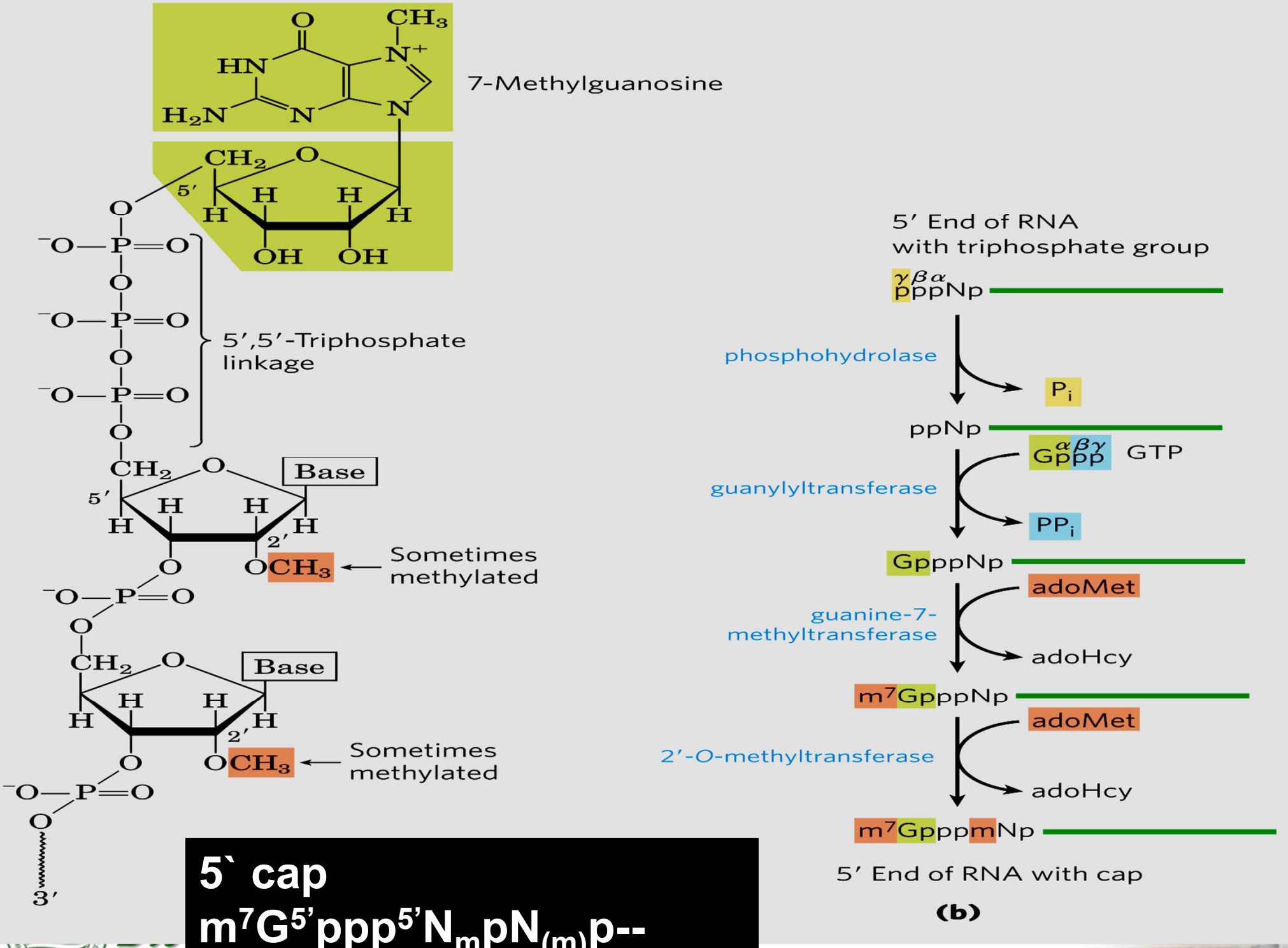
Intriguingly, the enzymes that catalyze these reactions sometimes consist of RNA rather than protein.

The discovery of catalytic RNAs, or ribozymes, has brought a revolution in thinking about RNA function and about the origin of life.



1. An eukaryotic mRNA precursor acquire a 5` cap shortly after transcription initiates

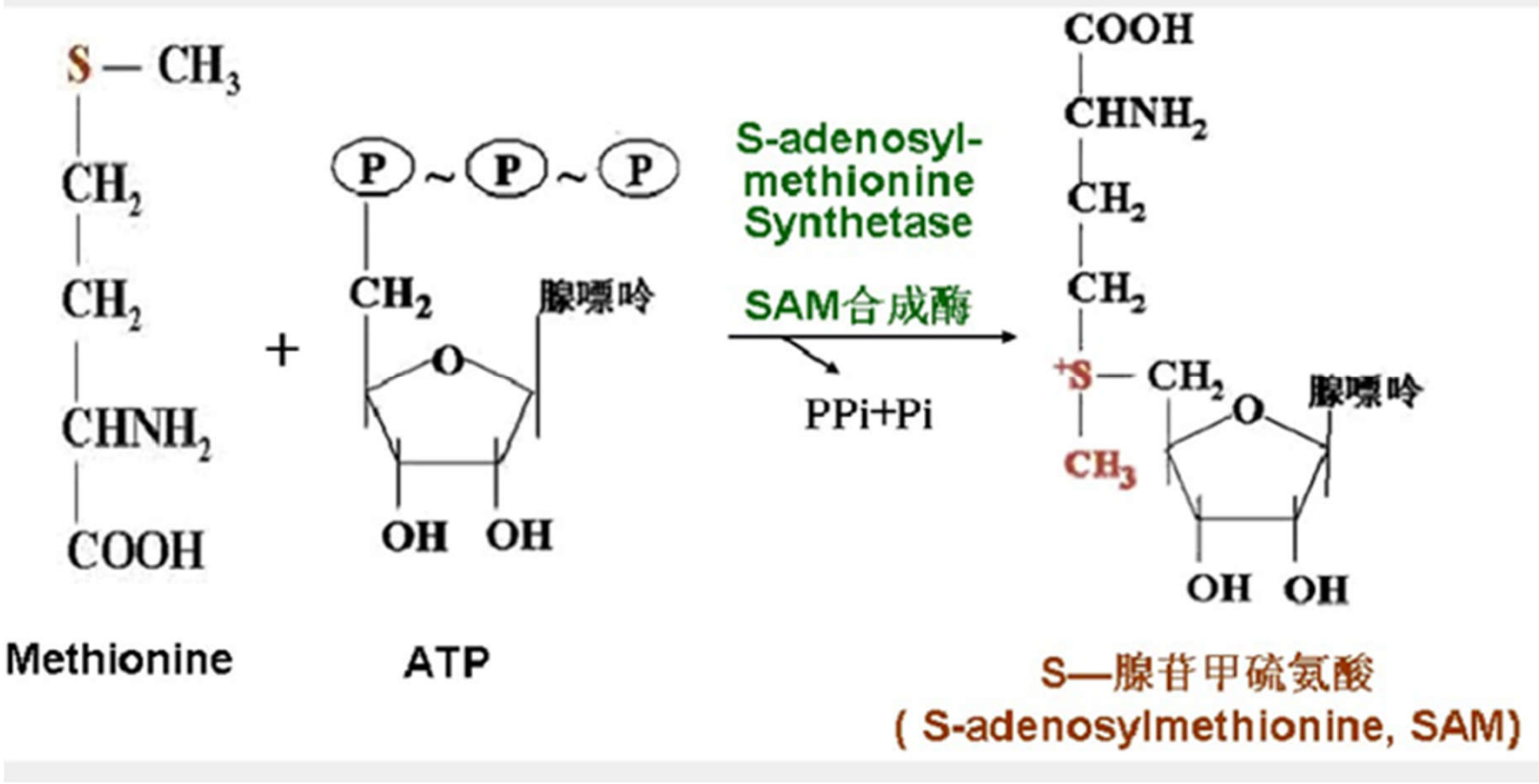


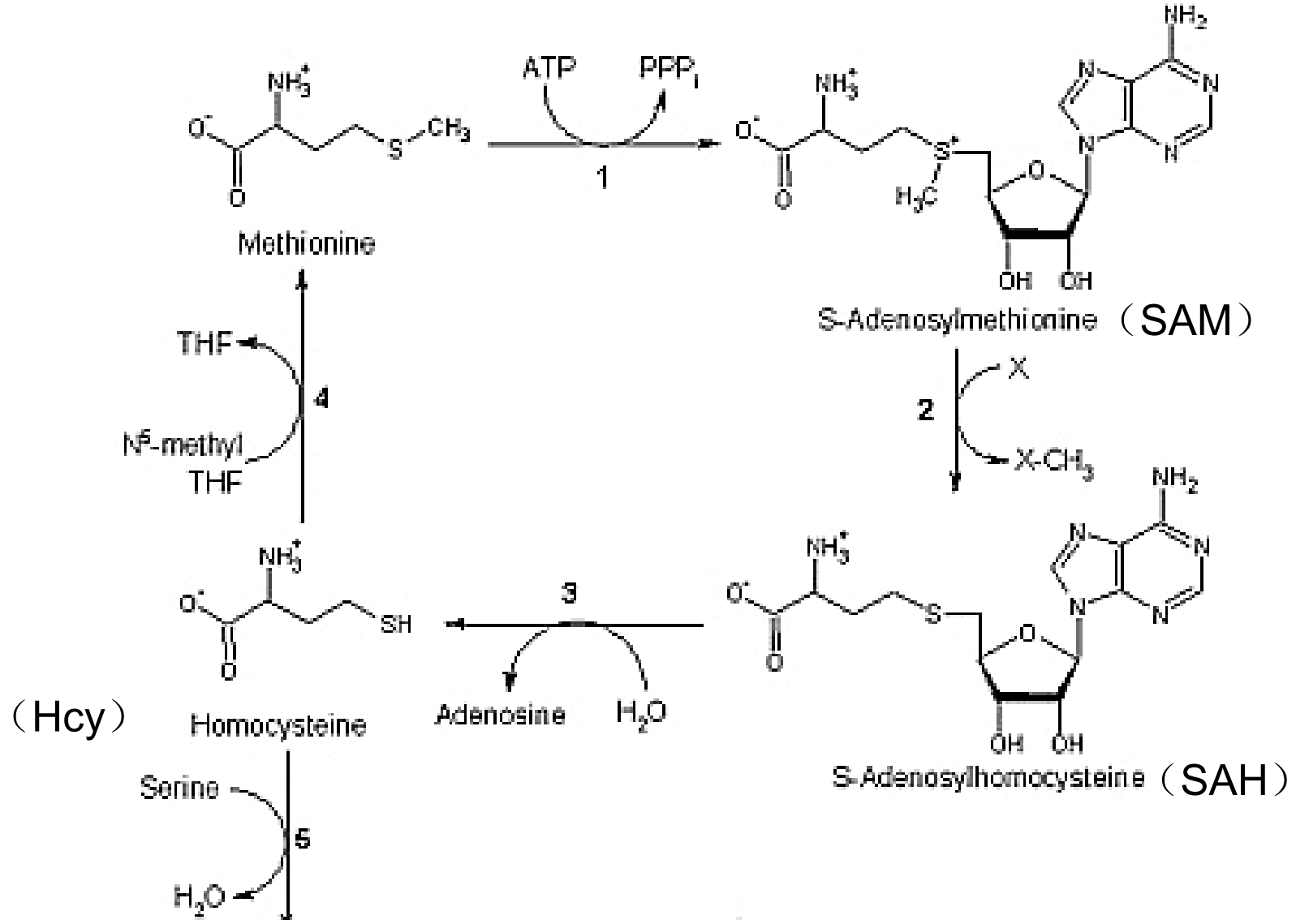


5` cap
$m^7G^{5'}ppp^{5'}N_mpN_{(m)}p$--

- The terminal 5` phosphate is first removed by a phosphatase.
- A **GMP** component (from a GTP) is joined to the 5` end of the mRNA catalyzed by guanosyl transferase in a novel **5`,5`-triphosphate linkage**.
- The guanine base is then methylated at the N-7.
- The 2`-OH groups of the 1st and 2nd nucleotides adjacent to the **7-methylguanine cap** may also be **methylated** in certain organisms.
- The methyl groups are transferred from S-adenosyl methionine（SAM S-腺苷甲硫氨酸）.



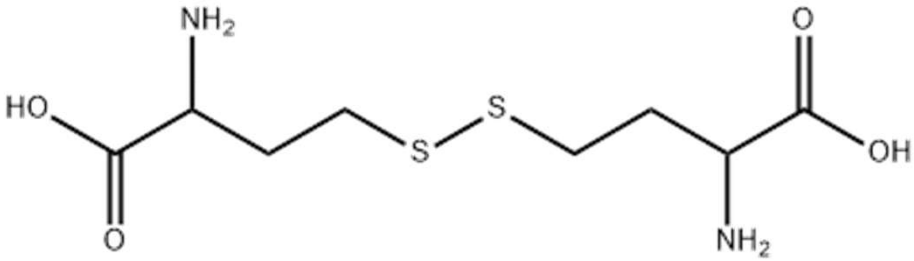

S — CH3
CH2
CH2
CHNH2
COOH
Methionine
+
P ~ P ~ P
CH2
腺嘌呤
O
OH OH
ATP
S-adenosyl-
methionine
Synthetase
SAM合成酶
PPi+Pi
COOH
CHNH2
CH2
CH2
+S — CH2
CH3
腺嘌呤
O
OH OH
S—腺苷甲硫氨酸
(S-adenosylmethionine, SAM)


ATP
PPPi
1
Methionine
S-Adenosylmethionine（SAM）
THF
4
N5-methyl
THF
2
X
X-CH3
（Hcy）
Homocysteine
3
Adenosine
H2O
S-Adenosylhomocysteine（SAH）
Serine
5
H2O

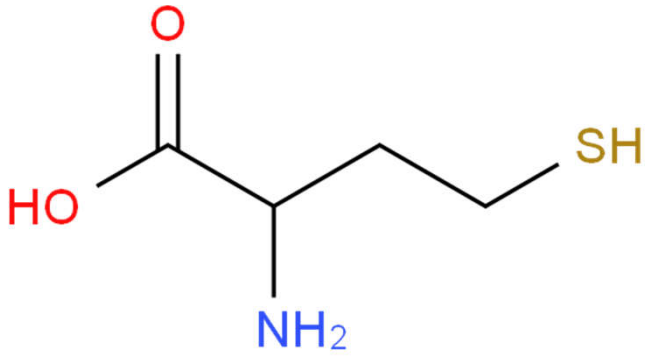
何为Homocysteine？

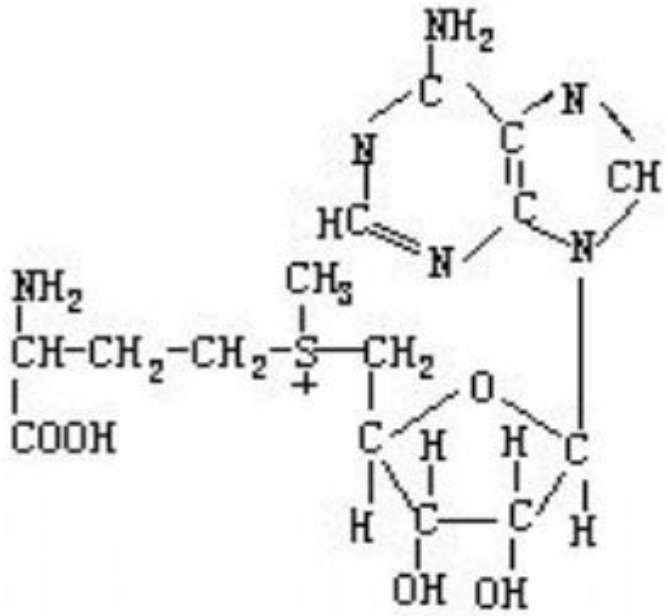


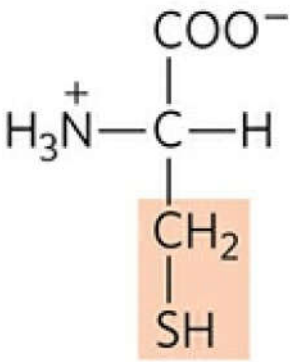
Homocystine



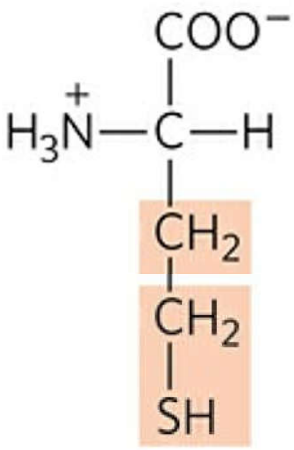
Homocysteine



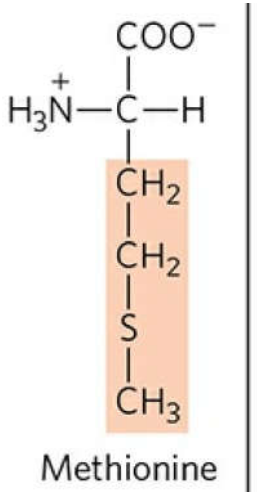




Cysteine



homocysteine

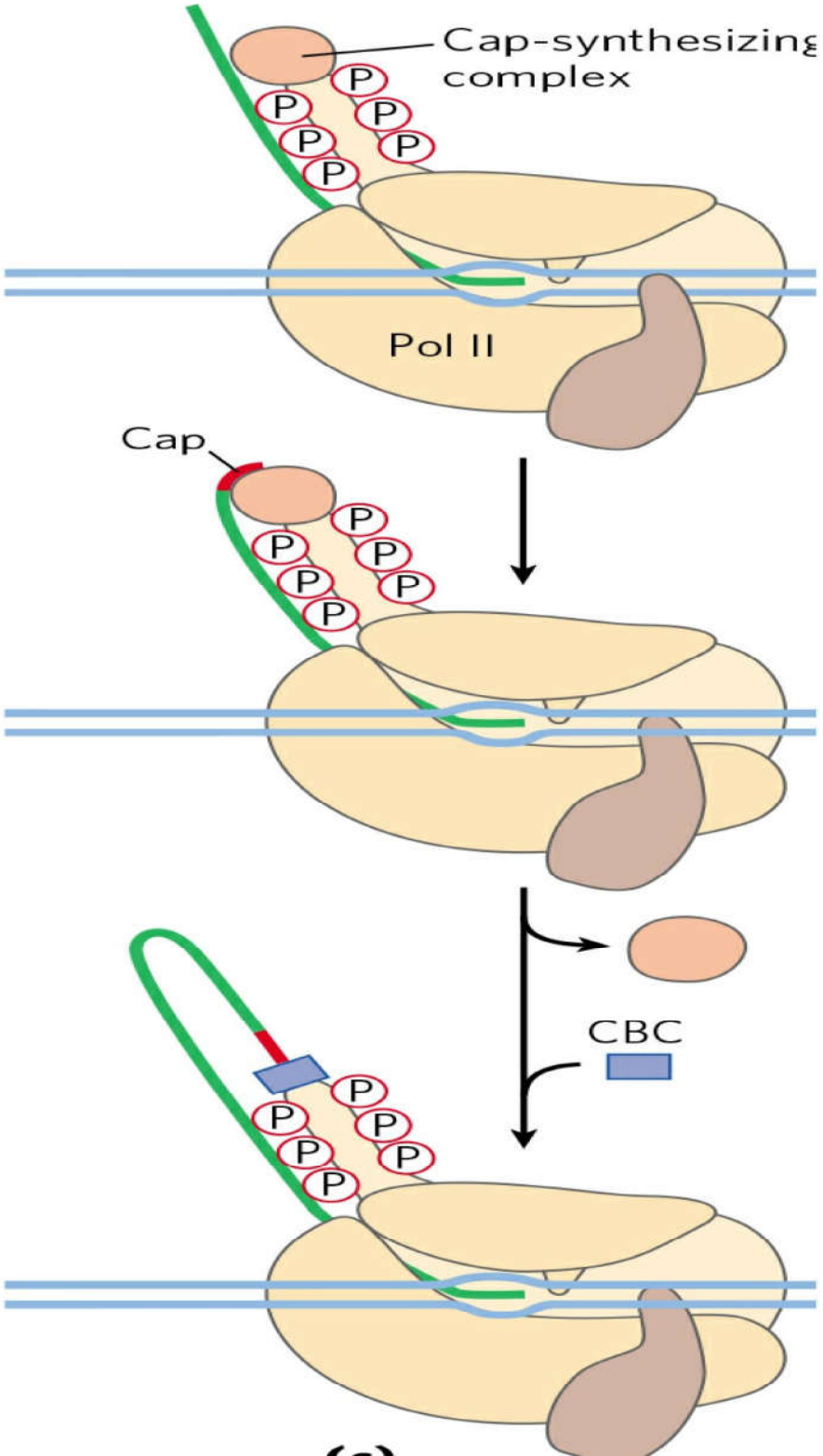


Methionine

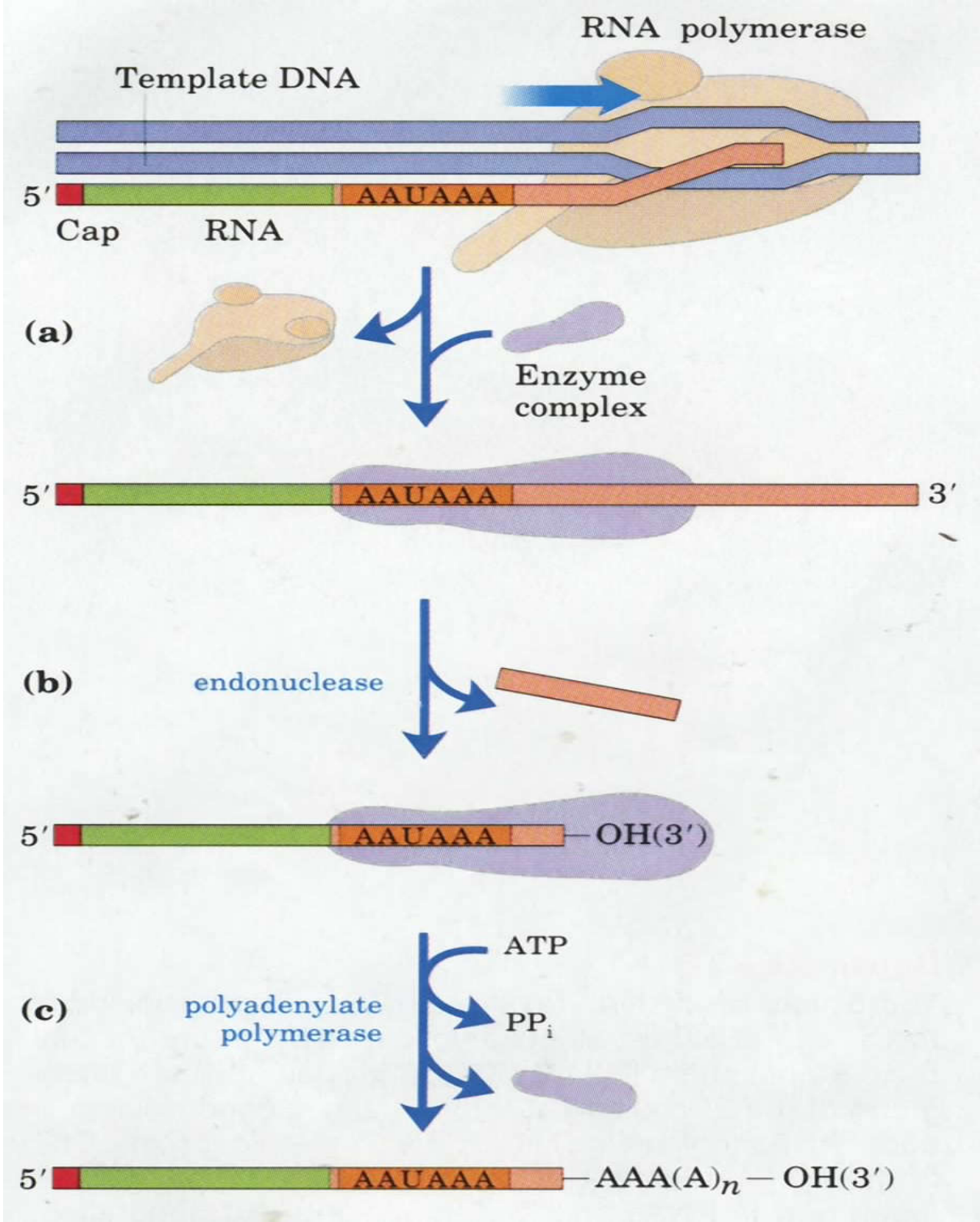


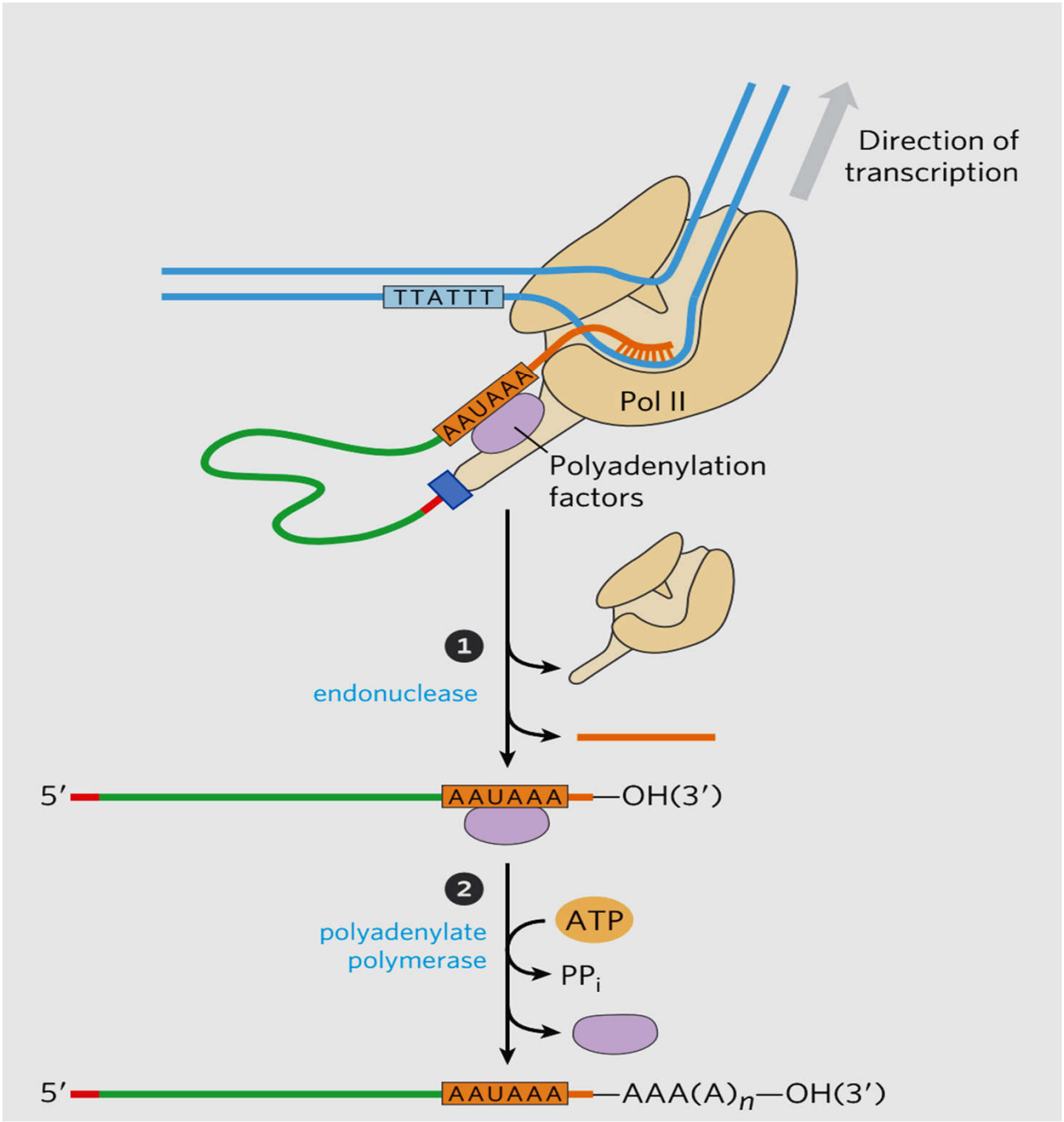
Biochemistry

very early in transcription, after the first 20 to 30 nucleotides of the transcript, the 5′ cap of mRNA have been added.

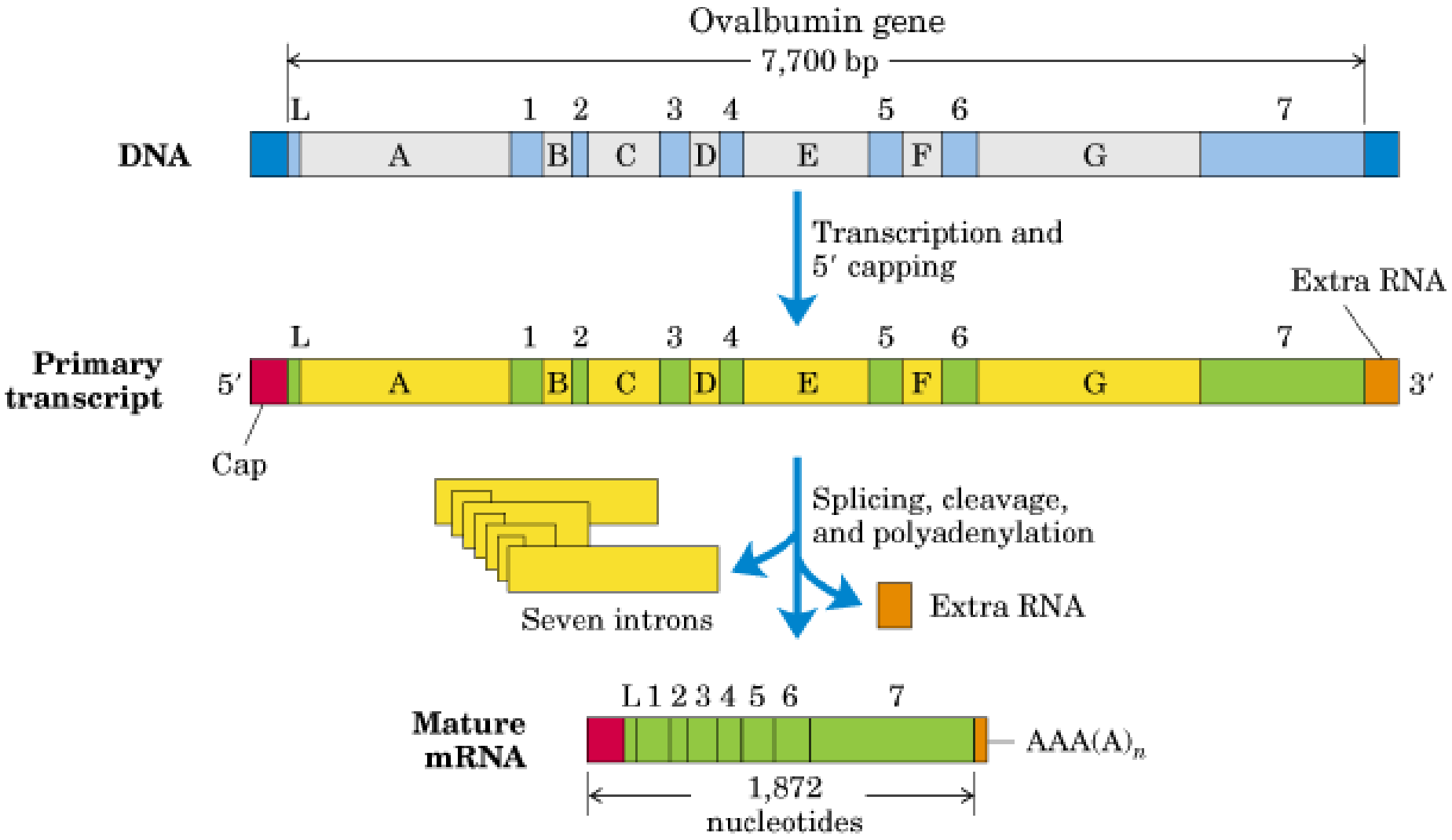


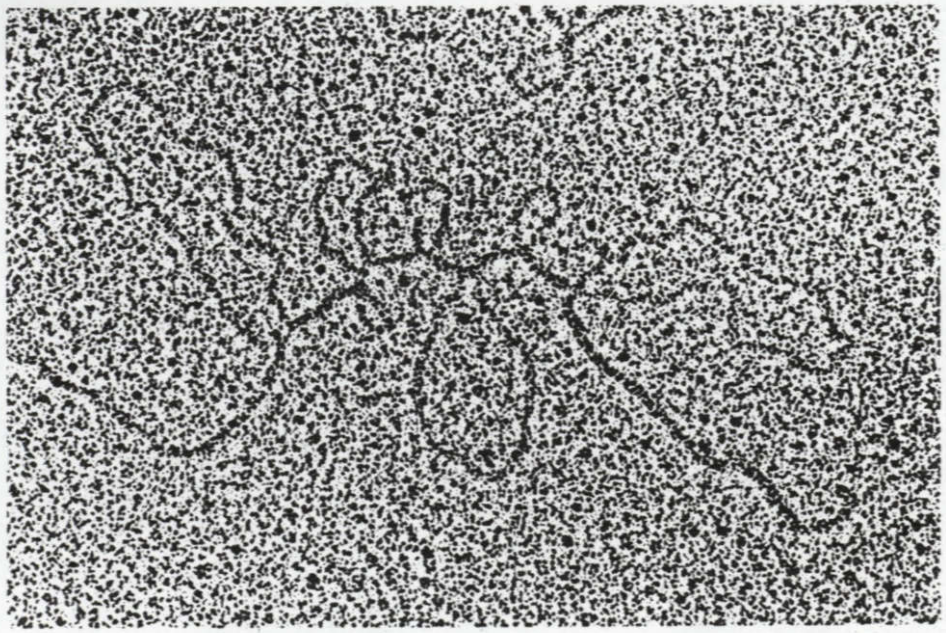
A poly(A) tail is usually added at the 3` end of an mRNA molecule via a processing step.

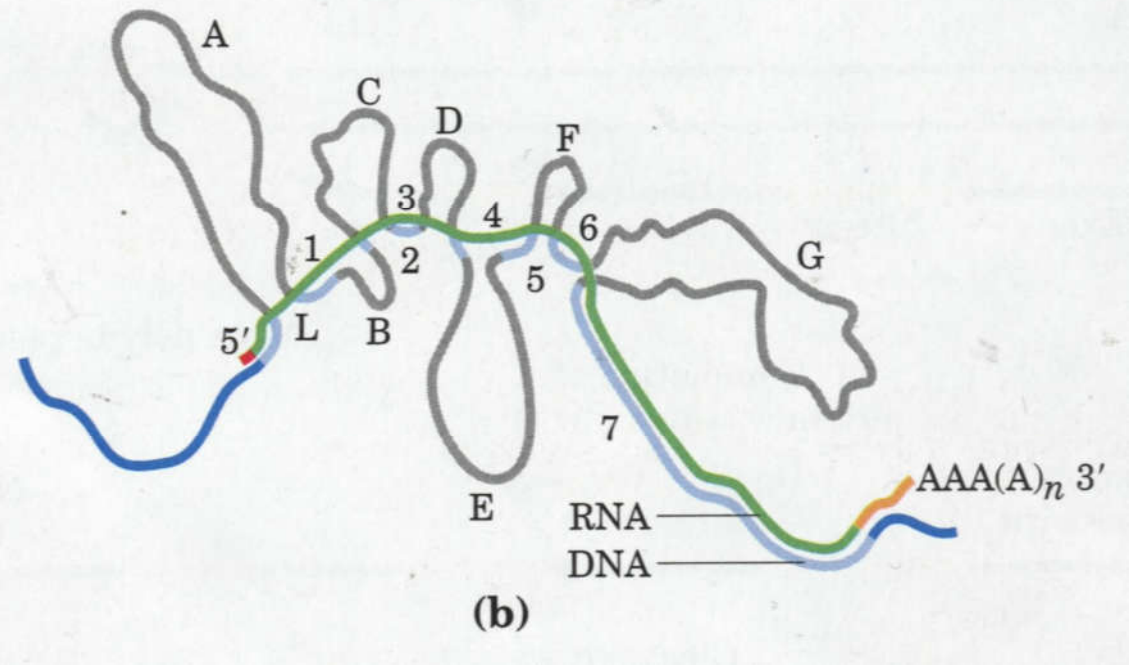


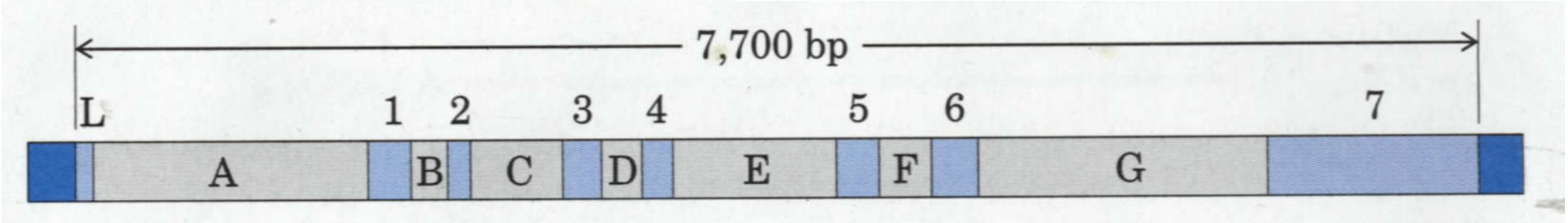

Direction of transcription
TTATTT
AAUAAA
Pol II
Polyadenylation factors
1
endonuclease
5′
AAUAAA
—OH(3′)
2
polyadenylate polymerase
ATP
PPi
5′
AAUAAA
—AAA(A)n—OH(3′)

3. The introns transcribed into RNA are removed by splicing









Demonstration of noncoding sequences in the chicken ovalbumin gene by RNA-DNA hybridization

The Nobel Prize in Physiology or Medicine 1993

"for their discoveries of split genes"

Press release
Illustrated presentation

Richard J. Roberts

Great Britain

New England Biolabs
Beverly, MA, USA

1943 -



Autobiography



Phillip A. Sharp

USA

Center for Cancer Research
Massachusetts Institute of Technology
Cambridge, MA, USA

1944 -

Autobiography



4. Introns are self-splicing and the ends of the exon sequences ligate together

- The intron is removed and the two exons precisely linked to produce a functional mRNA molecule in Eukaryotes.



Four classes of introns

Group I introns nuclear, mitochondrial, and chloroplast genes that code for rRNAs, mRNAs, and tRNAs.

Group II introns mitochondrial or chloroplast mRNAs in fungi, algae, and plants.

self-splicing

no high-energy (ATP) for splicing

Group III introns

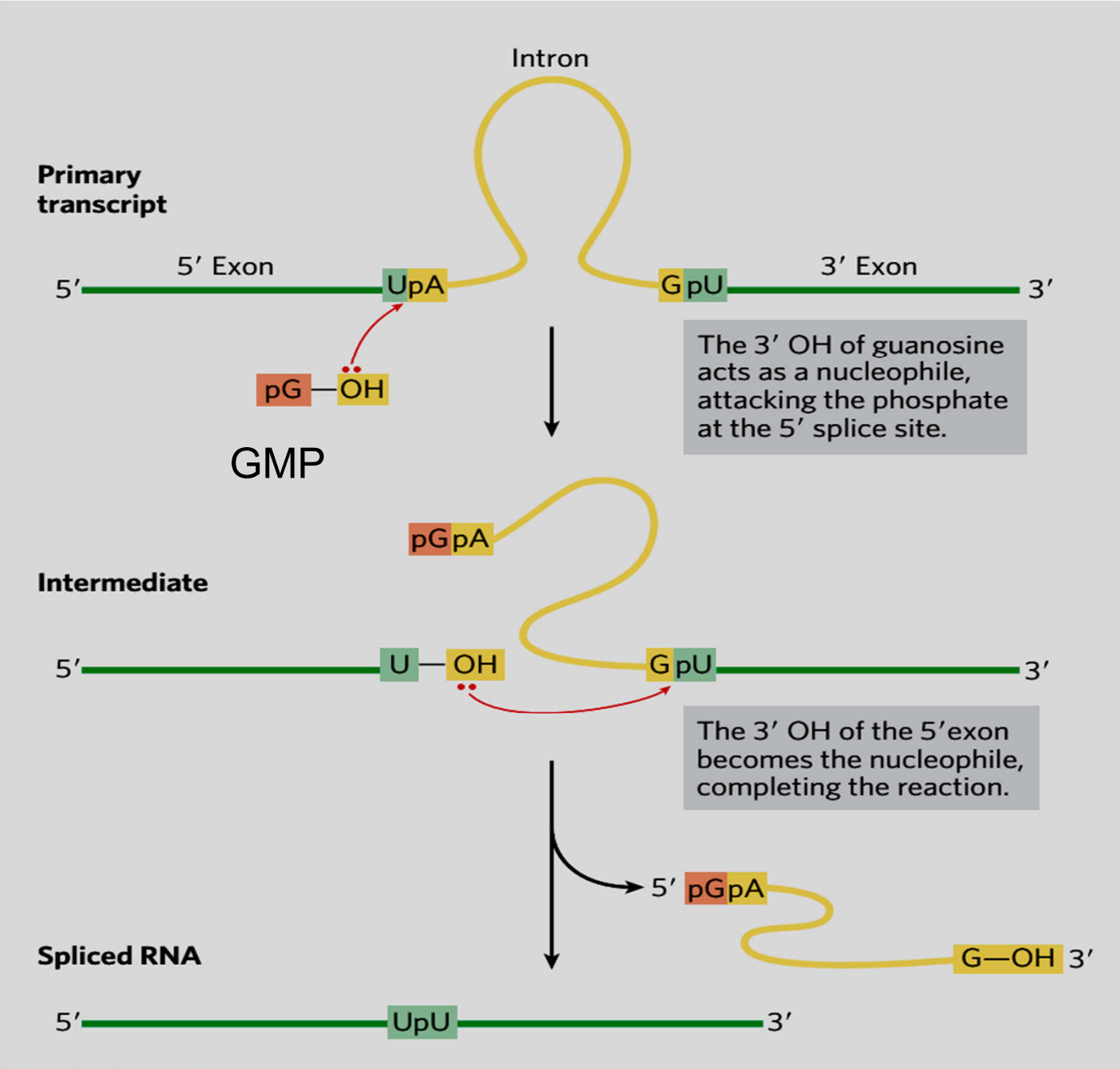
Group IV introns

Group III: the **spliceosomal introns**
spliceosome (protein complex) , containing:
small nuclear RNAs (snRNAs)：Five snRNAs (U1, U2, U4, U5, and U6)
small nuclear ribonucleoproteins (snRNPs)

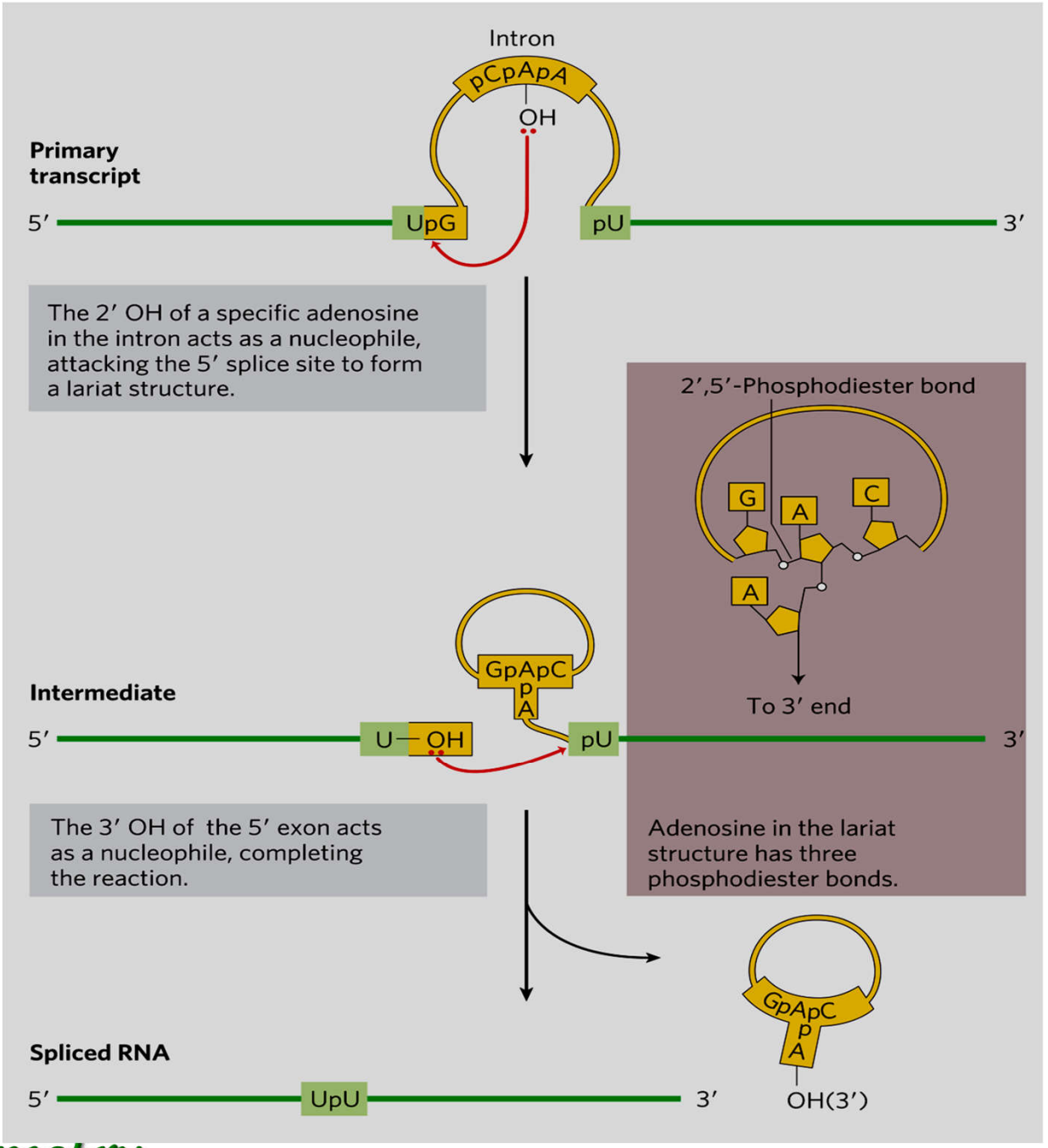
Group IV: found in certain tRNAs, requires ATP and an endonuclease.



Splicing mechanism of group I introns



Splicing mechanism of group II introns



Splicing mechanism with spliceosome complexes

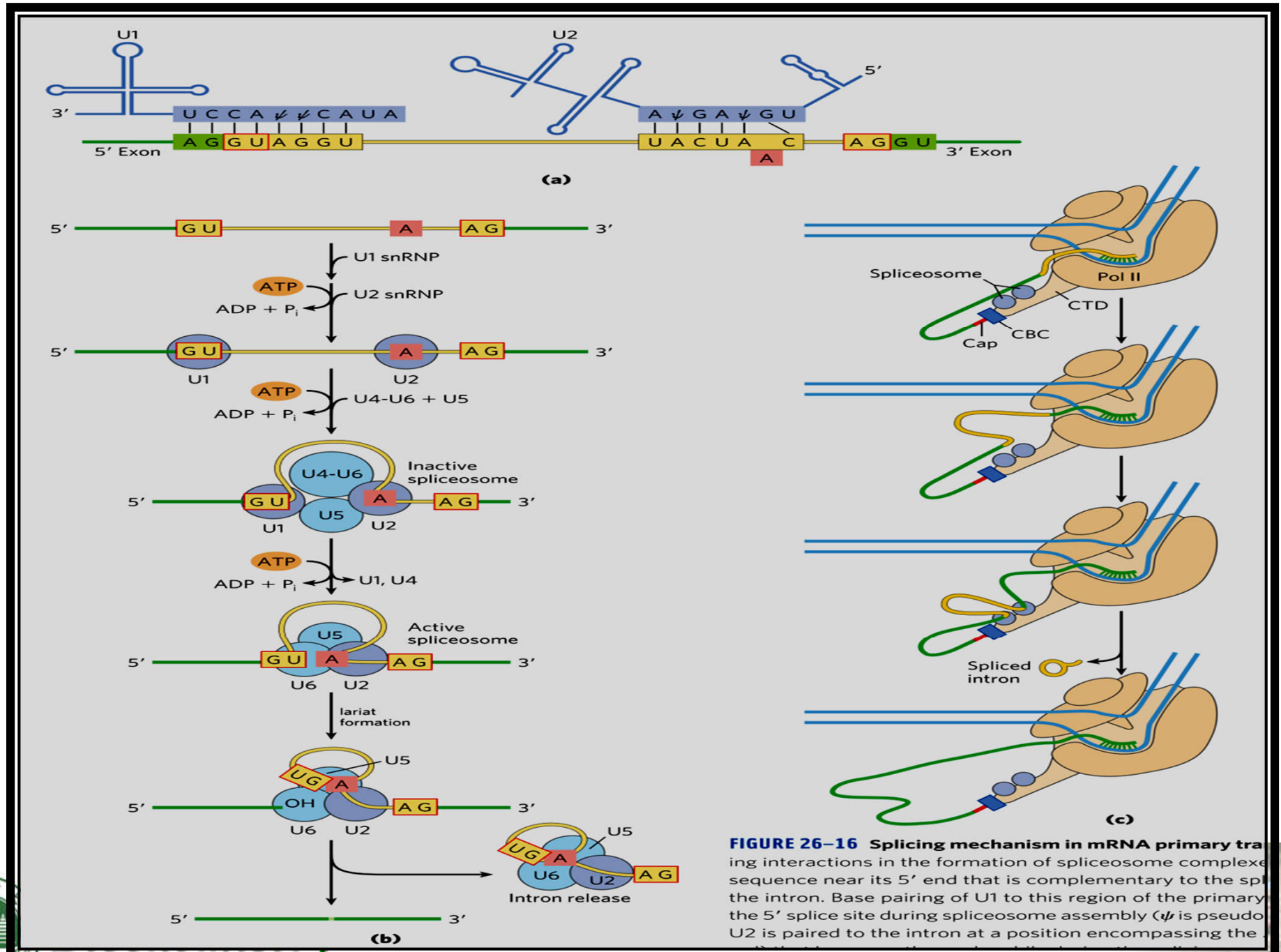


FIGURE 26–16 **Splicing mechanism in mRNA primary tra**
ing interactions in the formation of spliceosome complexe
sequence near its 5′ end that is complementary to the spl
the intron. Base pairing of U1 to this region of the primary
the 5′ splice site during spliceosome assembly (ψ is pseudo
U2 is paired to the intron at a position encompassing the

R- purine 嘌呤
Y- pyrimidine 嘧啶

Group III

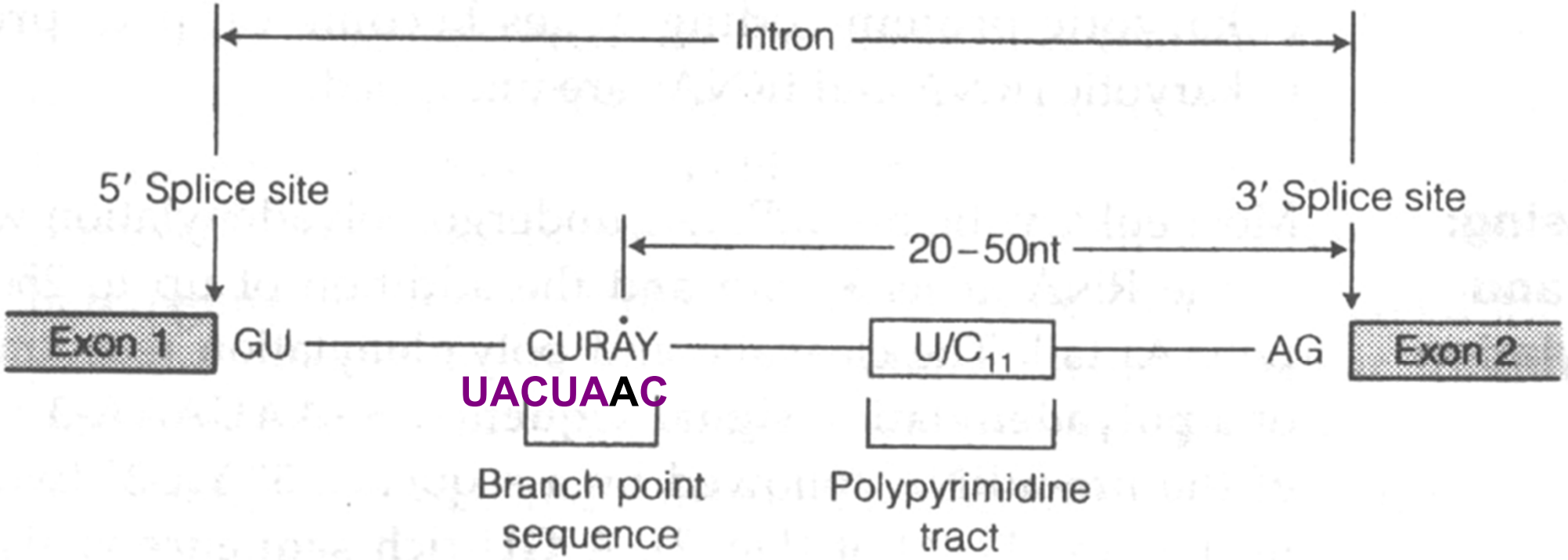


Fig. 3. Conserved sequences for RNA splicing. The residue marked as Ȧ in the branchpoint sequence is the site of formation of the 2′5′ branch.

Group III

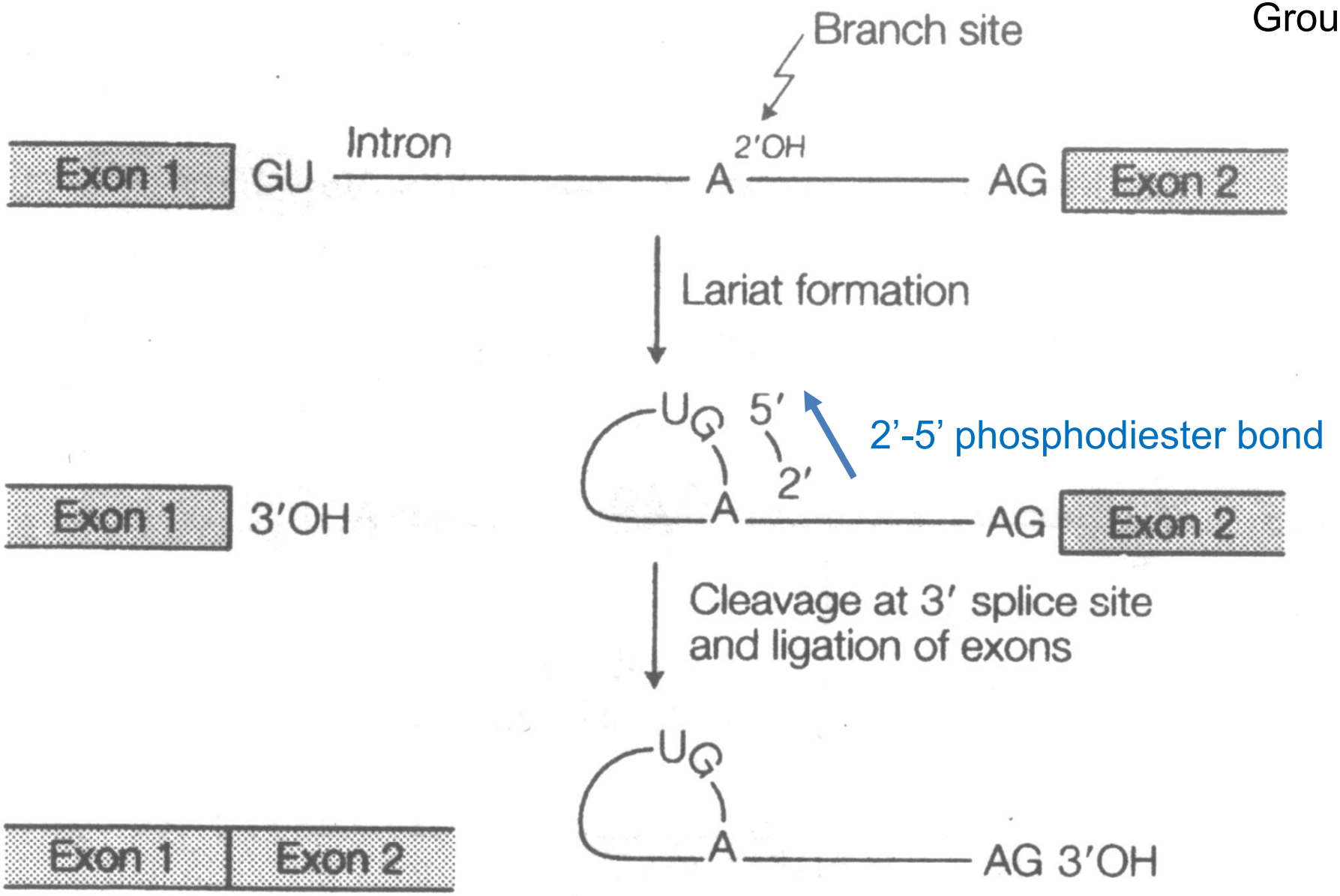


Fig. 4. The two steps of RNA splicing.

RNA splicing occurs in two steps:

•The 2'OH of a specific adenosine in the intron acts as a nucleophile, attacking the 5' splice site to form a lariat structure.

•The 3'OH of 5'exon acts as a nucleophile, completing the reaction

In each of the two splicing reactions, one phosphate-ester bond is exchanged for another.



RNA splicing requires the involvement of several small nuclear RNAs (snRNA) each of which is associated with several proteins to form a small nuclear ribonucleoproteins particle or snRNP(a large protein complex called a **spliceosome**)

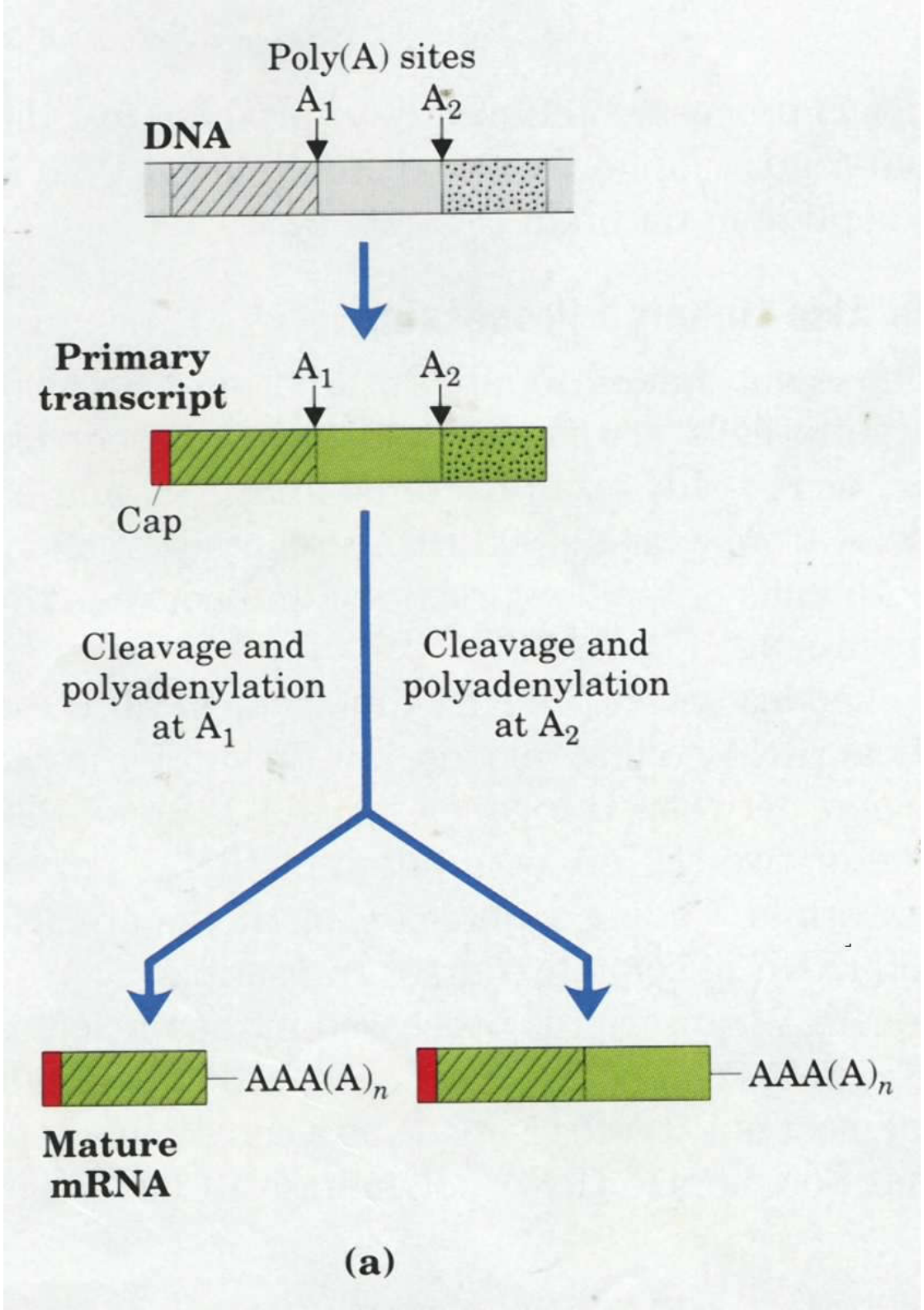


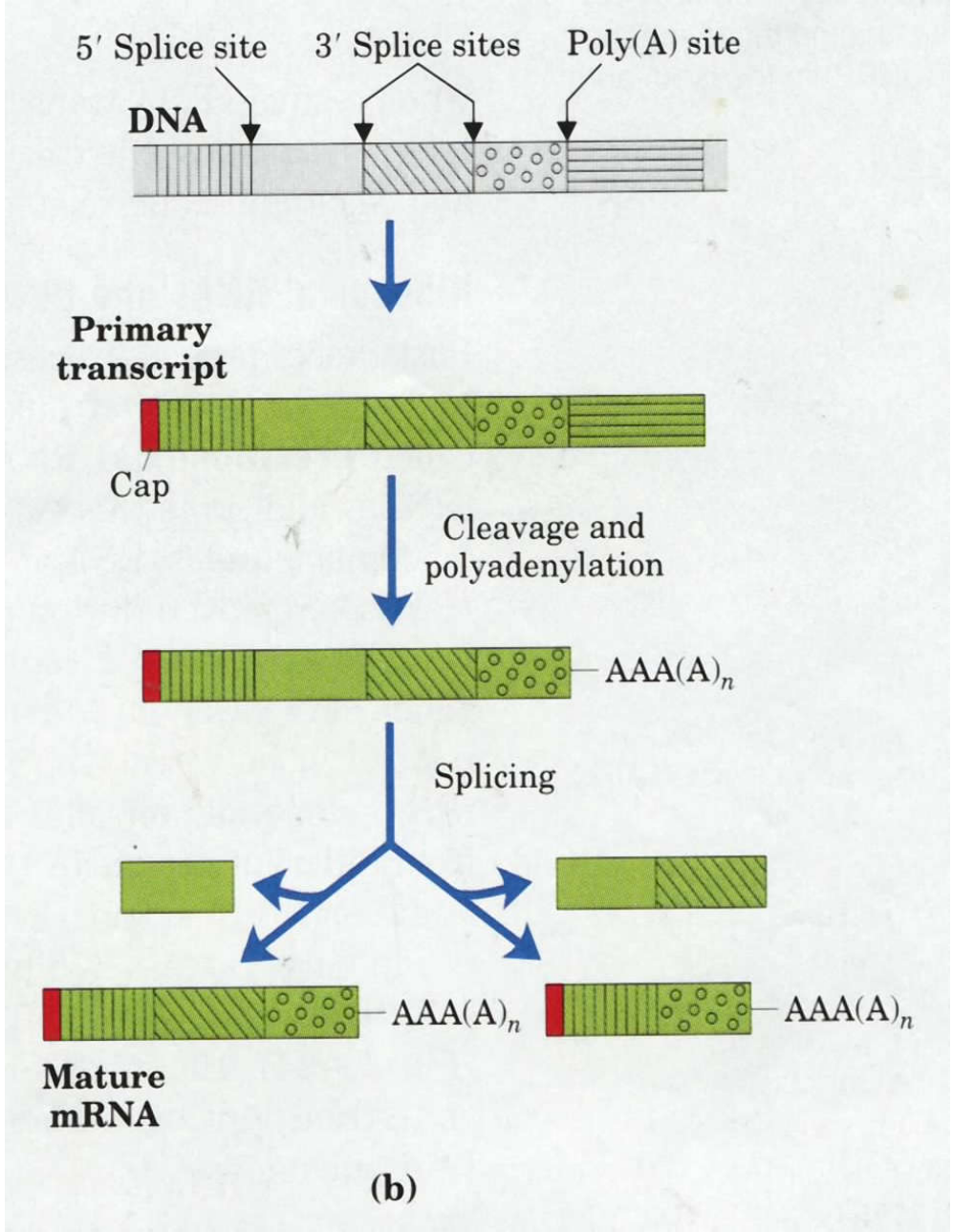
5.Mutiple products are derived from one gene by differential RNA processing

The transcription of introns consumes cellular resources and energy without returning any obvious benefit to the organism, but introns may confer an advantage not yet appreciated by scientists.

Although most eukaryotic mRNA transcripts produce only one mature mRNA and one corresponding polypeptide, some can be processed in more than one way to produce **different** mRNAs and thus different polypeptides.







Multiple mRNAs can be produced via differential RNA processing.

In Fig. a, alternative cleavage and polyadenylation patterns.

Two poly(A) sites, A1 and A2, are shown.

In Fig. b, alternative splicing patterns.

Two different 3' splice sites are shown. In both mechanisms, different mature mRNAs are produced from the same primary transcript.



6. The sequence of an RNA molecule may be changed after synthesis and processing

Individual nucleotides within the mRNA may be changed to other mucleotides, deleted entirely or additional nucleotides inserted. The effect of RNA editing is to change the coding capacity of the mRNA so that it encodes a different polypeptide than that originally encoded be the gene.



CAA =gln

UAA =stop

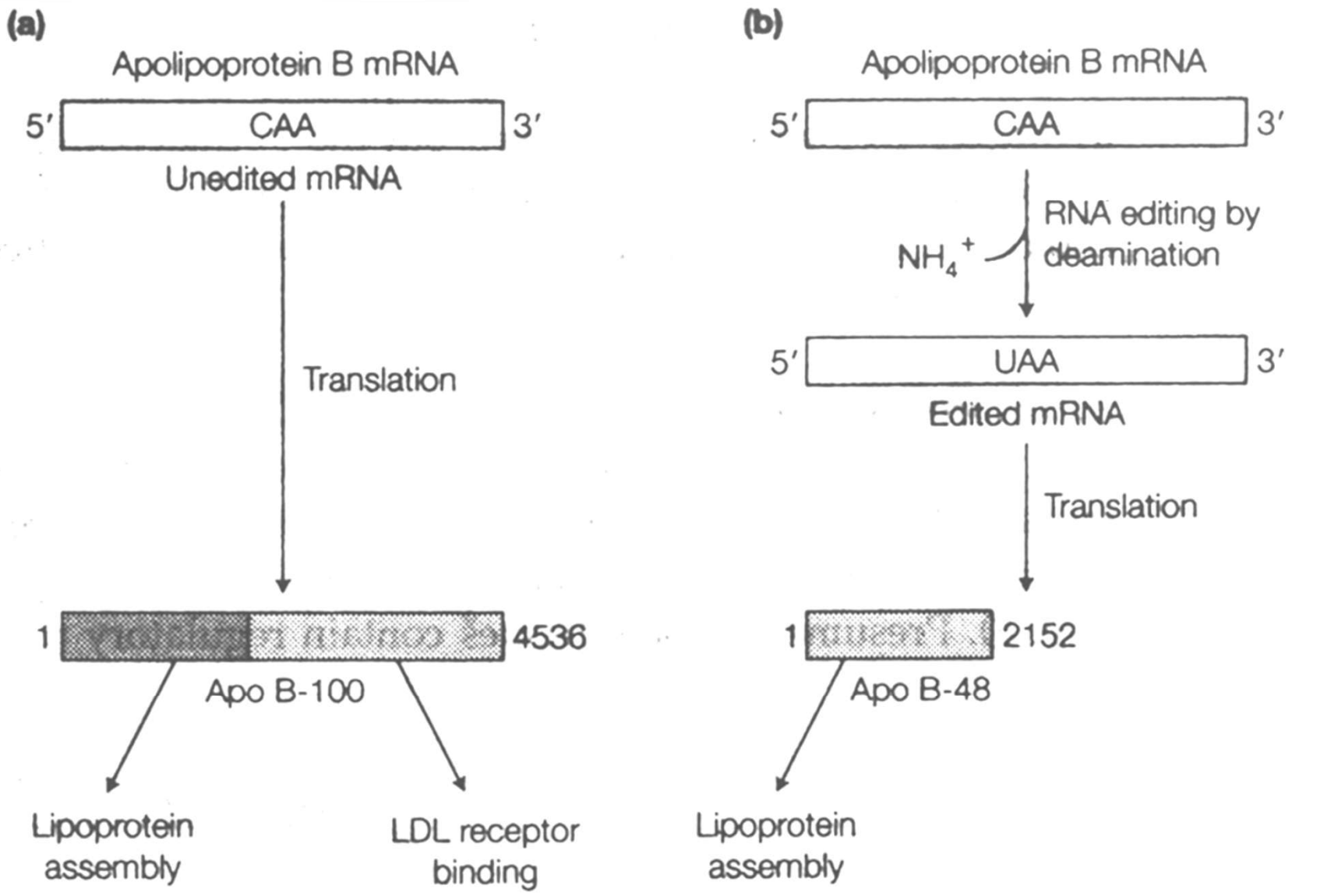
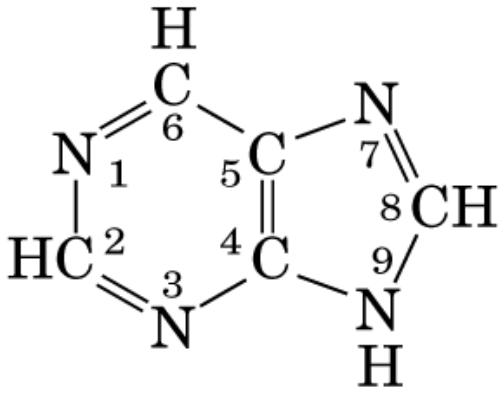
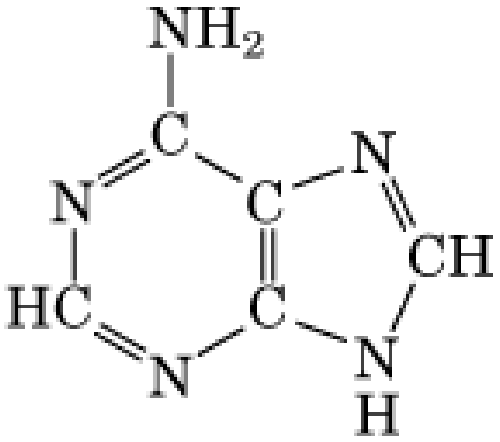


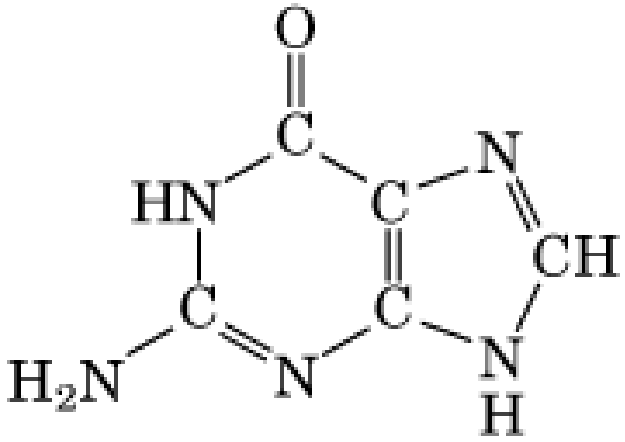
Fig. 9. RNA editing. (a) Unedited apolipoprotein B mRNA is translated to yield ApoB-10 a 4536-amino acid long polypeptide with structural domains for lipoprotein assembly and receptor binding functions; (b) translation of the edited mRNA yields the shorter ApoB-48 which lacks the receptor binding domain.



Purine

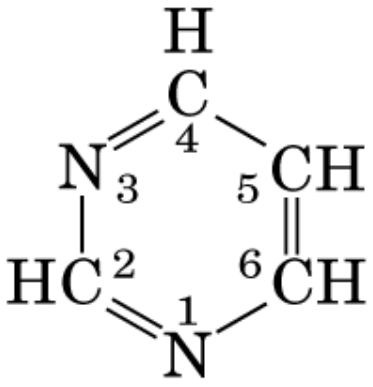


Adenine

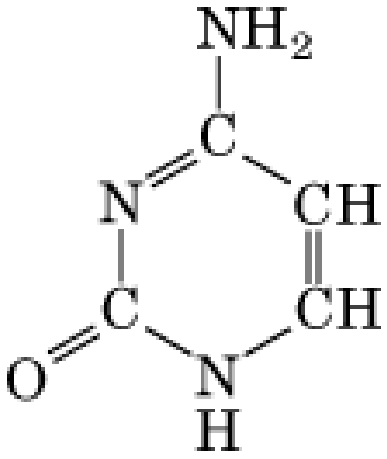


Guanine

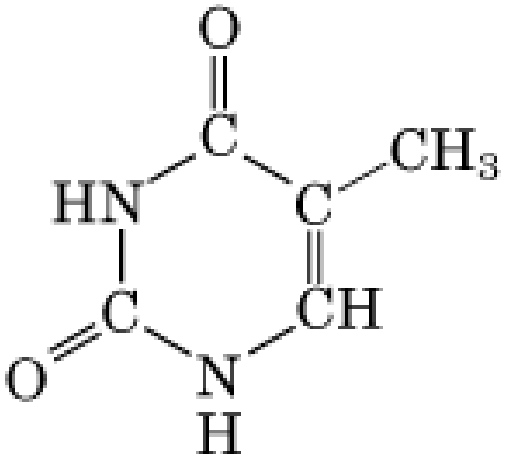
Purines



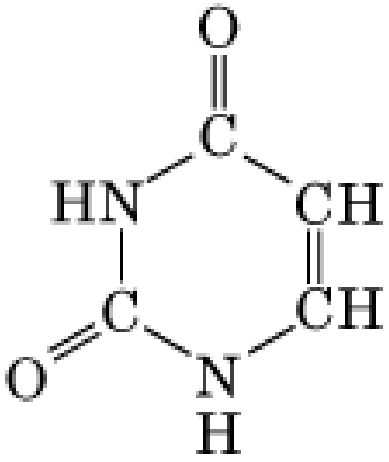
Pyrimidine



Cytosine



Thymine
(DNA)



Uracil
(RNA)

Pyrimidines



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Research | 28 February 2018

The SMAD2/3 interactome reveals that TGFβ controls m^6A mRNA methylation in pluripotency

...signalling to diverse effectors including m6A methyltransferase, which has a...

Alessandro Bertero, Stephanie Brown [...] Ludovic Vallier

Nature **555**, 256–259

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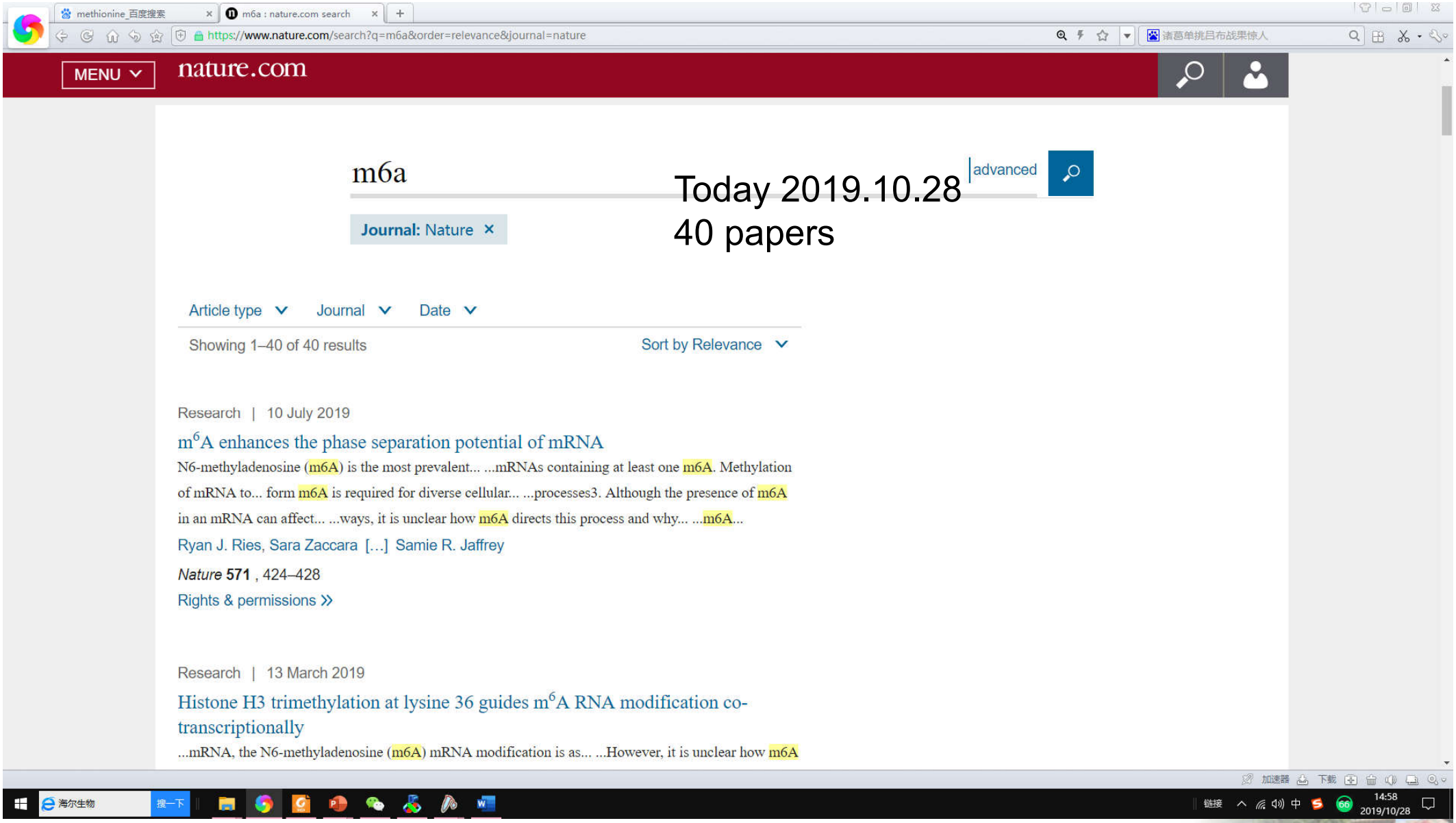
Amendments and Corrections | 29 November 2017

Corrigendum: RNA m^6A methylation regulates the ultraviolet-induced DNA damage response

Yang Xiang, Benoit Laurent [...] Yang Shi

Biochemistry





Biochemistry

Keyword Detail | 新标签页 | m6A_百度搜索 | sam 甲基_百度 | ovalbumin_百 | 硫酸_百度百科 | M6A : nature.c | RNA甲基化修饰 | DNA甲基化_百 | DNA甲基化_百 | 姜大卫回应周杰

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Histone H3 trimethylation at lysine 36 guides m^{6}A RNA modification co-transcriptionally

...mRNA, the N6-methyladenosine (m6A) mRNA modification is as... ...However, it is unclear how m6A is specifically and dynamically deposited... ...marker for transcription elongation, guides m6A deposition globally. We show... that m6A modifications are enriched in the... ...the m6A... ...the m6A...

Huilin Huang, Hengyou Weng [...] Jianjun Chen

Nature **567**, 414–419

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Research | 10 July 2019

m^{6}A enhances the phase separation potential of mRNA

N6-methyladenosine (m6A) is the most prevalent... ...mRNAs containing at least one m6A. Methylation of mRNA to... form m6A is required for diverse cellular... ...processes3. Although the presence of m6A in an mRNA can affect... ...ways, it is unclear how m6A directs this process and why... ...m6A...

Ryan J. Ries, Sara Zaccara [...] Samie R. Jaffrey

Nature **571**, 424–428

nature reviews molecular cell biology
20th Anniversary Collection
20 YEARS

正在建立安全连接...

21:35 周四 2020/10/29

Biochemistry

2022.10.27
767 Papers

m6A | Nature Search Results

https://www.nature.com/search?q=m6A&journal=

Sort by: Relevance | Date published (new to old) | Date published (old to new)

Showing 1–50 of 767 results

Research Highlights
06 Oct 2022
Nature Cell Biology
Volume: 24, P: 1448

m^{6}A in planarian cell fate

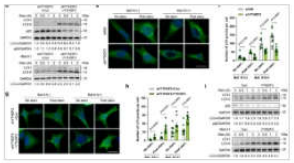
Daryl J. V. David

Research
Open Access
04 Oct 2022
Nature Communications
Volume: 13, P: 1-23

Autophagy induction promoted by m^{6}A reader YTHDF3 through translation upregulation of FOXO3 mRNA

The role of eiptranscriptomic modifications in autophagy is unclear. Here, the authors show that the m^{6}A reader YTHDF3 functions as a nutrient responder to recognize upregulated m^{6}A modification, promoting FOXO3 translation to subsequently initiate autophagy.

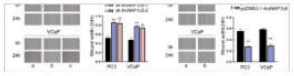
WeiChao Hao, MeiJuan Dian ... Dong Xiao



Research
Open Access
04 Oct 2022
Cancer Gene Therapy
P: 1-10

LncNAP1L6 activates MMP pathway by stabilizing the m6A-modified NAP1L2 to promote malignant progression in prostate cancer

Yuxiao Zheng, Feng Qi ... Xiao Li



Research
07 Sept 2022
Nature Genetics
Volume: 54, P: 1427-1437

RNA m^{6}A regulates transcription via DNA demethylation and chromatin accessibility

METTL3-dependent RNA N^6-methyladenosine (m^{6}A) deposition can lead to DNA demethylation. The m^{6}A reader F recruits DNA 5-methylcytosine dioxygenase TET1 to chromatin, which is linked to chromatin accessibility and gene transcription.

Shuang Deng, Jialiang Zhang ... Jian Zheng

Research

Mettl3-dependent m^{6}A modification attenuates the brain stress response in *Drosophila*

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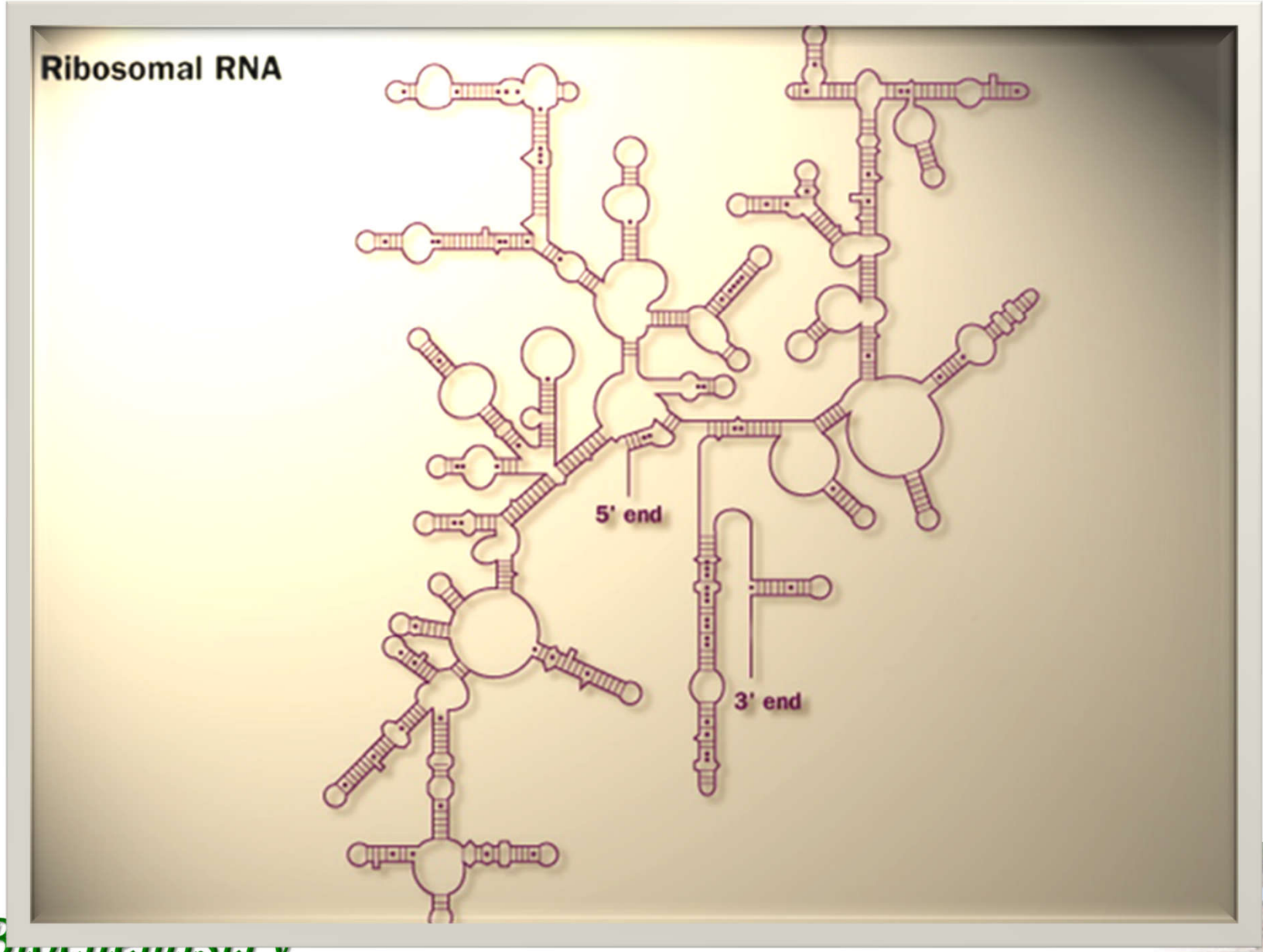
腾讯会议
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43% 19°C 15:58 2022/10/27

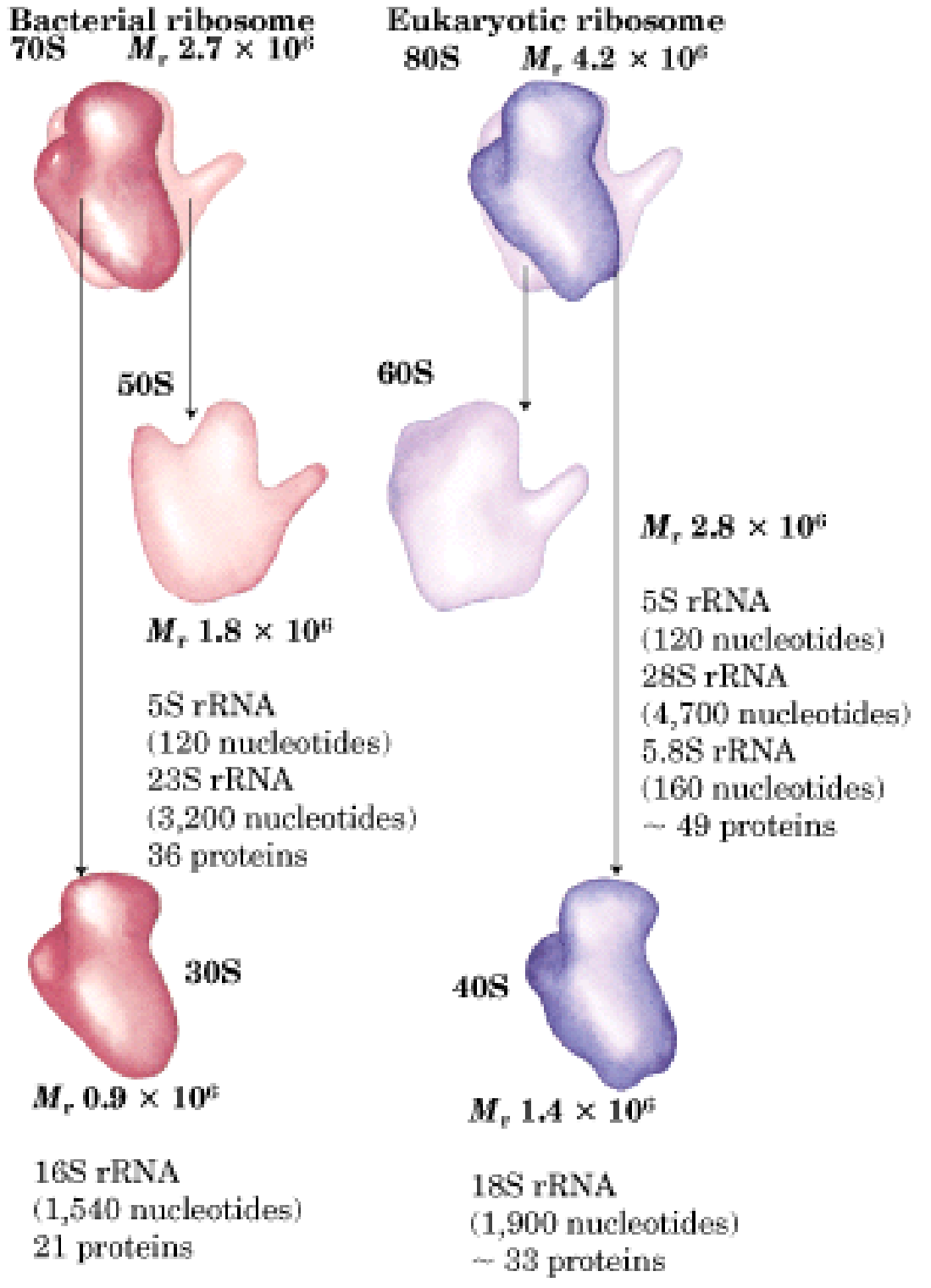
Biochemistry

G8 Ribosomal RNA



1. Ribosomes

沉降系数**S**

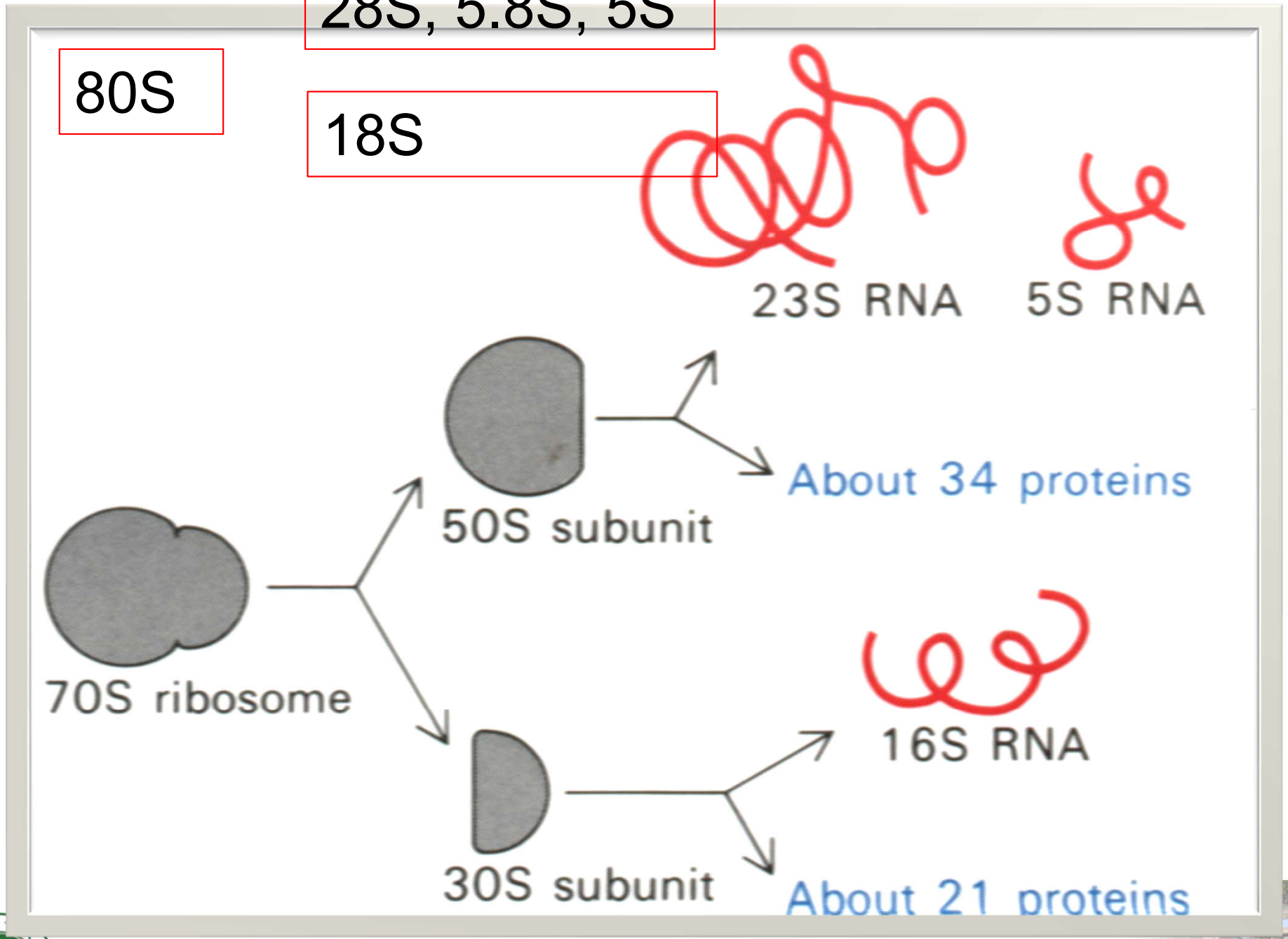


(b)

28S, 5.8S, 5S

80S

18S



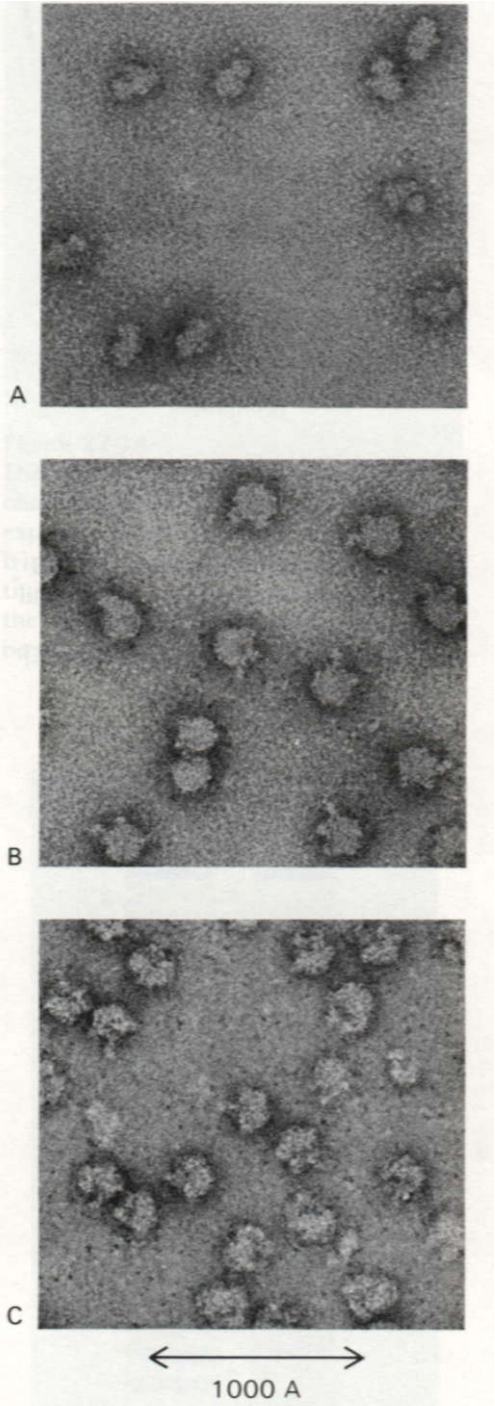
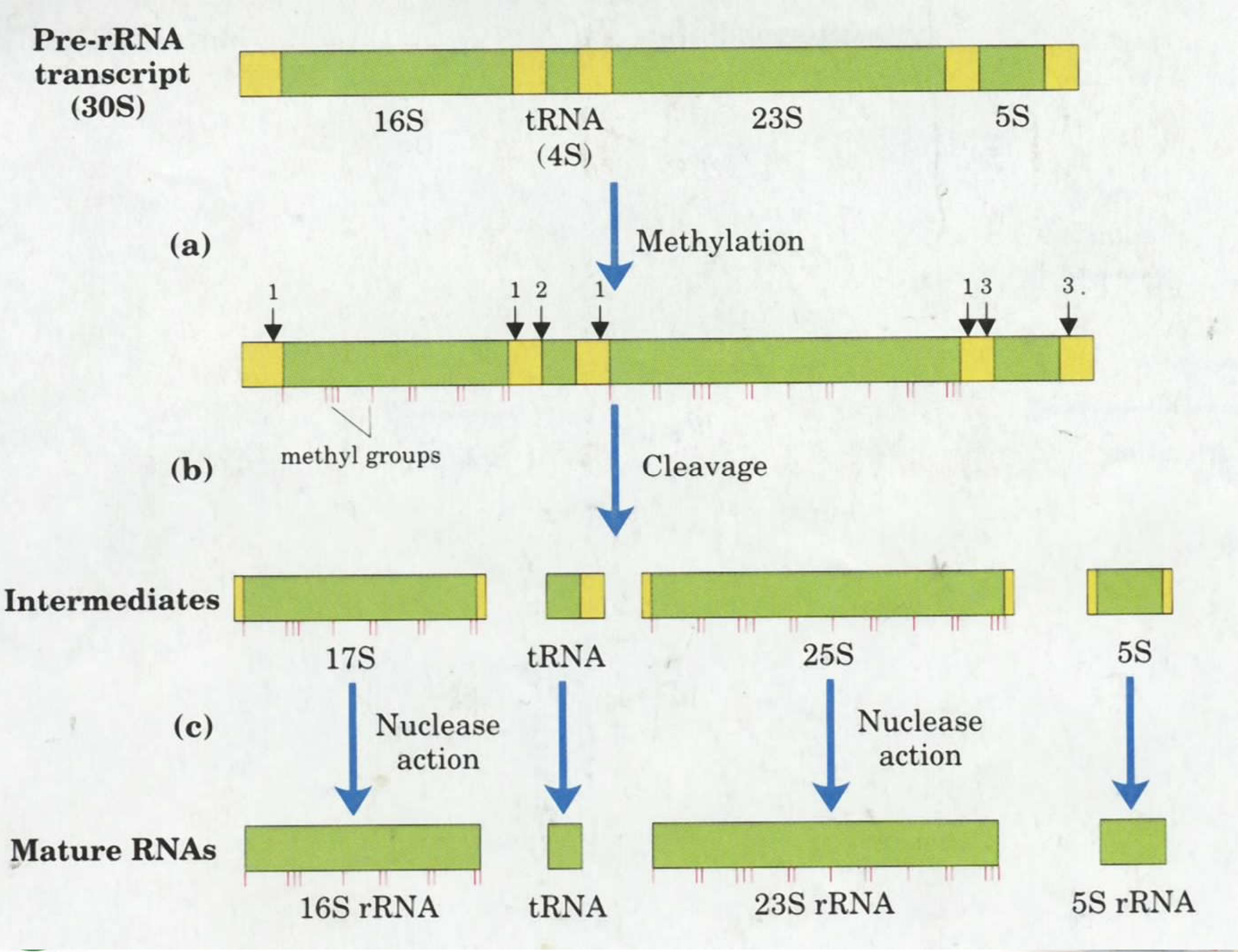


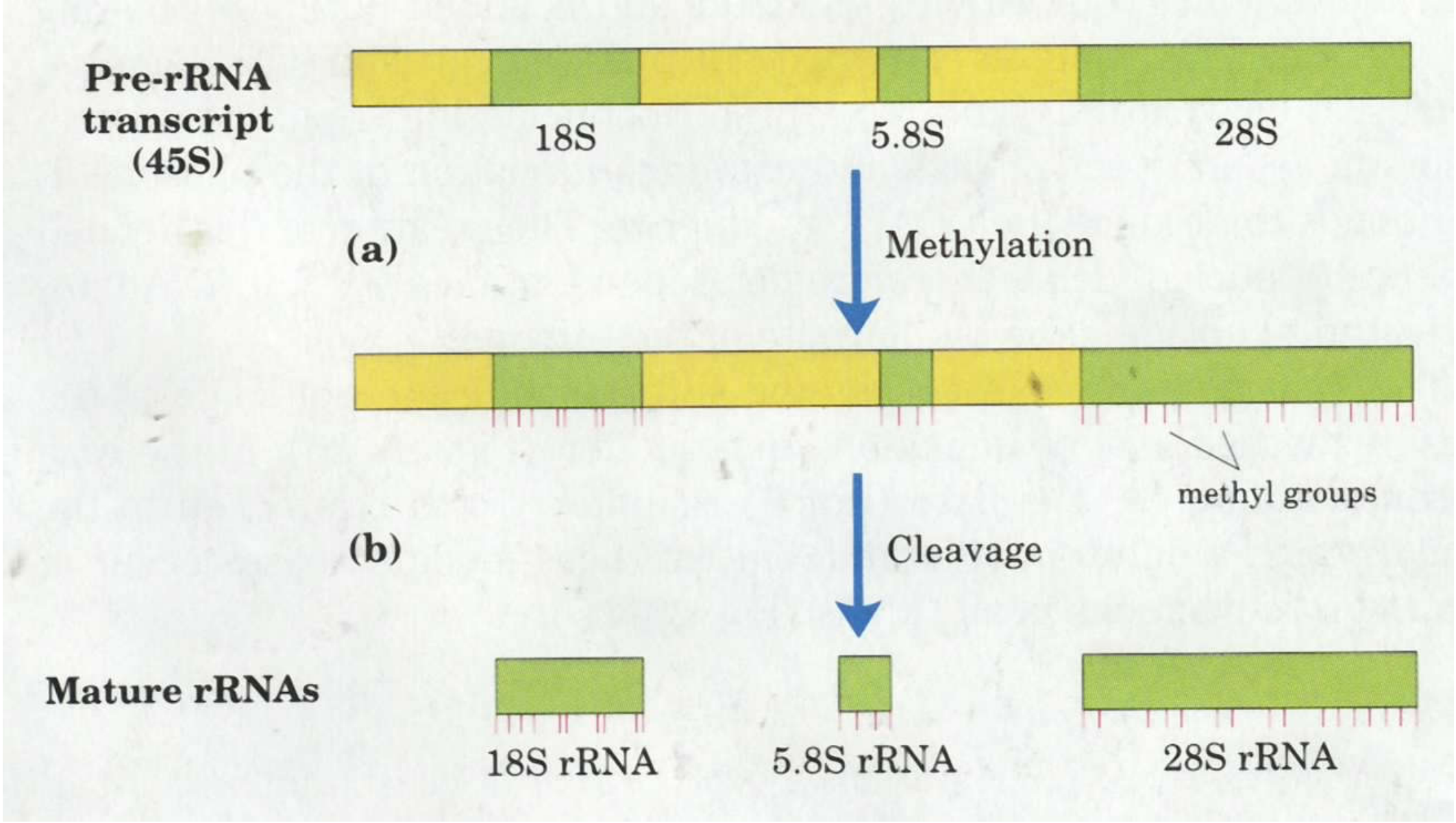
Figure 27-10
Electron micrographs of (A) 30S subunits, (B) 50S subunits, and (C) 70S ribosomes. [Courtesy of Dr. James Lake.]

2. The different rRNA molecules of both prokaryotes and eukaryotes are generated from single pre-rRNAs

- The 16S, 23S and 5S rRNAs in bacteria are all generated from a single 30S pre-rRNA (*about 6.5 kb*).
- The 18S, 28S and 5.8S rRNAs in eukaryotes are generated from a single 45S pre-rRNA (~14 kb).
- The 5S rRNA in eukaryotic cells is generated separately (*transcribed by RNA polymerase III*).




Pre-rRNA transcript (30S)
16S
tRNA (4S)
23S
5S
(a)
Methylation
1
1 2
1
1 3
3
methyl groups
(b)
Cleavage
Intermediates
17S
tRNA
25S
5S
(c)
Nuclease action
Nuclease action
Mature RNAs
16S rRNA
tRNA
23S rRNA
5S rRNA



The 18S, 5.8S, and 28S rRNAs in eukaryotic cells are derived from one pre-rRNA molecule (the processing needs small nucleolar RNA-containing proteins).

3. More RNA molecules (ribozymes) were found to be catalytic



The Nobel Prize in Chemistry 1989

"for their discovery of catalytic properties of RNA"

Press release



Sidney Altman

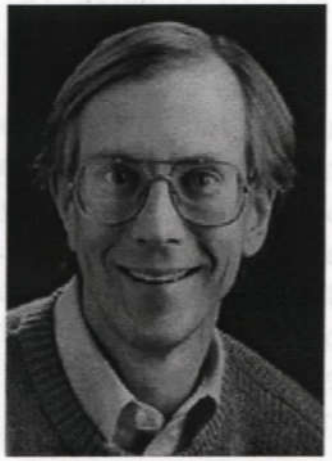
USA and Canada

Yale University
New Haven, CT, USA

1939 -

Autobiography

The M1 RNA in ribonuclease P is catalytic



Thomas R. Cech

USA

University of Colorado
Boulder; CO, USA

1947 -

Autobiography
Other Resources

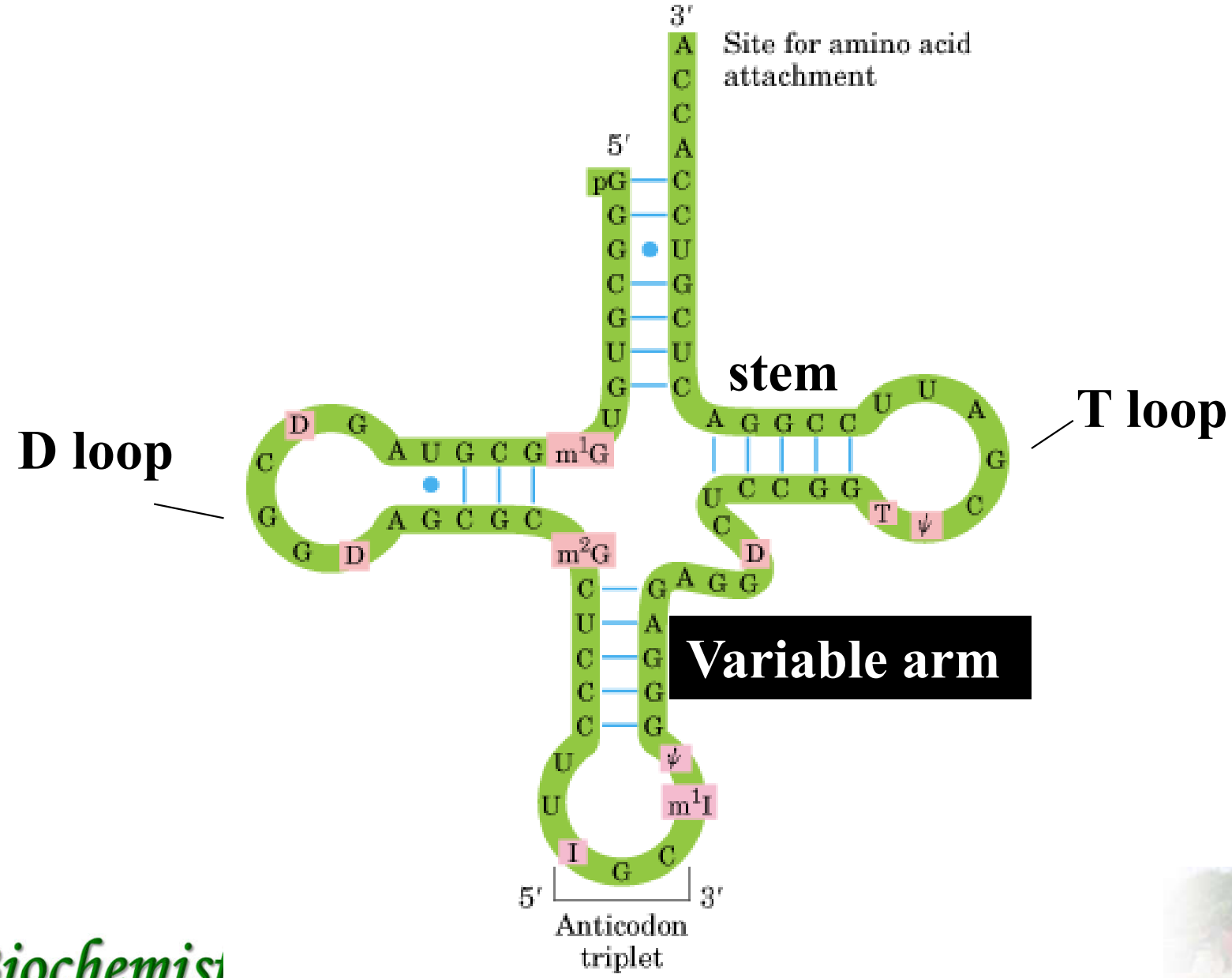
The intron in the pre-rRNA of *Tetrahemena* is self-spliced

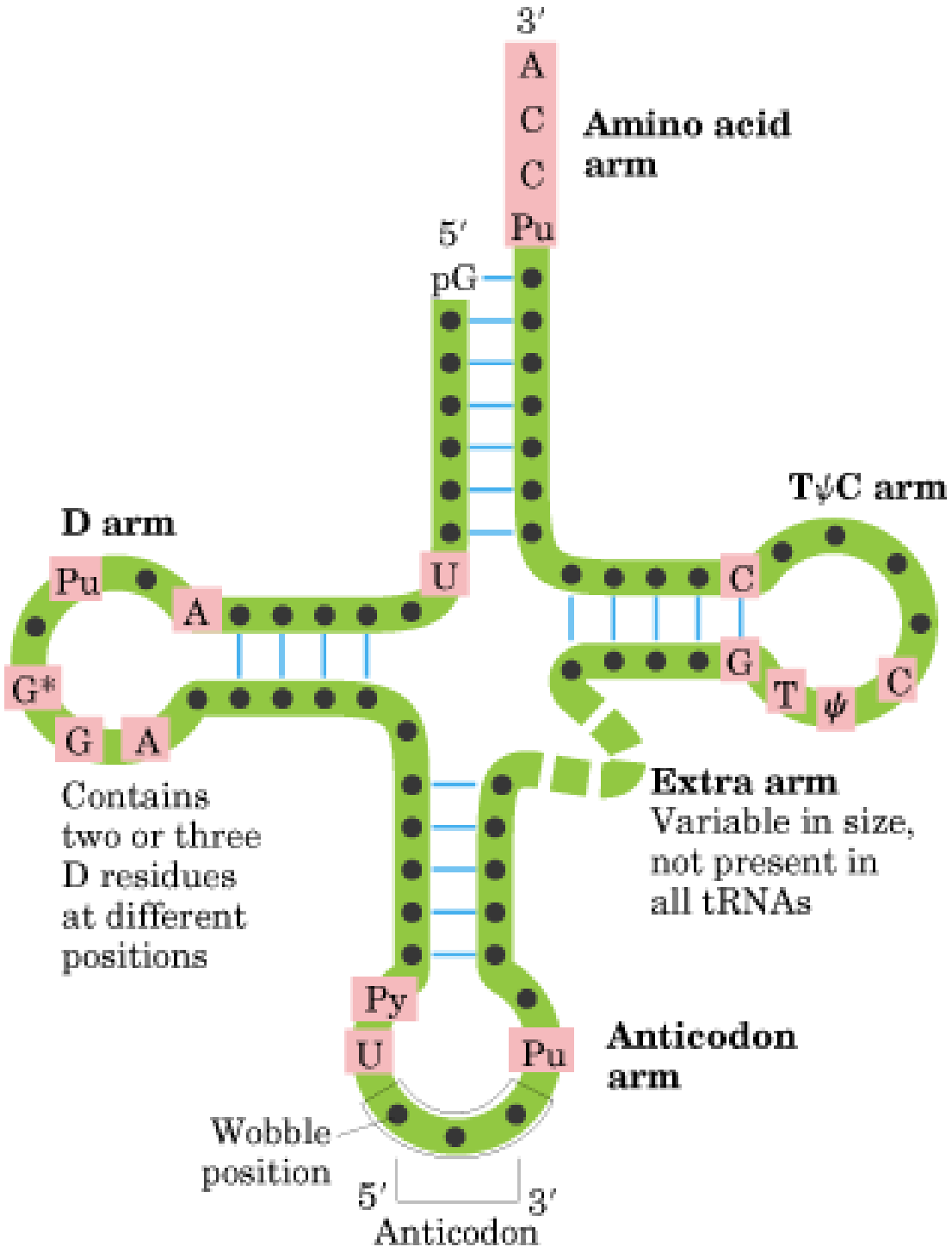


G9 Transfer RNA

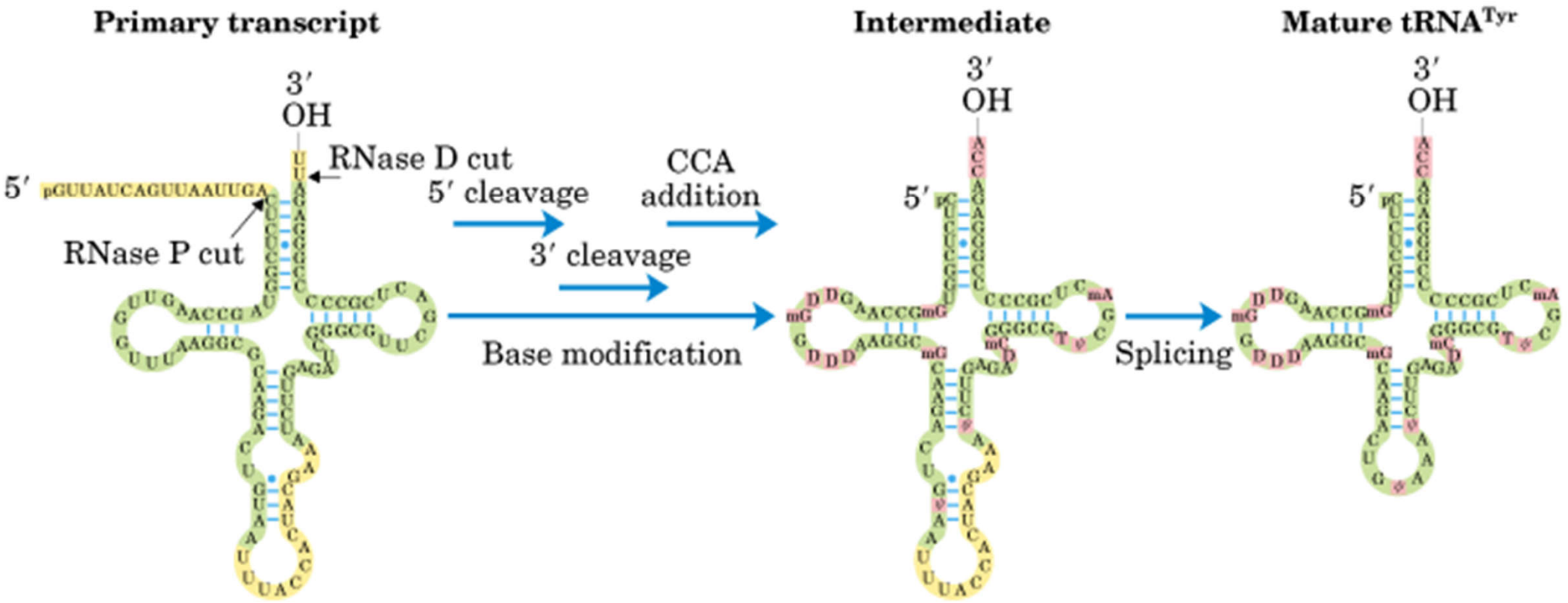


Cloverleaf secondary structure of tRNA




3′
A
C
C
Pu
Amino acid arm
5′
pG
D arm
Pu
A
U
G*
G
A
Contains two or three D residues at different positions
TψC arm
C
G
T
ψ
C
Extra arm
Variable in size, not present in all tRNAs
Py
U
Pu
Anticodon arm
Wobble position
5′
3′
Anticodon

1. Primary tRNA transcripts undergo a series of posttranscriptional processing

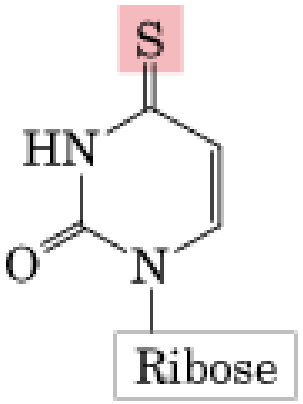


Biochemistry

Primary tRNA posttranscriptional processing

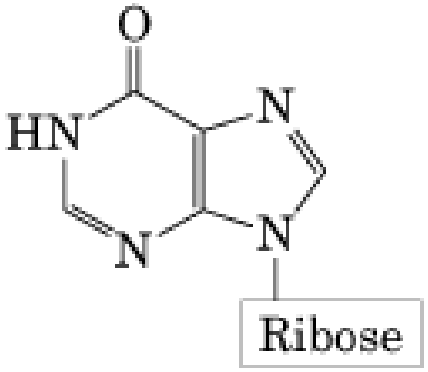
- The extra sequences at the 5` and 3` ends are removed by **RNase P** and **RNase D** respectively. **Intron sequences are spliced**.
- The **CCA** sequence is generated at the 3` end.
- Some of the bases in tRNA molecules are modified by methylation, deamination, reduction and others.





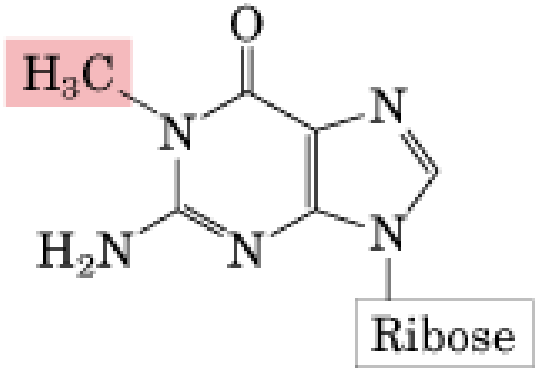
4-Thiouridine (S^4U)

4-硫尿苷

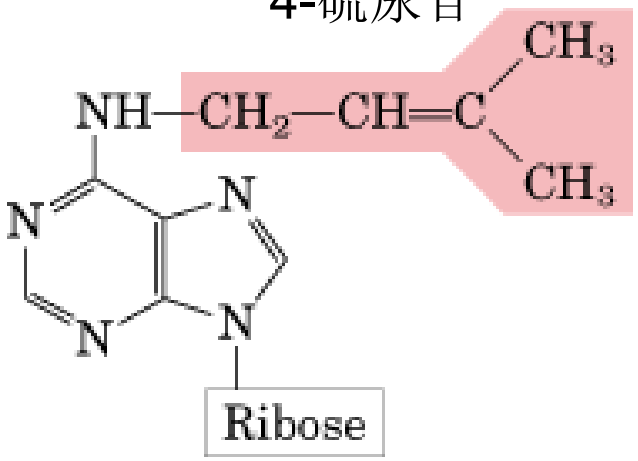


Inosine (I)

次黄（嘌呤核）苷，肌苷

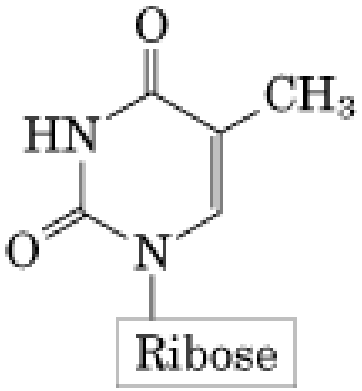


1-Methylguanosine (m^1G)

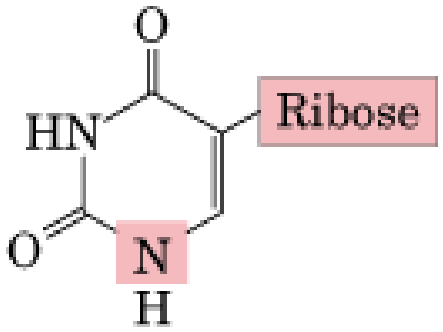


N^6-Isopentenyladenosine (i^6A)

异戊烯腺苷

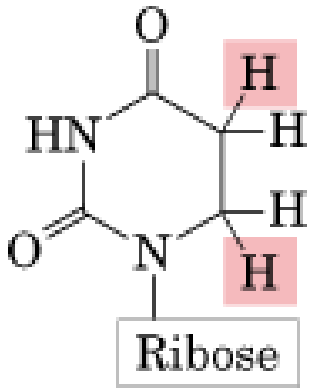


Ribothymidine (T)



Pseudouridine (ψ)

. 假尿（嘧啶核）苷



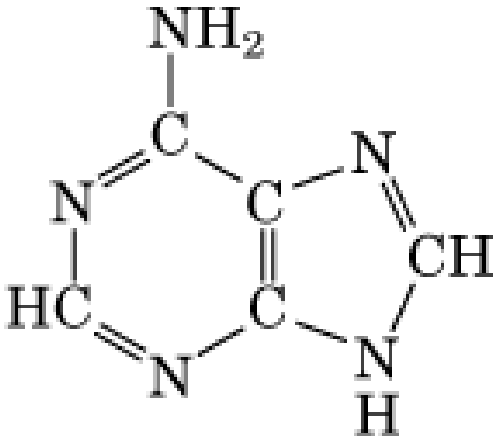
Dihydrouridine (D)

二氢尿嘧啶核苷

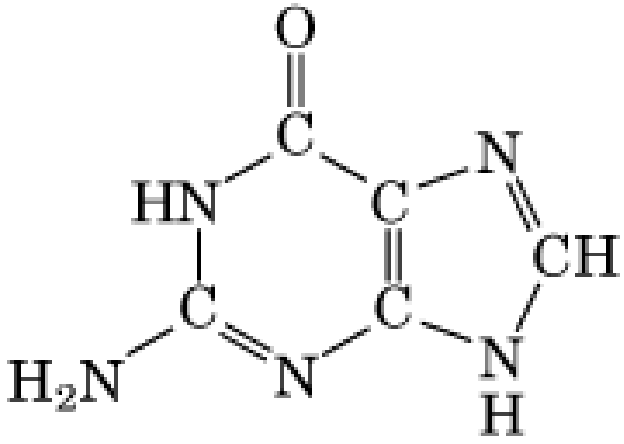
Some typical modificied bases found In mature tRNA molecules.

Biochemistry

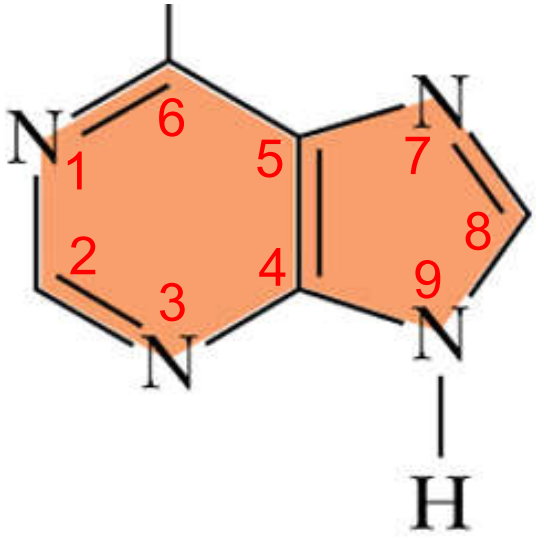




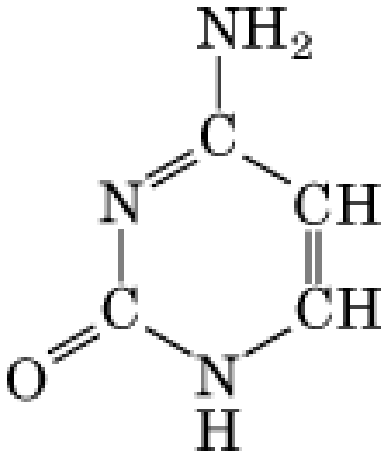
Adenine



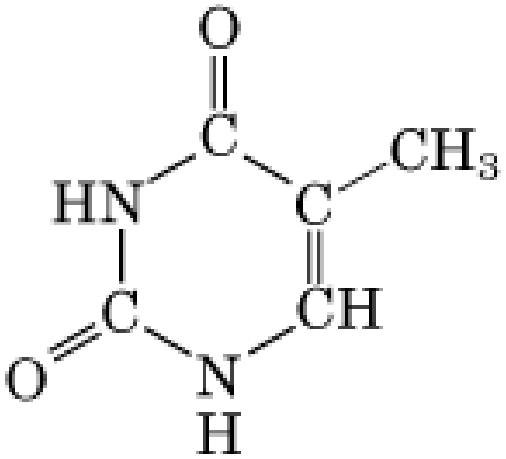
Guanine



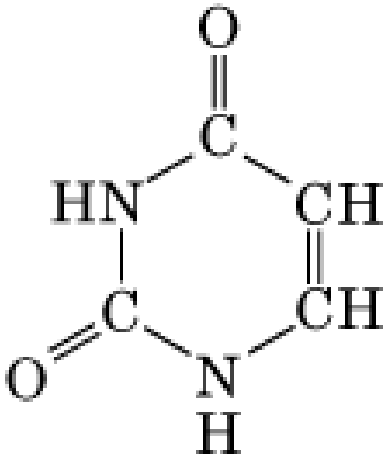
Purines



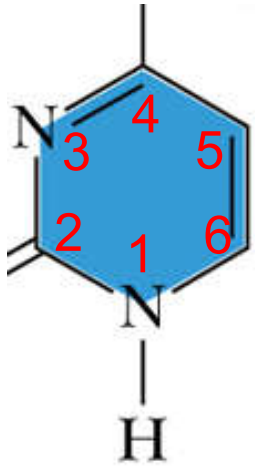
Cytosine



Thymine
(DNA)



Uracil
(RNA)



Pyrimidines

RNA 指导下的RNA和DNA的合成

RNA dependent synthesis of DNA and RNA

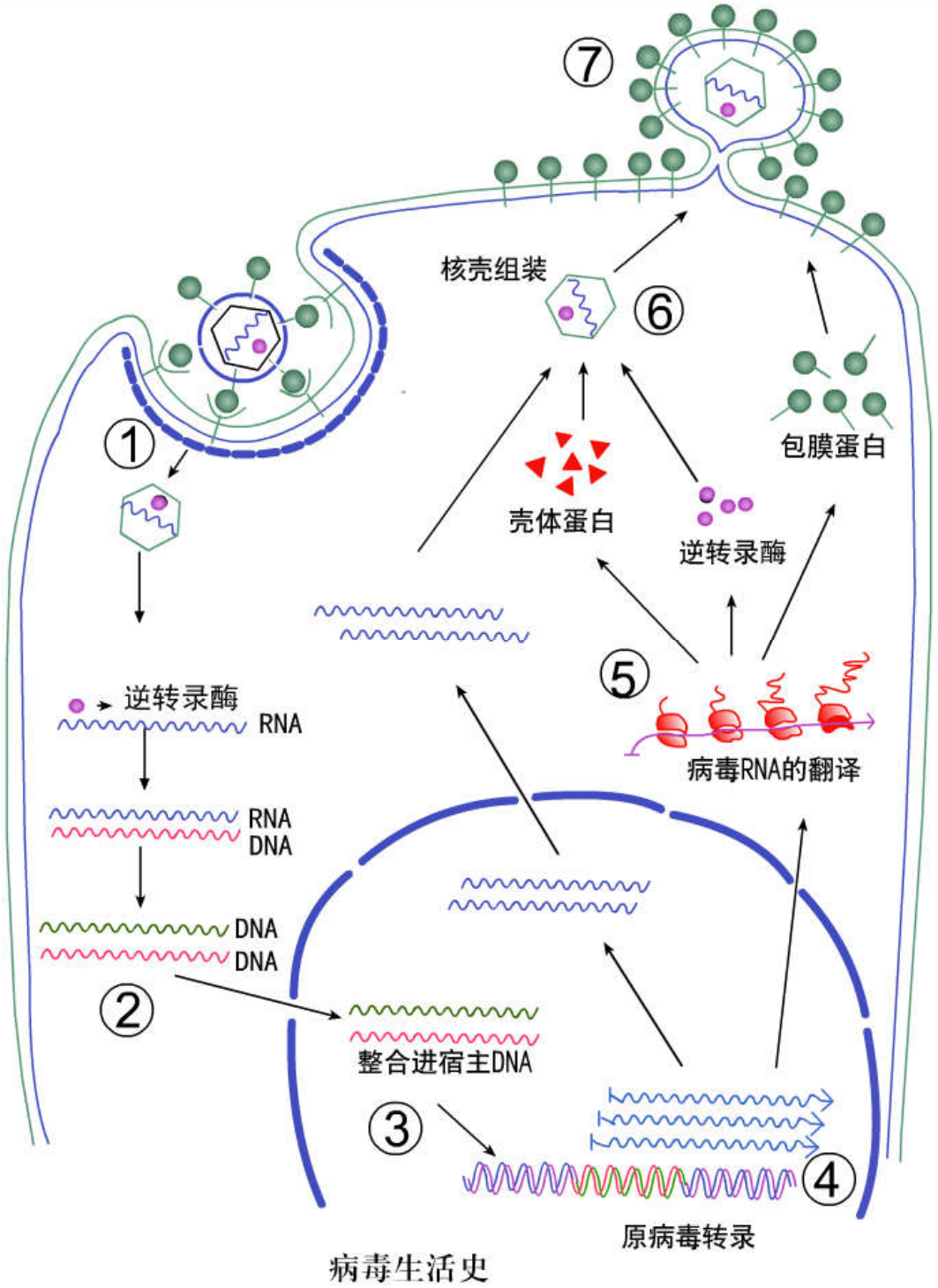


Reverse transcriptases catalyze the production of DNA from RNA

- This enzyme catalyzes three reactions:
  - RNA-directed DNA synthesis using tRNAs as primers;
  - Degradation of the RNA template;
  - DNA-directed DNA synthesis;
- The enzyme has **no 3`- 5`** proofreading exonuclease activity, thus generating high rate of mutations.
- This enzyme is widely used to synthesize **complementary DNAs (cDNAs)** from mRNAs.



逆转录病毒


⑦
核壳组装
⑥
包膜蛋白
①
壳体蛋白
逆转录酶
⑤
逆转录酶
RNA
RNA
DNA
DNA
DNA
②
病毒RNA的翻译
整合进宿主DNA
③
④
原病毒转录

病毒生活史

Some viral RNAs are replicated by RNA-directed RNA polymerase

- The RNA genomes of some viruses are replicated using RNA-directed RNA polymerases (or **RNA replicases**).
- The RNA replicase consists of subunits encoded both by the viruses and the host genome.
- They have features similar to that of DNA-directed RNA polymerases, but are usually specific for the RNA of the specific viruses.



Summary

- Transcription shares the basic chemical mechanisms with replication except: no primers required, no proofreading activity exist.
- Transcription begins at specific promoter sequences and ends at specific terminator sequences (being ρ -independent or dependent in bacteria).
- One multimeric RNA polymerase catalyzes the synthesis of all the RNA molecules in bacteria and different RNA polymerases are used to synthesize the different types of RNAs in eukaryotes.



- Primary RNA transcripts are further processed: capped, tailed, spliced and sometimes edited for the mRNAs; ends removed and modified, bases modified for the tRNAs; cleaved and sometimes spliced for the rRNAs.
- DNA can be synthesized by using RNA as templates in a reaction catalyzed by reverse transcriptase.
- RNA is very likely the first type of biomacromolecules produced during biochemical evolution.
- RNA replicase catalyzes the synthesis of RNA from RNA templates.



Summary of RNA transcription

G1 RNA structure: tRNA cloverleaf
G2 RNA Transcription in Prokaryotes
- **1. similar and different aspects between DNA replication and RNA transcription**
- **2. RNA polymerase**
- **3. transcriptional bubble**
- **4. promoter**
- **5. RNA transcription: initiation, elongation, termination**

G3 the Lac operon
G5,G6 Transcription in Eukaryotes
- **1. Three RNA polymerases**
- **2. Pol II binding promoter, transcription factor**

G7 Processing of eukaryotic pre-mRNA
- **1. mRNA processing**
- **2. rRNA processing**
- **3. tRNA processing**



END

Thank you

Biochemistry



2006年诺贝尔医学奖



美国 安德鲁·菲尔



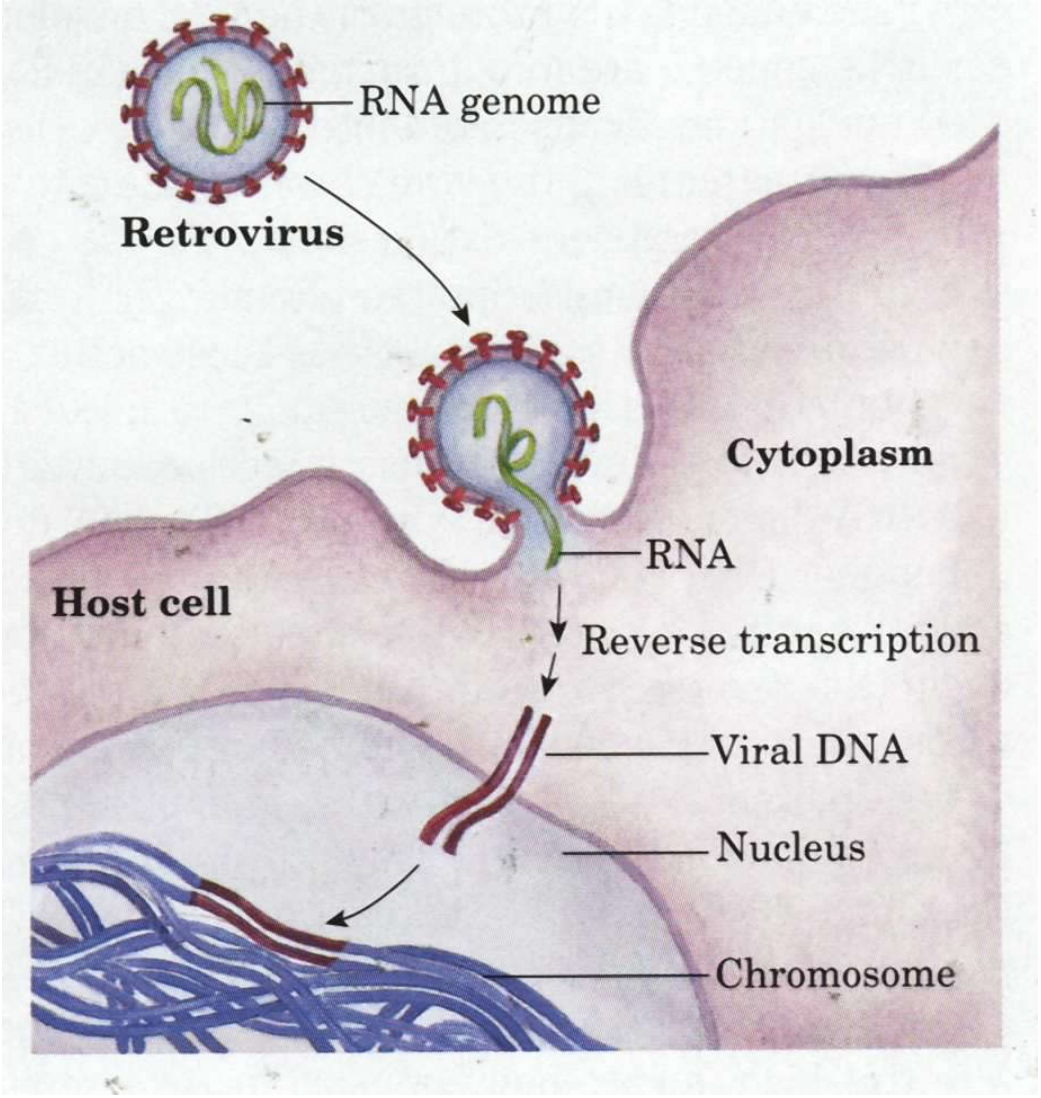
美国 克雷格·米洛

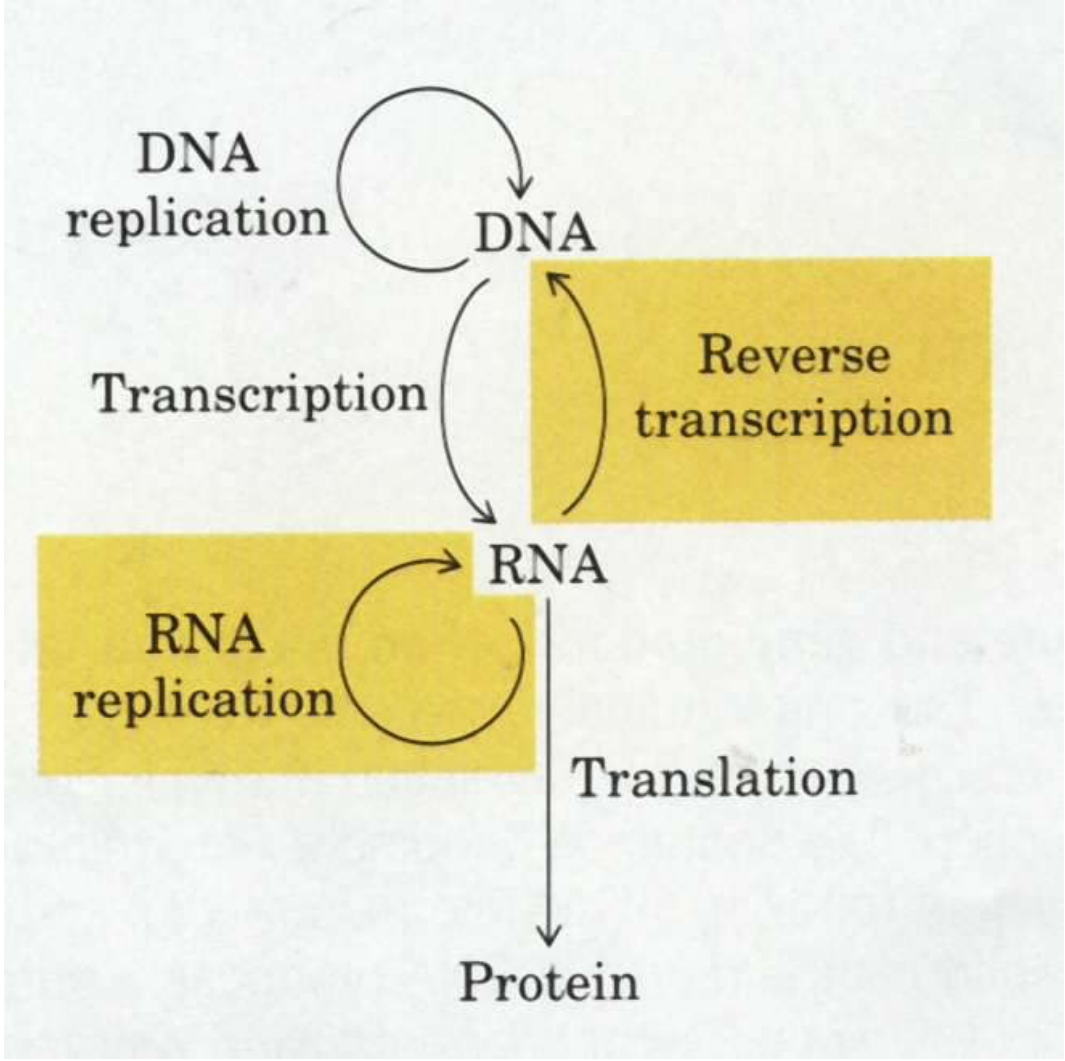
2006年诺贝尔医学奖得主：

- [美] 安德鲁·菲尔
- [美] 克雷格·米洛

获奖原因：

- 发现了控制基因信息流动的基本机制，因RNA干扰的发现而入选。RNA干扰是一个生物过程，在这个过程中双链RNA以一种非常明确的方式抑制了基因表达。自1998年发现以来，RNA干扰已经作为一种强大的“基因沉默”技术而出现。这项技术被用于全球的实验室来确定各种病症中哪种基因起到了重要作用。[详细]





Reverse transcriptases catalyzes the sythesis of DNA from RNA template.



The Nobel Prize in Physiology or Medicine 1975

"for their discoveries concerning the interaction between tumour viruses and the genetic material of the cell"

Press release

David Baltimore

USA

Massachusetts Institute of Technology
Cambridge, MA, USA

1938 -



Autobiography

Renato Dulbecco

USA

Imperial Cancer Research Fund Laboratory
London, Great Britain

1914 -





Howard Martin Temin

USA

University of Wisconsin
Madison, WI, USA

1934 - 1994



关于沉降系数S

$$F_c(\text{离心力}) = m\omega^2 x$$

$$F_b(\text{浮力}) = \mathrm{V}\rho\omega^2 x = m\bar{v}\rho\omega^2 x$$

$$F_f(\text{摩擦力}) = fv = f\frac{dx}{dt}$$

$$F_c - F_b = F_f$$



$$F_c - F_b = F_f$$

$$\frac{dx/dt}{\omega^2 x} = \frac{m(1-\bar{v}\rho)}{f}$$

m 是分子颗粒的质量（g）；ω 是转头的角速度；x 是旋转中心至界面的径向距离（cm）；$\omega^2 x$ 是离心加速度（也称离心场强度或离心场，是单位质量的力）；V 是分子颗粒体积；ρ 是溶剂密度（g/cm3）；$\bar{v}$ 是蛋白质的偏微比容；$V\rho$ 或 $m\bar{v}\rho$ 是被分子颗粒排开的溶剂质量；f 是摩擦系数；v 是沉降速率，即 dx/dt 。



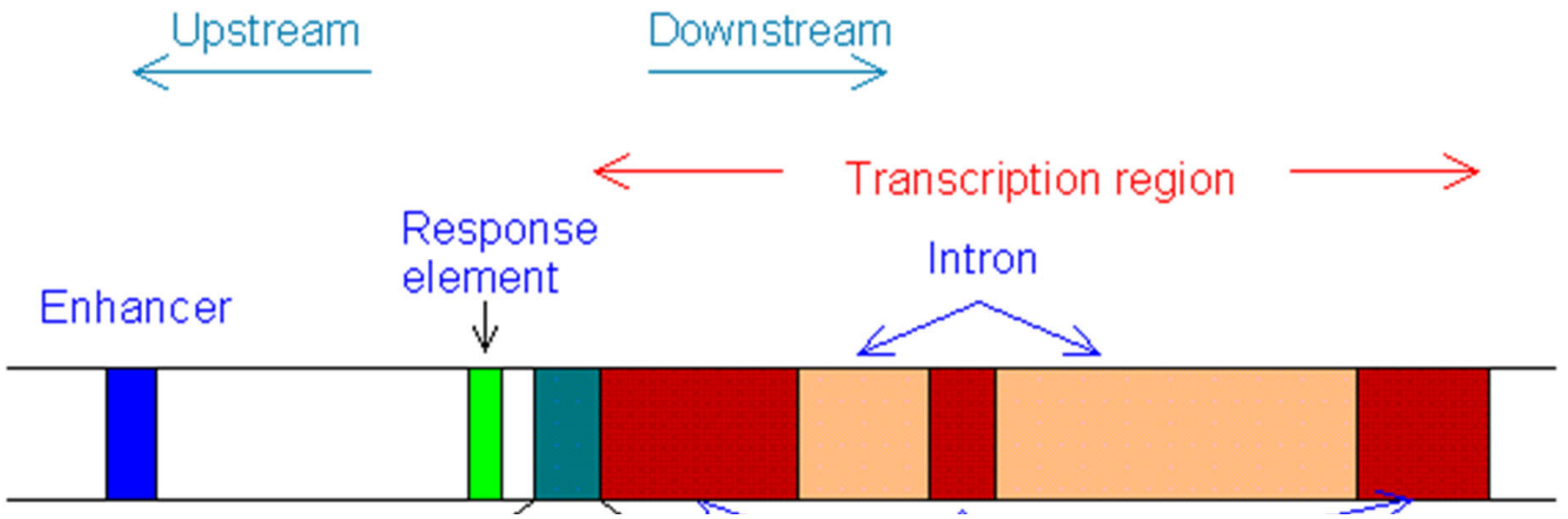
可见单位离心场的沉降速度是个定值，称为沉降系数（sedimentationcoefficient）。用 s（小写）表示。

蛋白质、核酸核糖体和病毒等的沉降系数介于1×10^{-13}到200×10^{-13}秒的范围。为方便起见，10^{-13}作为一个单位，称为（Svedberg unit）或沉降系数单位，用 S（大写）表示。



Enhancers (upstream activator sequences (UASs) in yeast)

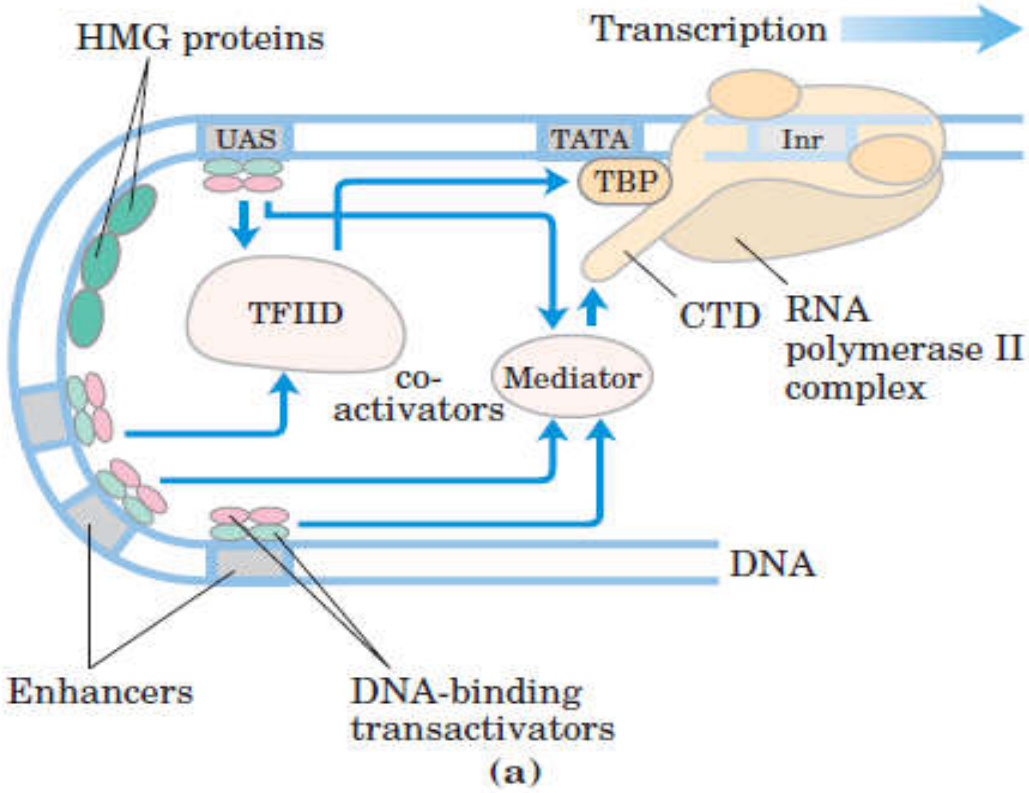
- Although many positive control elements lie close to the gene they regulate, others can be located long distances away (10-50kb) either upstream or downstream of the gene.



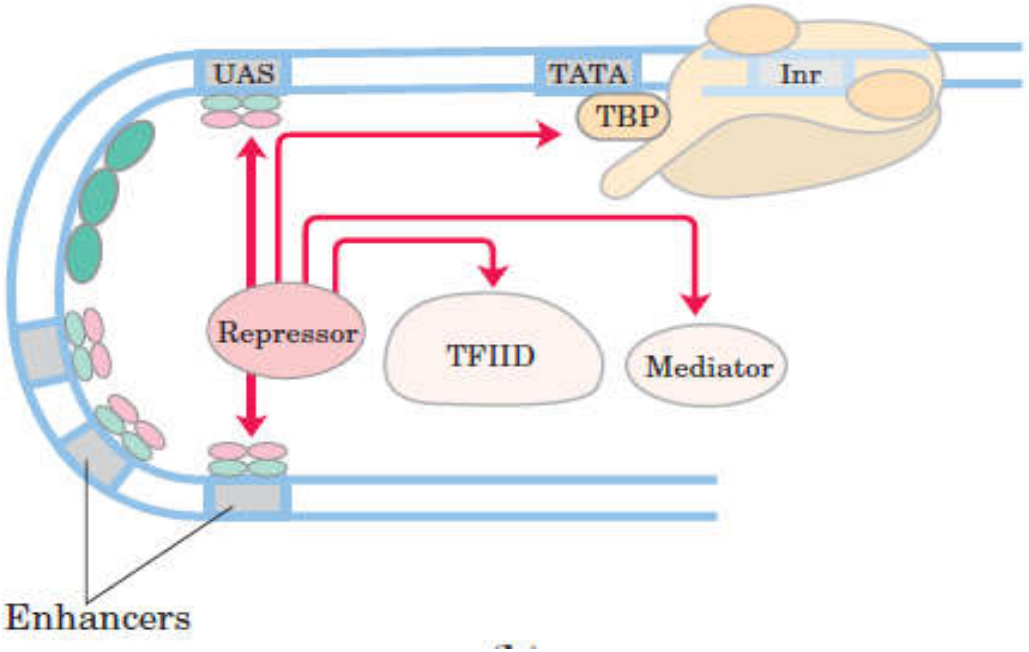
- Repressors near the gene called negative regulators.

- Repressors long distance away from the gene called silencers (沉默子）.

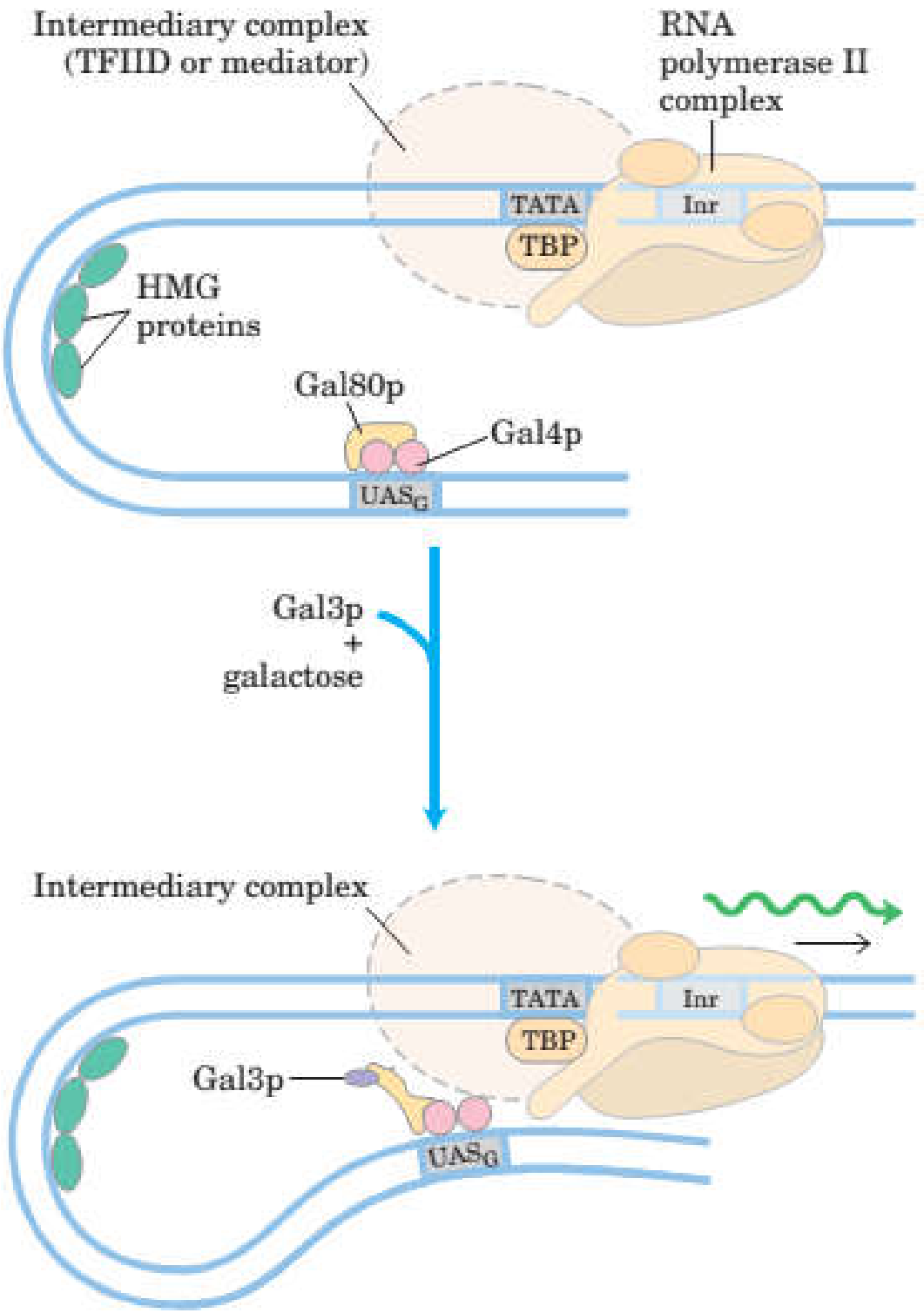




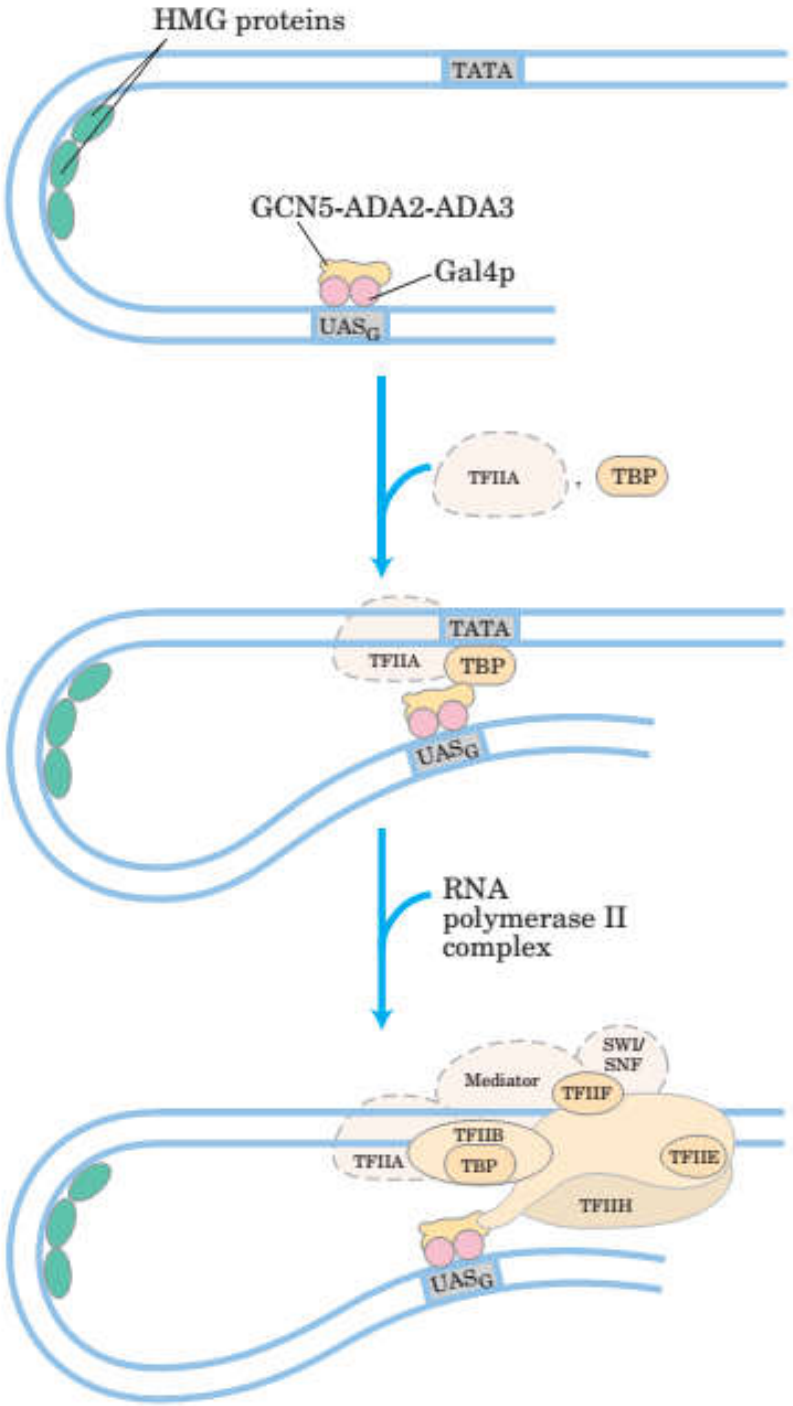
TBP: TATA-Binding Protein;



Regulation of transcription at genes of galactose metabolism in yeast.



Protein complexes involved in transcription activation of a group of related eukaryotic genes



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